


Master's Thesis

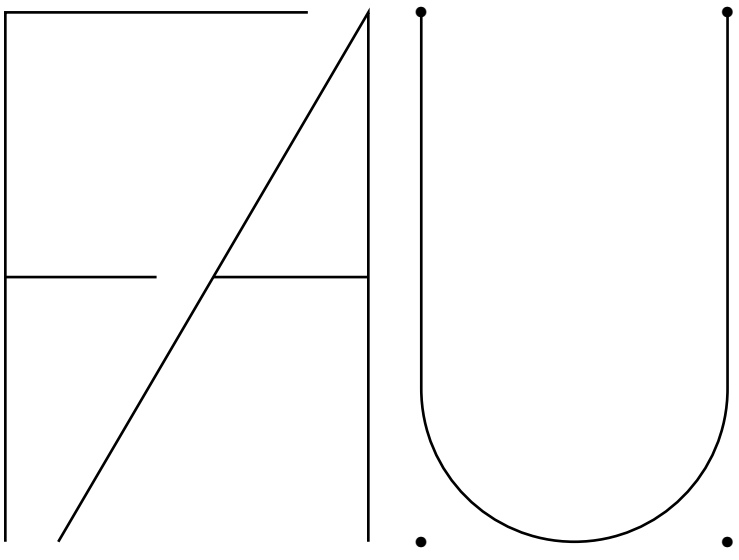
# Learning Contact Representations from Zero-Cost Proprioceptive Effort Signals for Contact-Rich Visuomotor Imitation Learning

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# Learning Contact Representations from Zero-Cost Proprioceptive Effort Signals for Contact-Rich Visuomotor Imitation Learning


**Keywords:** Low-Cost Robotics, Contact Representation Learning, Multimodal Fusion, Visuomotor Imitation Learning, Proprioceptive Effort Signals, Contact-Phase Understanding.


**Motivation.** Teaching robots to perform manipulation tasks that require precise physical contact remains a major challenge. Examples include assembling parts, grasping fragile objects, or inserting components into tight tolerances. In such tasks, success does not only depend on moving the robot to the right position, but also on whether the robot actually touches, holds, presses, or releases an object in the right way. Imitation learning (IL) is a promising approach for teaching such skills. Instead of manually programming every movement, the robot learns a control policy from human demonstrations. Recent policies such as the Action Chunking Transformer (ACT) [1] and Diffusion Policy [2] can learn complex manipulation behaviours from comparatively few demonstrations. However, these methods usually rely mainly on camera images and joint positions. Both sources of information can become unreliable when physical contact matters most: the gripper or the object may hide the relevant contact area from the camera [3], and joint positions alone do not reveal whether an object was actually grasped or whether the robot merely moved to the commanded position.

Recent research shows that adding contact-related information can substantially improve robot learning in such situations. For example, tactile sensing, joint-torque information, and force feedback have been shown to improve success rates in contact-rich manipulation tasks [4], [5], [6]. These results suggest that contact information is an important missing signal for visuomotor policies. However, most existing solutions require additional sensing hardware, calibrated force or tactile sensors, or expensive robot platforms with built-in torque sensing. This makes them difficult to use in low-cost and open-source robot learning setups. A possible alternative is to use signals that are already available on many low-cost robot platforms. For instance, bus servos often expose motor-current or effort readings through their communication registers at no additional hardware cost. These readings are not precise force measurements and are expected to be noisy and low-dimensional. Nevertheless, preliminary logging indicates that they differ visibly between successful grasps and contact misses, even when the commanded joint positions are identical [7].

This creates an open research gap: while contact-aware representations are known to improve imitation learning, it remains unclear whether weak and uncalibrated proprioceptive effort signals can provide a useful learned contact representation. This thesis addresses this gap by investigating whether such zero-cost signals can improve contact-rich visuomotor imitation learning on a low-cost robot platform.

**Related Work.** Prior work on contact-aware imitation learning shows that non-visual contact signals can substantially improve manipulation performance. Haptic-ACT [6], FTACT [8], tactile imitation learning [4], and reactive contact-aware policies [9] demonstrate that haptic, tactile, or force/torque information helps in tasks where vision and robot state are insufficient. FACTR [5] shows similar benefits for ACT-style policies using joint torque, while FoAR [10] demonstrates phase-aware force fusion for diffusion-based policies. Surveys by Xie and Correll [11] and Tsuji et al. [3] consolidate these findings and identify force and tactile sensing as important modalities for assembly, insertion, and other contact-rich tasks. However, these approaches typically assume dedicated sensors, calibrated force/torque measurements, or robot platforms that already provide reliable torque information. The literature also shows that the

way contact signals are fused with visuomotor policies matters. FACTR [5] highlights the risk of modality collapse, where a policy receives an additional modality but effectively ignores it. FoAR [10] shows that contact information is not equally useful throughout the whole task and that phase-aware fusion can outperform static fusion. Other approaches, such as Haptic-ACT and FTACT, use direct concatenation of haptic or force/torque features. These results leave open whether simple fusion is sufficient for weak, uncalibrated effort readings, or whether such signals require a learned representation to become useful for the policy. A related line of work studies sensorless contact estimation. Classical methods recover contact torques from motor commands and known robot dynamics [12]; motor-current-based approaches can substitute for direct torque sensing when motor constants and mechanical properties are known [13]; and learning-based estimators often require force/torque-labelled data [14], [15]. Yamane et al. [16] similarly rely on identified manipulator dynamics for sensorless bilateral teleoperation. These works show that motor-side signals can contain contact information, but they do not directly address raw bus-servo registers on low-cost platforms, where accurate dynamics, calibration, and force/torque supervision are unavailable.

The remaining gap is whether weak, noisy, zero-cost effort signals from low-cost bus servos can be used directly as learned contact representations in visuomotor imitation learning. In particular, it is still unclear how such signals should be fused with vision and joint state, whether policies actually use them rather than ignore them, and during which manipulation phases they provide measurable benefit.

**Overall Goal.** This thesis investigates whether low-dimensional proprioceptive effort signals serve as a scalable contact representation for visuomotor IL on low-cost robot hardware.

The student will first characterise the STS3215 motor-current signals as policy input features. This includes analysing their signal-to-noise ratio, temporal behaviour, and correlation with task phases such as approach, grasp, insertion, and release. The student must determine whether these signals contain sufficient contact-discriminative information to be useful for end-to-end policy learning. The student will then implement and compare several policy configurations. As a baseline, the student will train a vision-and-position variant of ACT. This baseline must be compared against a multimodal ACT variant that additionally receives effort signals. The student will first evaluate direct concatenation as a simple fusion strategy and then implement a learned effort encoder as an ablation. To test whether observed improvements are caused by the effort signal itself rather than by an architecture-specific effect, the student will replicate the same comparison with Diffusion Policy. The student will further analyse whether the policies actually use the effort modality. For transformer-based models, this must include modality-use diagnostics such as attention-weight analysis or an equivalent ablation-based diagnostic. This step is necessary to distinguish genuine contact-signal utilisation from modality collapse, where the additional input is present but effectively ignored by the policy.

The evaluation must include repeated real-world trials and phase-conditioned success annotations. The student will analyse which phases of contact-rich manipulation benefit from proprioceptive fusion and which phases remain primarily driven by vision. This provides empirical evidence on when weak effort signals add useful information beyond standard visuomotor observations. All policy variants must be trained and evaluated on a single multimodal dataset collected via leader–follower teleoperation on the SO-101. The dataset will contain synchronised camera observations, joint states, actions, and bus-servo effort readings using an extended LeRobot-compatible dataset schema [17]. Feature subsetting will ensure that all policy condi-

tions are compared on the same demonstrations.

**Timetable (6 months, in person-weeks [PW]).**

- [2 PW] Literature review: contact representations in IL, multimodal fusion, and force-aware policy learning.
- [2 PW] Learnability analysis of bus-servo registers: signal-to-noise ratio, temporal dynamics, normalisation, and relation to task phases.
- [4 PW] Task setup and multimodal dataset collection: leader, follower teleoperation with synchronised camera, joint-state, action, and effort-register logging using an extended LeRobot [17] schema.
- [5 PW] Implementation and training of policy variants: ACT and Diffusion Policy baselines, effort-aware variants, direct concatenation, and learned-encoder ablations.
- [4 PW] Evaluation: $2 \times 2$ cross-architecture comparison, repeated real-world trials, per-phase success annotation, fusion ablations, and modality-use diagnostics.
- [7 PW] Writing of the final thesis and documentation.

**Expected Results and Scientific Contributions.**

- A documented software framework for recording and using bus-servo effort signals in visuomotor imitation learning.
- An extended LeRobot-compatible dataset schema for synchronised camera observations, joint states, actions, and effort-register readings [17]; a multimodal demonstration dataset for a contact-rich manipulation task, including task-phase annotations for evaluation.
- Implemented and trained policy variants based on ACT and Diffusion Policy, each with and without proprioceptive effort input.
- Documentation of an experimental evaluation showing whether effort signals improve task success compared to vision-and-position baselines; an analysis of fusion strategies and modality use, showing whether direct concatenation is sufficient or whether a learned effort representation is beneficial; a phase-wise analysis showing during which parts of the task effort signals are most useful.
- A final thesis and accompanying documentation describing the setup, dataset, implementation, experiments, and results.

## References

[1] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning Fine-Grained Bimanual Manipulation With Low-Cost Hardware," in *Proc. Intl. Conf. Robotics: Science and Systems (RSS)*, Daegu, Republic of Korea, Jul. 2023. DOI: 10.15607/RSS.2023.XIX.016

[2] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. C. M. Burchfiel, and S. Song, "Diffusion Policy: Visuomotor Policy Learning via Action Diffusion," in *Proc. Intl. Conf. Robotics: Science and Systems (RSS)*, Daegu, Republic of Korea, Jul. 2023. DOI: 10.15607/RSS.2023.XIX.026

[3] T. Tsuji, Y. Kato, G. Solak, H. Zhang, T. Petrič, F. Nori, and A. Ajoudani, "A Survey on Imitation Learning for Contact-Rich Tasks in Robotics," *The Intl. J. of Robotics Research*, 2026, OnlineFirst. DOI: 10.1177/02783649261417694

[4] N. Funk, C. Chen, T. Schneider, G. Chalvatzaki, R. Calandra, and J. Peters, *On the Importance of Tactile Sensing for Imitation Learning: A Case Study on Robotic Match Lighting*, 2025. DOI: 10.48550/arXiv.2504.13618 arXiv: 2504.13618 [cs.RO].

[5] J. J. Liu, Y. Li, K. Shaw, T. Tao, R. Salakhutdinov, and D. Pathak, "FACTR: Force-Attending Curriculum Training for Contact-Rich Policy Learning," in *Proc. Intl. Conf. on Robotics: Science and Systems (RSS)*, Los Angeles, CA, USA, 2025. DOI: 10.15607/RSS.2025.XXI.079

[6] P. M. Uriguen Eljuri, H. Shibata, K. Maeyama, Y. Jia, and T. Taniguchi, *Haptic-Informed ACT With a Soft Gripper and Recovery-Informed Training for Pseudo Oocyte Manipulation*, Accepted at IEEE/RSJ Intl. Conf. on Intelligent Robots and Systems (IROS), 2025. DOI: 10.48550/arXiv.2506.18212 arXiv: 2506.18212 [cs.RO].

[7] J. Ranpariya, "Preliminary STS3215 Current and Load Register Characterisation on the SO-101 During Contact Events," Unpublished preliminary measurement, Friedrich-Alexander-Universität Erlangen-Nürnberg, 2026.

[8] R. Watanabe, M. Alvarez, P. Ferreiro, P. Savkin, and G. Sano, *FTACT: Force Torque Aware Action Chunking Transformer for Pick-and-Reorient Bottle Task*, 2025. DOI: 10.48550/arXiv.2509.23112 arXiv: 2509.23112 [cs.RO].

[9] H. Xue, J. Ren, W. Chen, G. Zhang, F. Yuan, G. Gu, H. Xu, and C. Lu, "Reactive Diffusion Policy: Slow-Fast Visual-Tactile Policy Learning for Contact-Rich Manipulation," in *Proc. Intl. Conf. on Robotics: Science and Systems (RSS)*, Los Angeles, CA, USA, 2025. DOI: 10.15607/RSS.2025.XXI.052

[10] Z. He, H. Fang, J. Chen, H.-S. Fang, and C. Lu, "FoAR: Force-Aware Reactive Policy for Contact-Rich Robotic Manipulation," *IEEE Robotics and Automation Letters*, 2025. DOI: 10.1109/LRA.2025.3564197 arXiv: 2411.15753 [cs.RO].

[11] W. Xie and N. Correll, *Towards Forceful Robotic Foundation Models: A Literature Survey*, 2025. DOI: 10.48550/arXiv.2504.11827 arXiv: 2504.11827 [cs.RO].

[12] A. De Luca and R. Mattone, "Sensorless Robot Collision Detection and Hybrid Force/Motion Control," in *IEEE Proc. Intl. Conf. on Robotics and Automation (ICRA)*, Barcelona, Spain, 2005, pp. 999–1004. DOI: 10.1109/ROBOT.2005.1570247

[13] A. Wahrburg, J. B"os, K. D. Listmann, F. Dai, B. Matthias, and H. Ding, "Motor-Current-Based Estimation of Cartesian Contact Forces and Torques for Robotic Manipulators and Its Application to Force Control," *IEEE Trans. on Automation Science and Engineering*, vol. 15, no. 2, pp. 879–886, 2018. DOI: 10.1109/TASE.2017.2691136

[14] S. Shan and Q.-C. Pham, "Fine Robotic Manipulation Without Force/Torque Sensor," *IEEE Robotics and Automation Letters*, vol. 9, no. 2, pp. 1206–1213, 2024. DOI: 10.1109/LRA.2023.3341770

[15] J. Lu, X. Jiang, and T. Zou, "Learning-Driven Sensorless Interaction Force Estimation for Low-Cost Robot Arm With Limited Dynamic Features," *Mechatronics*, vol. 111, p. 103 396, 2025. DOI: 10.1016/j.mechatronics.2025.103396

[16] K. Yamane, Y. Li, M. Konosu, K. Inami, J. Oaki, S. Sakaino, and T. Tsuji, *Fast Bilateral Teleoperation and Imitation Learning Using Sensorless Force Control via Accurate Dynamics Model*, 2025. DOI: 10.48550/arXiv.2507.06174 arXiv: 2507.06174 [cs.RO].

[17] R. Cadene et al., *LeRobot: An Open-Source Library for End-to-End Robot Learning*, 2026. DOI: 10.48550/arXiv.2602.22818 arXiv: 2602.22818 [cs.RO].

# Abstract

Contact-rich manipulation requires information about interactions that camera images and joint positions may not reveal. Force/torque and tactile sensors provide such information but require additional sensing hardware. Low-cost bus servos already expose motor-current readings; however, these uncalibrated signals combine external contact with pose-dependent loading, motion, friction and transmission effects. This thesis investigates whether they can support useful learned representations for contact-rich visuomotor imitation learning.

The study combines signal characterisation on a low-cost SO-101 arm with policy training from 100 multimodal peg-in-hole demonstrations. Vision-and-position and current-aware policies share the same demonstrations. We examine direct concatenation and learned temporal encoding in Action Chunking Transformer (ACT), alongside a Diffusion Policy system comparison. Input-replacement diagnostics and physical trials assess policy dependence on current and task success.

Our results show that motor current exhibits contact-related variation and influences the predicted actions of the tested current-aware policies. Direct concatenation shows lower observed ACT success. The strongest favourable result comes from joint-selective temporal ACT, where success rises from 23.3% to 40.0%, a 71.4% relative increase. The current-aware Diffusion Policy system also records higher observed success than its control.

The thesis contributes platform-specific signal evidence, a multimodal dataset and a software framework for separating available information, learned policy dependence and closed-loop benefit. The findings favour further investigation of learned current representations while leaving their reliable performance advantage open.

**Keywords:** multimodal imitation learning; proprioceptive effort; contact-rich manipulation; Action Chunking Transformer; Diffusion Policy; SO-101.

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# Declaration on the Use of AI Tools

I disclose the following use of generative AI tools during the preparation of this thesis.

- **Literature review:** FhGenie, a Fraunhofer-internal interface to several language models, was used to help organise the literature taxonomy and identify claims requiring source verification.
- **Planning, programming, analysis, and writing:** Anthropic Claude Code and OpenAI Codex were used to review plans and source code, suggest analyses, draft or revise prose, and check consistency across the manuscript and repository. Their outputs were treated as suggestions and were accepted only after review against the experimental evidence and thesis scope. Prose drafting and structural revision covered the English Abstract and Chapters 1–9; assistance with front matter and accompanying source-archive documentation is also included in this disclosure. AI-assisted drafting also supported the German summary supplied with the source archive.
- **Image inspection:** Representative frames were selected from the recorded videos using reproducible frame indices, signal traces, and direct visual inspection, and the author determined their interpretation and captions, AI tools were not used for image inspection.
- **Verification:** numerical claims, plots, model settings, and evaluation summaries were checked against the retained data, code, checkpoints, and commands. Its added values come from checkpoint execution; generative AI supplied no unobserved experimental outcomes.

I reviewed and edited all submitted text and take responsibility for the manuscript, its interpretations, and its cited claims.

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# List of Abbreviations

**ACT** Action Chunking Transformer.

**ACT-M** motor-current-aware Action Chunking Transformer.

**ACT-V** vision-and-position Action Chunking Transformer baseline.

**CI** confidence interval.

**CNN** convolutional neural network.

**DDIM** Denoising Diffusion Implicit Model.

**DDPM** Denoising Diffusion Probabilistic Model.

**DP** Diffusion Policy.

**DP-M** motor-current-aware Diffusion Policy.

**DP-V** vision-and-position Diffusion Policy baseline.

**EMA** exponential moving average.

**FiLM** feature-wise linear modulation.

**GroupNorm** group normalisation.

**KL** Kullback–Leibler.

**LayerNorm** layer normalisation.

**MDP** Markov decision process.

**POMDP** partially observable Markov decision process.

**ReLU** rectified linear unit.

**ResNet** residual network.

**RGB** red, green, blue.

**RMS** root mean square.

**U-Net** U-shaped convolutional neural network.

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# 1 Introduction

## 1.1 Motivation

Most people learn a manual skill by watching someone else, trying it, or by guidance from an expert. They are not given fixed instructions that specify the angle of every joint or a rule for every possible scenario upfront. A robot, on the contrary, has traditionally required an engineer to describe much of that behaviour in advance. Robot learning offers a more scalable approach of letting the machine learn behaviour from examples. In imitation learning, a person demonstrates a task and a policy learns the relationship between what it observes in environment and how it should act in order to achieve success on the task. If this works reliably, teaching a robot a new manipulation skill becomes similar to teaching by showing, just like training a trainee. For assembly, laboratory handling, and other varied physical tasks, this offers a practical alternative to writing task-specific rules for every object and exception.

The promise of learning from demonstrations is easiest to realise when the task can be inferred from geometry. In free space, a camera can show location of objects and joint positions can show the configuration of the arm. Contact-rich manipulation is harder because the decisive state may be physical rather than purely geometric. In reality, gripper can close without securing an object or object can slip mid-way, a peg can reach a hole while pressing against its rim, and two nearly identical camera images can precede either a smooth insertion or a jam. In each case, the robot must respond to an interaction that is only partly visible. This loss of direct visual evidence near contact is a central observability problem in contact-rich imitation learning [1].

Sequence models such as the Action Chunking Transformer (ACT) [2] and Diffusion Policy [3] can learn complex manipulation behaviours from demonstrations. Their success, however, still depends on whether the robot can observe the information needed when an action is chosen.

Images and joint positions provide strong information about appearance, configuration, and motion, but they do not directly measure interaction effort. Contact-aware policies therefore augment vision and kinematics with tactile, force, or torque measurements. These modalities can reveal contact onset, load, deformation, or slip, but they usually require additional hardware, calibration, or a robot that already provides reliable torque estimates [1], [4], [5], [6], [7].

Low-cost bus servos offer another possibility. For example, the Waveshare ST3215 control table specifies a read-only motor-current register [8]. Reading an existing current register requires no additional sensing hardware. Its interpretation is nevertheless difficult because motor-side current is affected by pose, motion, friction, and the high-ratio transmission, and it does not directly specify the direction or location of an external force [9], [10].

## 1.2 Problem Statement and Research Questions

Research on contact-aware imitation learning has shown how dedicated force, torque, and tactile observations can be incorporated into visuomotor policies [4], [5], [6], [7]. Sensorless estimation research has likewise shown that motor-side signals can support force or contact inference when suitable dynamics, calibration, or labelled reference measurements are available [9], [11], [12]. Prior policy studies also investigate motor-current-derived torque histories and report task-dependent benefits from raw current [13], [14]. The question I'm aiming to investigate is how weak bus-servo current signals can be represented and used effectively without explicit dynamics estimation or force/torque calibration. The scientific problem is therefore not only about the characterisation of the available motor-current as discriminative signal but also whether a weak, mechanically entangled signal can be converted into policy information that improves contact-rich behaviour without adding sensing hardware.

The thesis addresses the following overarching question:

> *Can weak, uncalibrated motor-current signals already available on a low-cost robot provide a useful learned contact representation for contact-rich visuomotor imitation learning?*

These questions are central to this investigation.

**Does motor current contain task-relevant information?**
This requires the registers to be available at the policy rate and to exhibit repeatable variation under contact-relevant conditions beyond their quiet-state variation.

**Do trained policies respond to current?**
This asks whether changing the current input while holding image and position fixed changes the actions predicted by a trained policy.

**Does adding current improve closed-loop task performance?**
This compares policies with and without current during repeated physical peg-in-hole trials and examines modes of failure and performance.

These questions separate three claims that could otherwise be confused: that the signal contains task-relevant information, that a trained model uses that information, and that its use improves closed-loop behaviour. Motor current makes this separation necessary because it mixes gravity, inertial and Coriolis effects, friction, and external load, while high-ratio gearing can make the friction component substantial [9], [10]. An instantaneous sample may be ambiguous even when a short history across several joints contains a recognisable interaction pattern. The camera and joint state may also be easier for the policy to exploit, allowing optimisation to minimise imitation loss while largely ignoring current [15], [16], [17].

Peg-in-hole insertion task provides a concrete test of the problem. The policy must approach and grasp a peg, transport it to a tight opening, respond to possible rim interaction, seat it, and

release it. Vision is informative during approach, but the wrist view becomes partly occluded near insertion and joint position cannot alone identify the contact condition. Current is recorded throughout the task, so the policy can use it during any phase in which it complements the visual and kinematic observations.

## 1.3 Approach

The study follows the three-stage design shown in Figure 1.1. The physical platform is an SO-101 arm fitted with Waveshare ST3215 bus servos. The study begins by characterising all six current channels with respect to acquisition rate, distribution, static variation, pose dependence, and contact-related temporal behaviour. It then records image, joint position, action, and current in one synchronised leader-follower demonstration dataset. The same demonstrations provide controlled position-only and current-aware views for policy training.

Two representation strategies are examined in ACT. The first appends instantaneous current directly to joint position, providing the smallest possible change to the baseline observation. The second uses a causal current history and a learned joint-selective temporal encoder. Additional bounded conditioning mechanisms explore how current can enter an already trained visual policy and are reported as auxiliary methods.

A Diffusion Policy comparison examines current conditioning under a different action-generation and execution scheme. For selected current-aware models in both policy families, counterfactual input replacement tests policy dependence and repeated physical trials test closed-loop performance.

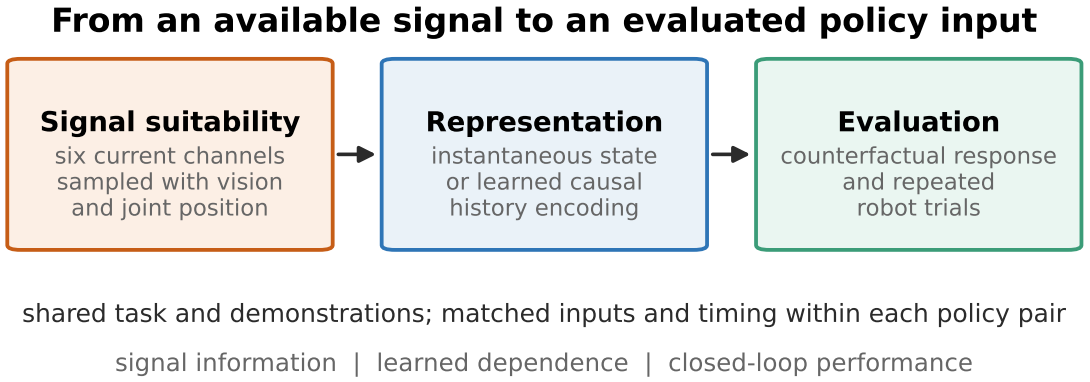



**Figure 1.1:** Study design for evaluating motor current as an additional policy observation. Signal suitability, learned policy dependence, and closed-loop performance are assessed as separate questions.

## 1.4 Contributions

The thesis contributes platform-specific empirical evidence, a shared multimodal dataset and an implementation framework for controlled current-aware policy comparisons:

1. a reproducible characterisation of all six ST3215 current channels as candidate learning inputs, covering acquisition, distribution, pose dependence, contact discrimination, and temporal behaviour;
2. an extended LeRobot-compatible schema and a synchronised 100-episode multimodal peg-in-hole dataset containing camera observations, follower joint positions, leader-derived actions, and follower current readings, together with human-reviewed phase annotations for the analysed subset;
3. controlled ACT implementations for direct current concatenation, training-derived current preprocessing, and a learned joint-selective temporal representation, supplemented by clearly separated auxiliary conditioning methods;
4. a Diffusion Policy vision-and-position system and a learned-current counterpart evaluated under a shared dataset and execution timebase;
5. a model-agnostic counterfactual diagnostic that measures how predicted actions change when one observation modality is replaced while the remaining inputs are held fixed; and
6. a software and verification framework that reproduces the numerical analyses, plots, observation transformations, and evaluated policy settings.

## 1.5 Scope

The investigation covers one low-cost manipulator, one primary peg-in-hole task, one demonstration dataset, and the evaluated ACT and Diffusion Policy implementations. Here, *proprioceptive effort* refers to the motor-current input, and *zero-cost* denotes the absence of additional sensing hardware. Motor current is treated as a policy observation and is not converted into an estimate of Cartesian force or joint torque. The robot is not augmented with tactile, acoustic, inertial, or force/torque sensors, and no simulated current model or sim-to-real transfer is introduced.

The resulting claims are specific to the evaluated task, operator, training runs, servo family, and robot embodiment. Historical task development and auxiliary architectures are reported separately from the principal policy comparisons.

## 1.6 Thesis Outline

Chapter 2 reviews contact observability, contact-related sensing, multimodal fusion, modality utilisation, and contact-aware ACT and Diffusion Policy research. Chapter 3 introduces the

learning principles needed to understand the implemented policies and diagnostics. Chapter 4 characterises the motor-current signal as a candidate policy input. Chapter 5 describes the peg-in-hole task, demonstration process, synchronised data, and controlled feature views. Chapter 6 specifies the evaluated policy architectures, training procedures, input transformations, diagnostics, and deployment rules. Chapter 7 presents the physical policy comparisons, statistical analysis, failure distributions, and modality-use evidence. Chapter 8 interprets the relationship between signal information, learned dependence, and control performance and examines the limits of the study. Chapter 9 answers the thesis questions and identifies directions for further research. Appendix A explains the code and data folders and how to run collection, training, evaluation and analysis. Appendix B reports auxiliary conditioning methods, development screens, and SmolVLA outcomes.

# 2 Related Work

This thesis lies between two established research directions: estimating physical interaction from signals inside a robot, and incorporating interaction measurements into learned manipulation policies. The first direction determines what motor current can reveal about contact and loading. The second determines how force, torque, tactile, or effort observations can influence action prediction. Across this literature, the physical signal, its policy representation, and the evidence used to establish its contribution are tightly coupled.

## 2.1 Motor-Side Effort and Sensorless Interaction Estimation

Motor current is related to actuator effort through the motor torque constant, but the current measured at the drive is not a direct measurement of external force. Motor dynamics, transmission losses, friction, configuration, acceleration, and the low-level controller all contribute to the recorded value. Model-based collision observers therefore estimate external torque by combining motor-side measurements with a robot dynamics model [18], [19]. Wahrburg et al. [9] estimate Cartesian contact force on an ABB YuMi from motor current after modelling the relevant mechanics. Shan and Pham [11] and Lu et al. [12] learn force estimators from internal robot signals using force/torque-labelled data. Yamane et al. [20] use identified manipulator dynamics to support sensorless bilateral teleoperation.

Transmission design has a strong influence on these estimates. In high-ratio mechanisms, gearbox friction and hysteresis can contribute materially to motor-side effort. Del Prete et al. [10] demonstrate torque control without joint-torque sensors on a high-ratio system using identified actuator and transmission models together with end-effector force/torque sensing. These results establish motor current as a plausible interaction-sensitive measurement while also showing why the raw register of a geared bus servo cannot be interpreted as calibrated force.

A separate line of work supplies sensed or estimated joint effort directly to learned manipulation policies [5], [14], [21]. Current remains less specific than contact-surface sensing because it does not localise pressure or directly observe relative motion between the peg and gripper. Tactile, acoustic, and vibration measurements provide local or high-frequency cues relevant to slip [1], [22].

## 2.2 Interaction Signals in Imitation Learning

Contact-aware imitation policies have used wrist force/torque sensors, tactile arrays, estimated joint torque, and motor-side observations. The sensor and representation usually change together, so the closest precedents are most informative when both choices are considered.

### 2.2.1 Force, Torque, and Tactile Observations

Wrist force/torque sensors provide a calibrated wrench close to the interaction point. FTACT [6] concatenates force/torque measurements with robot state before an ACT encoder and evaluates cumulative completion over pick, press, and place stages. FoAR [7] uses force/torque measurements in a contact-guided policy and a runtime refinement stage. AdmitDiff [23] combines force-conditioned diffusion with admittance control so that measured interaction affects both the learned condition and the commanded motion.

Robot-provided or estimated external joint torques retain a joint-wise description of interaction without adding a wrist sensor. FACTR [5] supplies robot-provided external joint torques from the Franka Panda as force tokens to an action-chunking policy. FACTR 2 [14] learns a free-motion torque model for commodity manipulators and estimates external torque by subtraction.

Tactile sensors move the measurement to the contact surface. Funk et al. [4] encode tactile images through a separate branch in a visuomotor imitation policy. 3D-ViTac [24] combines visual and tactile point-cloud representations for fine-grained manipulation. PolyTouch [22] integrates tactile, acoustic, and peripheral visual sensing in one finger, while MimicTouch [25] transfers multimodal human tactile demonstrations to a robot. These systems provide contact geometry and local deformation that are unavailable from a joint-level effort signal, while requiring contact hardware and a sensor-specific representation pipeline.

### 2.2.2 ACT-Based Contact Policies

ACT predicts an action chunk from visual and robot-state tokens through a transformer-based conditional variational autoencoder [2]. Its state projection and token sequence support several forms of contact fusion. FTACT uses direct fusion by concatenating a calibrated wrist wrench with robot state before the ACT encoder [6]. This arrangement is the closest architectural precedent for the direct current-concatenation comparison in Section 6.3.2, where instantaneous current is appended to joint position before the state projection.

FACTR maps robot-provided external joint torque to a separate force token and initially reduces the visual input during training so that the force route acquires a predictive role [5]. FACTR 2 adds a learned free-motion torque estimate and increases the sampling frequency of pre-contact and contact segments [14]. Its raw-current ACT ablation improves NIST Insertion task progress but does not improve Cap Screwing; learned external torque performs better on both tasks [14, Appendix D.6]. Haptic-ACT [26] conditions an ACT-derived policy on position, force, and image observations and includes recovery demonstrations following failed grasps.

Bi-ACT [21] uses observer-estimated torque with position and velocity in a bilateral-control formulation, then predicts leader-side motion and torque targets.

These architectures place effort information in a shared state vector, a separate token, or an action quantity. The joint-selective temporal route in Section 6.3.3 preserves joint identity and encodes a causal current history before forming additional Transformer tokens. The bounded auxiliary route in Appendix B.1 uses the same histories to condition visual features. Together, the prior architectures motivate direct and structured fusion as distinct model designs.

### 2.2.3 Diffusion-Based Contact Policies

Diffusion Policy generates an action trajectory through iterative denoising conditioned on the robot observation [3]. Contact-aware variants place interaction information in the denoising condition, in an explicit phase model, or in a faster correction path. FoAR predicts impending contact and changes the use of force information as contact approaches [7]. FARM [27] uses tactile-conditioned feature modulation in a diffusion policy that predicts motion and grip-force trajectories. PhaForce [28] combines a task-phase belief with slow visual planning and faster force-guided correction. Reactive Diffusion Policy [29] and ImplicitRDP [30] introduce faster contact-conditioned pathways for brief interactions.

The current-aware Diffusion Policy in Section 6.4 encodes recent current histories and uses the resulting condition to modulate the denoising network. TA-VLA derives torque from motor current using manufacturer torque constants and encodes its history in a decoder token [13]. Other vision–language–action models incorporate explicit force pathways [31], [32].

## 2.3 Learning and Measuring Modality Use

Multimodal networks can concentrate predictive capacity in the representation that learns fastest or already explains much of the training target [15], [16], [17]. In contact-aware manipulation, several methods reshape the learning opportunity for the interaction branch. FACTR reduces visual fidelity early in training [5]. FACTR 2 samples contact-relevant segments more frequently [14]. FoAR gates force use with predicted contact [7], and PhaForce schedules visual and force processing through a learned phase belief [28]. Haptic-ACT changes the demonstrated state distribution by including recovery behaviour [26].

The evaluation combines the counterfactual dependence diagnostic in Sections 6.5 and 7.4 with the principal direct and temporal current comparisons; bounded feature conditioning is explored in auxiliary screens (Appendix B.1).

Table 2.1 summarises the closest approaches across signal preparation, fusion, and control role.

**Table 2.1:** Closest contact-aware learning approaches and the unresolved combination examined in this thesis.

| Work | Contact-related input | Required preparation | Fusion or use | Role in control |
|---|---|---|---|---|
| FACTR [5] | external joint torque | robot with joint-torque sensing | force token and visual curriculum | policy observation |
| FACTR 2 [14] | external torque; raw-current ablation | contact-free data for learned torque | contact-aware resampling | policy observation |
| FTACT [6] | wrist force/torque | calibrated sensor | state concatenation | policy observation |
| FoAR [7] | wrist force/torque | calibrated sensor | predicted-contact gating | policy and refinement |
| Funk et al. [4] | tactile images | gripper sensors | separate tactile encoder | policy observation |
| Shan and Pham [11] | motor current and joint state | force/torque labels | learned force estimator | estimated wrench |
| Bi-ACT [21] | estimated torque, position, and velocity | disturbance observer and bilateral demonstrations | joint-state input and action prediction | bilateral policy input and output |
| **This thesis** | **raw joint current** | **no force/torque calibration** | **direct and learned fusion** | **policy observation** |

## 2.4 Research Gap

Prior work has established three relevant capabilities. Motor-side signals can support force or collision estimation when combined with models, calibration, or force-labelled supervision. Force, torque, and tactile observations can improve learned contact-rich manipulation through direct fusion, structured tokens, conditional modulation, and corrective control. Contact-aware sampling, gating, and curricula can strengthen the influence of an interaction branch during learning.

The closest approaches couple the meaning of the interaction signal to its policy representation. FTACT supplies a calibrated wrist wrench through the robot-state route, while FACTR uses robot-provided external joint torque and FACTR 2 learns a free-motion model to estimate it [5], [6], [14]. FoAR and PhaForce introduce an explicit estimate of contact or task phase before deciding how force information should affect action generation [7], [28]. FARM and the reactive diffusion methods give contact information a conditional or corrective route inside an iterative policy [27], [29], [30]. Contact-sensing performance therefore depends jointly on the physical quantity, its preprocessing, its temporal context, and its point of fusion.

Their evaluations also address different levels of the learning problem. Force-estimation studies quantify agreement with a reference sensor, contact-prediction methods assess event or phase estimates, and imitation-policy studies report task or stage completion. These endpoints establish sensor fidelity, contact awareness, or behavioural performance respectively. The manipulation studies also use different tasks and success definitions: FACTR evaluates

behaviours including lifting and pivoting, while FTACT reports cumulative pick, press and place completion in bottle reorientation [5], [6]. Their reported percentages therefore do not form a common benchmark for ranking policies against the present insertion task; the comparison here concerns sensing assumptions, fusion and control roles. The evaluation in this thesis therefore combines physical outcomes with evidence that the trained action function depends on the supplied current signal. This distinguishes an active current route from a model whose predictions are effectively carried by vision and joint position alone.

The surveys by Xie and Correll [33], Tsuji et al. [1], and Shan et al. [34] cover force-aware foundation models, contact-rich imitation learning, and tactile-and-force robot learning respectively, and provide the broader context for this investigation.

FACTR 2, released during this thesis, also evaluates raw motor current as an ACT input [14, Appendix D.6]. Building on these precedents, this thesis characterises uncalibrated ST3215 current on the SO-101 and evaluates instantaneous and learned temporal current representations for peg-in-hole imitation learning, with a further Diffusion Policy system comparison. Its input-replacement diagnostics distinguish policy sensitivity to current from closed-loop task performance.

# 3 Background Knowledge

Learning a manipulation policy from demonstrations requires a model of both the physical interaction and the information available to the robot. The robot observes that interaction through imperfect measurements, predicts a sequence of actions, and may reach situations that were not represented in the demonstrations. These issues are connected: contact creates hidden physical state, behavioural cloning can amplify observation errors during rollout, and action-sequence models determine when a new measurement can influence control.

## 3.1 Contact-Rich Manipulation and Interaction Sensing

Peg-in-hole insertion couples geometric alignment with forces created when the peg and fixture touch. Whitney [35] analyses the geometric and frictional conditions under which a cylindrical peg can jam or wedge. Mason [36] relates contact constraints to admissible force-controlled motion, while Hogan [37] formulates interaction through a controlled relationship between force and motion. Lozano-Pérez et al. [38] describe fine manipulation under uncertainty as a sequence of decisions shaped by contact. These formulations show why the pose of the manipulator and the state of its physical interaction are different quantities.

A visuomotor policy receives only measurements of that state. The wrist camera can show the peg and fixture during approach, but the gripper and peg can obscure their relative alignment near insertion [1]. Joint position describes robot configuration without distinguishing the same configuration in free motion and under load. An interaction-sensitive measurement can resolve part of this ambiguity during grasping, transport, insertion, and release. The required information changes across the episode, as summarised in Table 3.1.

Interaction sensing can be placed at the wrist, contact surface, joint, or motor [1, 33, 34]. A wrist force/torque sensor measures a calibrated Cartesian wrench close to the end effector. Tactile sensors measure local pressure, deformation, or contact geometry at the gripper. External joint torques describe interaction across the mechanism, while motor current records electrical effort at the actuator. The measurements differ in spatial resolution, physical interpretation, hardware integration, and dependence on calibration or a robot model. Table 3.2 summarises these properties for the sensor types used by the closely related methods in Chapter 2.

Combining several interaction sensors can broaden the information available to a policy, but it also requires the complete sensor stack during data collection and deployment and entangles the contribution of each measurement. Sensor selection therefore depends on the physical quantity required by the task, the integration budget, and the experimental comparison being made. The observation used in this study combines the existing wrist image and joint positions with six

**Table 3.1:** Observation demands across a peg-in-hole episode and the interaction information available to complement visual and kinematic measurements.

| **Phase** | **Visual or kinematic information** | **Remaining ambiguity** | **Interaction information** |
|---|---|---|---|
| Approach | peg, fixture, and arm geometry | depth and alignment uncertainty | unintended early contact |
| Grasp | gripper aperture and object location | whether the object is securely held | contact onset and sustained effort |
| Transport | arm motion and approximate object pose | hidden slip or object loss | change in grasp or support load |
| Insertion | alignment before occlusion | rim contact, jamming, and correction direction | contact timing and distributed loading |
| Release | open gripper and final scene | whether seating remains stable | loss of grasp load |

**Table 3.2:** Contact-related observations and their principal sensing trade-offs.

| **Observation** | **Information supplied** | **Integration or interpretation** | **Representative work** |
|---|---|---|---|
| Wrist force/torque | Cartesian force and moment near the end effector | added calibrated sensor and mechanical integration | FTACT, FoAR [6], [7] |
| External joint torque | joint-wise external loading | joint-torque sensing (FACTR) or learned free-motion model (FACTR 2) | FACTR, FACTR 2 [5], [14] |
| End-effector tactile sensing | local pressure, deformation, or contact geometry | gripper modification and sensor-specific representation | Funk et al., 3D-ViTac [4], [24] |
| Motor-side current | platform-native joint-wise electrical effort | combines load with pose, motion, friction, gearing, and control | Wahrburg et al., Shan and Pham [9], [11] |

current registers from the follower arm. This preserves the low-cost hardware configuration and provides a joint-wise effort measurement throughout the recorded motion. The current channels cannot supply the calibrated force direction of a wrist sensor or the contact location of a tactile surface. Their potential contribution is a repeatable change in actuator loading that complements visual geometry and robot configuration.

## 3.2 Sequential Decision-Making and Partial Observability

### 3.2.1 Markov and Partially Observable Decision Processes

A Markov decision process (MDP) describes sequential interaction through the tuple

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma), \tag{3.1}$$

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P(s_{t+1} \mid s_t, a_t)$ is the transition model, $R(s_t, a_t)$ is the reward, and $\gamma$ is a discount factor. The state is Markov when it contains all information from the past that is needed to determine the distribution of the next state after an action. A policy can then select an action from the current state without consulting the full interaction history [39].

In a partially observable Markov decision process (POMDP), the agent does not receive $s_t$ directly. The tuple becomes $\mathcal{P} = (\mathcal{S}, \mathcal{A}, P, R, \mathcal{O}, Z, \gamma)$, adding an observation space $\mathcal{O}$ and an observation model $Z(o_{t+1} \mid s_{t+1}, a_t)$. The policy therefore acts from a measurement $o_t$ that may correspond to more than one underlying state [39]. This is partial observability in plain terms: information that matters for choosing the action exists in the physical system but is absent or ambiguous in the current observation. An ideal POMDP agent can summarise the observation–action history as a belief distribution

$$b_t(s) = p(s_t = s \mid o_{1:t}, a_{1:t-1}), \tag{3.2}$$

and act from that belief rather than from the latest measurement alone [39].

The learned policies operate directly on recorded observations and histories, without identifying $P$, $R$, or $Z$ or maintaining an explicit belief state. The POMDP formulation identifies the resulting observation problem. During peg insertion, the physical state includes contact location, grasp security, and the relation between peg and hole. The wrist image and joint positions observe only part of that state: occlusion can hide the insertion geometry, and the same joint configuration can occur in free motion or under load. Adding current supplies another measurement of the latent interaction, while a causal current history offers a finite record from which the policy may distinguish a transient contact from ordinary motion. Neither addition guarantees a fully observable state; it can only reduce ambiguity if the recorded signal contains the missing task-relevant distinction.

### 3.2.2 Learning from Demonstrations

Imitation learning replaces explicit reward-based policy optimisation with examples of expert behaviour. For a demonstration $i$ of length $T_i$, let $o_t^{(i)}$ denote the observation available at time $t$ and let $a_t^{(i)}$ denote the corresponding expert action. The complete training set is

$$\mathcal{D} = \left\{ \left( o_t^{(i)}, a_t^{(i)} \right)_{t=1}^{T_i} \right\}_{i=1}^{N}. \tag{3.3}$$

The policy $\pi_\theta$ is parameterised by $\theta$ and maps the available observation, or an explicitly supplied observation history, to an action or action sequence.

### 3.2.3 Behavioural Cloning

Behavioural cloning treats imitation learning as supervised prediction from demonstrated observation and action pairs [40, 41]. For single-step actions, the training objective is

$$\theta^* = \arg\min_\theta \mathbb{E}_{(o,a)\sim\mathcal{D}} \left[ \mathcal{L}(\pi_\theta(o), a) \right], \tag{3.4}$$

where $\mathcal{L}$ measures the difference between the predicted and demonstrated actions. The loss is evaluated on observations visited by the demonstrator, whereas the trained policy is evaluated on states produced by its own earlier actions. Even small prediction errors can therefore move the robot away from the demonstration distribution and place it in situations for which the training set provides little guidance. This covariate shift can make errors accumulate over a rollout rather than remain independent from one step to the next [41]. Figure 3.1 illustrates the difference between supervised training on demonstrated states and deployment on states reached by the learned policy.

Interactive data collection can reduce this mismatch by asking the demonstrator for corrective actions in states reached by the learner [41]. That approach changes the training data after observing a policy's behaviour. The experiments in this thesis use one fixed offline demonstration set, providing the same training examples to each policy comparison.

Partial observability makes this rollout mismatch more consequential in contact-rich tasks. After an unnoticed slip or rim contact, a familiar-looking image and joint configuration can conceal a different physical state from those represented in the demonstrations.

## 3.3 Action Sequences and Closed-Loop Execution

Predicting one action at a time requires a fresh decision at every control step. Action chunking instead predicts a sequence of $H$ future actions from the current observation:

$$\hat{A}_t = \pi_\theta(o_t) = \left( \hat{a}_{t|t}, \hat{a}_{t+1|t}, \ldots, \hat{a}_{t+H-1|t} \right). \tag{3.5}$$

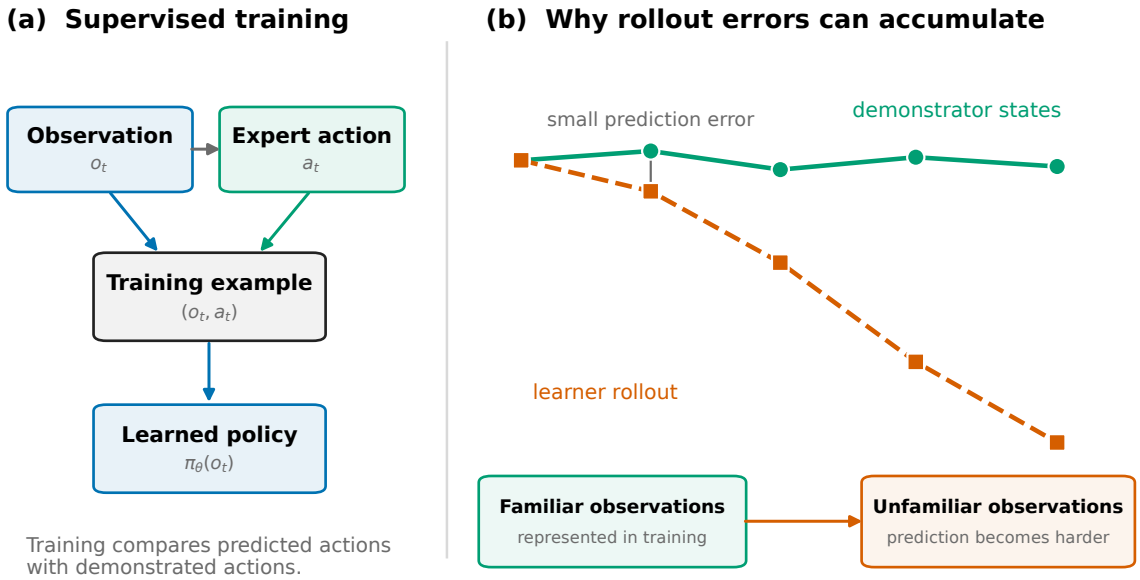



**Figure 3.1:** Behavioural cloning and covariate shift. Panel (a) shows supervised learning from observations visited by the demonstrator. Panel (b) shows how a small prediction error can move the deployed policy away from that distribution, after which later predictions are made from unfamiliar observations [41].

The second subscript indicates the observation time from which an action was predicted. Chunking lets the model represent coordinated short-horizon motion and reduces the number of independent decisions required over a task [2].

The execution strategy determines how predicted chunks become motor commands. Executing the whole chunk before updating the plan gives temporal consistency but delays feedback. Replanning after a short prefix uses feedback more frequently but can make consecutive plans disagree. Both ACT and Diffusion Policy therefore combine sequence prediction with a specified execution schedule. Depending on that schedule, a new observation may influence overlapping predictions, trigger replanning after a short prefix, or wait until a queued chunk has completed. These mechanisms can limit how long a policy continues with an outdated plan, but they do not supply expert actions for unfamiliar states and therefore do not eliminate distribution shift.

ACT can query the policy repeatedly and combine overlapping predictions for the same action time. A general exponentially weighted temporal ensemble orders the available predictions from oldest to newest as $(\hat{a}_t^{(0)}, \ldots, \hat{a}_t^{(n-1)})$ and computes

$$a_t^{\text{exec}} = \frac{\sum_{i=0}^{n-1} \exp(-mi)\, \hat{a}_t^{(i)}}{\sum_{i=0}^{n-1} \exp(-mi)}, \tag{3.6}$$

where the oldest prediction has index zero and $m \geq 0$ controls the decay towards newer predictions. Larger values favour older predictions, whereas smaller values flatten the weights and incorporate new observations more quickly [2]. Temporal ensembling smooths disagreements between overlapping chunks while retaining feedback from new observations [2]. Diffusion Policy commonly follows a receding-horizon rule instead: it predicts a chunk, executes a prefix, observes again, and generates a new plan [3]. Section 6.3.4 specifies the exact chunk lengths, replanning intervals, and execution rules used in the experiments.

## 3.4 Action Chunking Transformer

The Action Chunking Transformer, abbreviated as ACT, combines visual representation learning, transformer-based sequence prediction, and a conditional variational autoencoder [2]. Its purpose is to predict a coherent action chunk from the current visual and proprioceptive observation.

### 3.4.1 Observation Encoding

An image backbone, such as a residual network, first converts the camera observation into a spatial feature map [42]. Each spatial location becomes a visual token after projection to the transformer dimension and addition of positional information. A separate projection maps the low-dimensional robot state to one or more state tokens. The transformer encoder applies self-attention across these tokens, allowing the representation of the robot state to depend on the visual scene and vice versa. For query matrix $Q$, key matrix $K$, and value matrix $V$, scaled dot-product attention is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\mathsf{T}}}{\sqrt{d_k}}\right)V, \tag{3.7}$$

where $d_k$ is the key dimension [43]. The learned attention weights determine which encoded tokens contribute to each updated representation.

This tokenisation is important for multimodal policies. Concatenating several scalar channels before one projection creates a compact state token, but it also mixes those channels before the transformer sees them. Representing a modality through separate tokens or a separate encoder preserves more structure and gives the architecture a clearer route through which to use it. Sections 6.3.2 and 6.3.3 specify the direct and temporal ACT routes in the principal comparisons; Appendix B.1 describes an auxiliary feature-conditioning variant.

### 3.4.2 Chunk Decoder and Conditional Latent Variable

The ACT decoder receives one learned query for each position in the action chunk. Through cross-attention, the queries retrieve information from the encoded observation and produce

the complete action sequence in one forward pass. During training, a conditional variational encoder also observes the demonstrated action chunk and represents variations in demonstrated behaviour through a latent variable $z$ [2, 44]. The objective combines action reconstruction with regularisation of the latent distribution:

$$\mathcal{L}_{\mathrm{ACT}} = \left\|\hat{A} - A\right\|_1 + \beta D_{\mathrm{KL}}\big(q_\phi(z \mid s, A) \,\|\, p(z)\big)\,, \tag{3.8}$$

The first term is the element-wise $L_1$ action reconstruction loss, and $D_{\mathrm{KL}}$ denotes the Kullback–Leibler (KL) divergence. Here $q_\phi$ is the training-time latent encoder conditioned on robot state $s$ and demonstrated action chunk $A$, $p(z)$ is the prior, and $\beta$ controls regularisation [2]. The latent encoder is removed at deployment, where ACT uses the prior mean $z = 0$ and therefore produces a deterministic chunk for a fixed observation.

ACT has no recurrent hidden state between independent policy queries. A single-step scalar input therefore conveys only its current value unless the observation explicitly includes a temporal history or derived temporal feature. This constraint motivates the causal effort histories introduced later in the method chapter. Figure 3.2 summarises the training-only latent route, the deployed policy, and temporal ensembling.

## 3.5 Diffusion Policy

Diffusion Policy represents the conditional distribution of action sequences through an iterative denoising process [3]. Instead of predicting the final action chunk in one pass, the policy learns to remove noise from a corrupted action sequence while conditioning every denoising stage on the robot observation. This formulation can represent several distinct action modes without averaging them into one trajectory.

### 3.5.1 Training by Noise Prediction

Let $A^0$ be a demonstrated action chunk. A forward diffusion process produces a noisy version at diffusion step $k$:

$$A^k = \sqrt{\bar{\alpha}_k}\, A^0 + \sqrt{1 - \bar{\alpha}_k}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \mathrm{Id}), \tag{3.9}$$

where $\bar{\alpha}_k$ is determined by a fixed noise schedule. The noise-prediction objective follows Ho et al., while observation conditioning follows the robotic policy formulation of Chi et al. [3, 45]. The network receives $A^k$, the diffusion step $k$, and the observation $o$ and is trained to predict the sampled noise:

$$\mathcal{L}_{\mathrm{DP}} = \mathbb{E}_{A^0, o, k, \epsilon}\left[\left\|\epsilon - \epsilon_\theta(A^k, o, k)\right\|_2^2\right]. \tag{3.10}$$

Learning this reverse direction allows the policy to transform an initially noisy action sequence into one that is consistent with the observation. The network prediction also gives an estimate

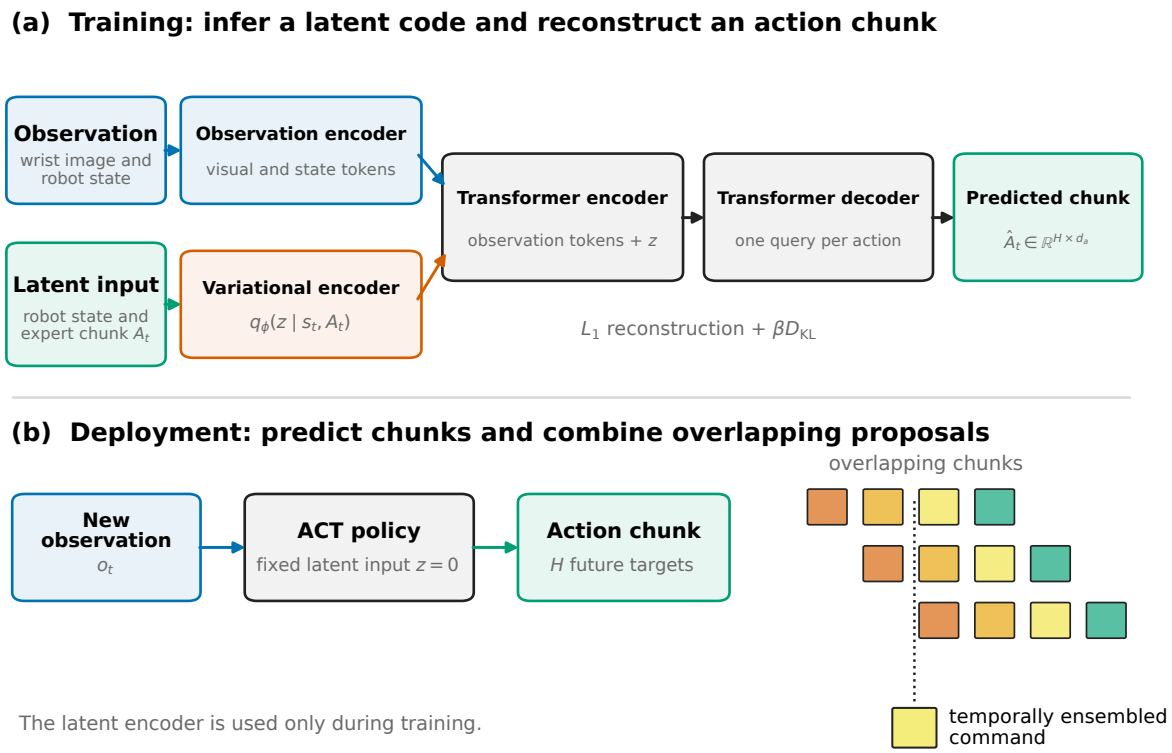



**Figure 3.2:** Conceptual ACT training and deployment flow, redrawn from the method described by Zhao et al. [2]. During training, the demonstrated action chunk and robot state define a latent distribution, while the policy predicts the complete future chunk from visual and state tokens. At deployment, the training-only latent encoder is removed and the latent input is fixed. Repeated policy queries produce overlapping chunks whose predictions for the current action can be combined by temporal ensembling.

of the clean action chunk:

$$\hat{A}_\theta^0 = \frac{A^k - \sqrt{1-\bar{\alpha}_k}\,\epsilon_\theta(A^k, o, k)}{\sqrt{\bar{\alpha}_k}}. \tag{3.11}$$

A reverse sampler combines this clean estimate with the predicted-noise direction to obtain a less corrupted action sequence.

### 3.5.2 Conditional Denoising and Receding Horizon

The convolutional neural network (CNN) variant of Diffusion Policy uses a conditional one-dimensional U-Net for action trajectories [3]. The U-Net processes information at several temporal resolutions, while the observation conditions its residual blocks. Feature-wise Linear

Modulation, abbreviated as FiLM, applies an observation-dependent affine transformation to intermediate features:

$$\mathrm{FiLM}(h \mid o) = \gamma_{\mathrm{F}}(o) \odot h + \beta_{\mathrm{F}}(o), \tag{3.12}$$

where $h$ is a feature tensor and the functions $\gamma_{\mathrm{F}}$ and $\beta_{\mathrm{F}}$ generate per-feature scale and bias values from the conditioning input [46]. This mechanism gives low-dimensional observations several opportunities to influence the denoising computation.

A Denoising Diffusion Probabilistic Model (DDPM) uses a stochastic reverse process that follows the training noise schedule [45]. A Denoising Diffusion Implicit Model (DDIM) is derived from non-Markovian forward processes with the same training objective and can use a shorter deterministic reverse sampling schedule [47]. For a deterministic DDIM transition from step $k$ to an earlier selected step $k'$, the update can be written as

$$A^{k'} = \sqrt{\bar{\alpha}_{k'}}\, \hat{A}^0_\theta + \sqrt{1 - \bar{\alpha}_{k'}}\, \epsilon_\theta(A^k, o, k). \tag{3.13}$$

Selecting a subsequence of diffusion steps reduces the number of network evaluations while using the same trained noise predictor. For a robot policy, fewer reverse steps reduce planning latency, although the chosen schedule must still preserve acceptable action quality. Section 6.4 specifies the training schedule, deployment sampler, and latency checks used in this thesis. Figure 3.3 contrasts the training objective with the receding-horizon deployment loop.

## 3.6 Comparison of ACT and Diffusion Policy

ACT and Diffusion Policy are both behavioural-cloning methods that predict short action sequences, but they represent and generate those sequences differently. ACT encodes the current observation once and decodes all positions in the chunk through learned queries in a single policy pass. Its conditional variational encoder shapes a latent representation during training; deployment fixes that latent input and obtains feedback by querying the policy again and combining overlapping predictions. Diffusion Policy learns a conditional noise predictor in action-sequence space and generates a plan through several denoising evaluations. It then executes a prefix and conditions a new denoising process on the next observation. Thus ACT places much of its temporal structure in the Transformer decoder and the overlap between chunks, whereas Diffusion Policy places it in the action-trajectory U-Net, the reverse process, and receding-horizon execution [2, 3].

This difference makes the two families useful complementary settings for the same observation question. ACT provides the clearest minimal comparison: instantaneous current can be appended to joint position before the existing state projection, changing the observation while retaining the action-chunk architecture. A separate temporal current encoder can instead preserve joint identity and recent changes as additional tokens. Diffusion Policy offers a structurally different route in which an encoded current history can condition several residual blocks during iterative action generation.

The architectures also lead to different expectations before evaluation. Direct ACT fusion

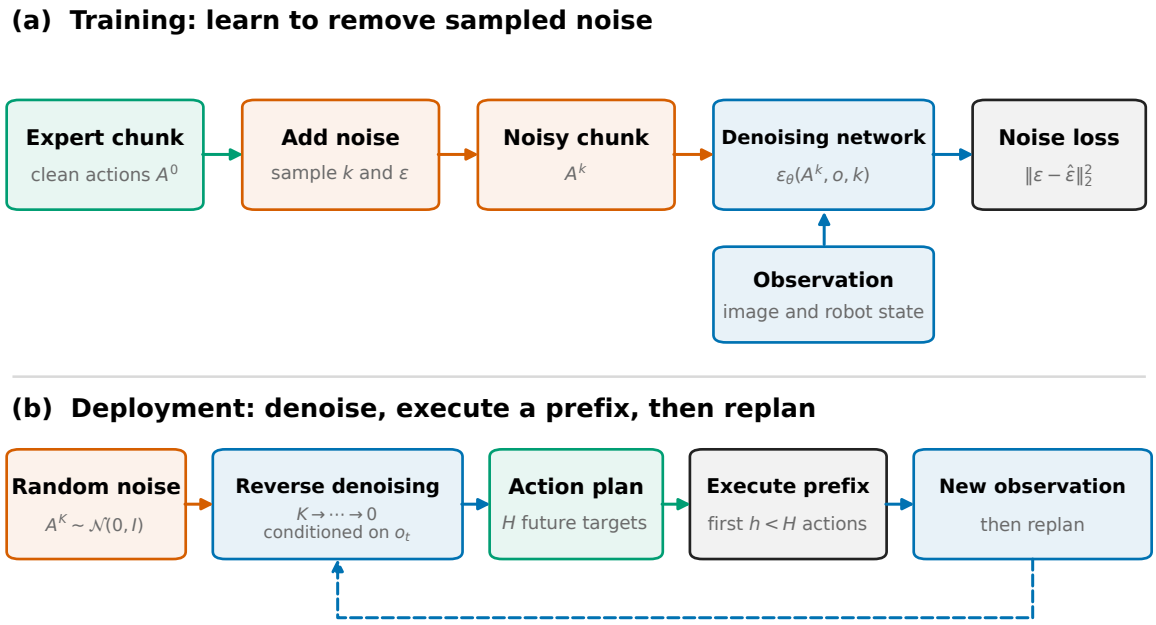



**Figure 3.3:** Diffusion Policy training and deployment. Training corrupts a demonstrated action sequence at a sampled diffusion step and teaches the conditional network to predict the added noise. Deployment begins with a noisy action sequence, repeatedly denoises it while conditioning on the current observation, executes only a prefix of the resulting plan, and then observes and replans [3, 45, 47].

changes few components, but an instantaneous scalar can be ambiguous and can be compressed with position into one state token. Joint-selective temporal ACT fusion exposes recent patterns more explicitly, at the cost of a larger learned route that must be trained from the same demonstrations. Diffusion conditioning gives the current representation repeated opportunities to modify a developing action plan, but iterative inference and prefix execution constrain how quickly a newly observed contact can change the commanded action.

## 3.7 Multimodal Observation and Fusion

A multimodal policy receives observations produced by different measurement processes. Images provide spatial context, joint positions describe robot configuration, and motor-current registers provide low-dimensional effort readings. These channels differ in dimension, scale, noise, sampling behaviour, and the task information they can reveal. Multimodal learning must therefore address both representation and fusion: each modality needs a useful feature space, and the features must be combined so that they can influence the prediction [48, 49].

### 3.7.1 Direct Fusion

The simplest strategy concatenates aligned features before passing them to a shared model. For an image representation $\mathbf{v}_t$, joint positions $\mathbf{q}_t$, and current values $\mathbf{I}_t$, a direct feature vector can be written as

$$c_t^{\text{direct}} = [\mathbf{v}_t; \mathbf{q}_t; \mathbf{I}_t] \,. \tag{3.14}$$

Direct fusion changes little beyond the input dimensionality and is therefore well suited to a controlled first comparison. Its weakness is that a low-dimensional or noisy modality enters the same optimisation path as features that may already predict most demonstrations well. The shared network can reduce training loss while assigning little practical influence to the additional input.

### 3.7.2 Learned Representations and Conditional Fusion

A separate encoder can first map a modality to an embedding:

$$e_t^I = f_\psi(\mathbf{I}_{t-L+1:t}), \qquad c_t^{\text{learned}} = \left[\mathbf{v}_t; \mathbf{q}_t; e_t^I\right], \tag{3.15}$$

where $f_\psi$ is learned jointly with the policy and the window contains only the present and past $L-1$ samples. The encoder can preserve per-joint structure, summarise temporal patterns, and map the signal to a scale that is easier for the policy trunk to use. Separate tokens offer another route by letting the transformer attend to modality or joint-specific features. Feature modulation instead uses the added modality to scale or shift intermediate features, as in Equation 3.12.

These choices change more than model capacity. They determine where modalities interact, how directly gradients reach each encoder, and whether the added signal competes with or conditions the dominant representation. The principal comparisons examine concatenation and learned temporal encoding (Sections 6.3.2 and 6.3.3), with bounded feature conditioning assessed in auxiliary screens (Appendix B.1).

## 3.8 Temporal Representation of Effort Signals

Contact is an event, whereas a single current reading is only an instantaneous sample. Its interpretation may depend on what the joint was doing immediately before the sample and on whether the value persists, rises, or decays. A causal history provides this context without using future information that would be unavailable during deployment.

For six joints and a history length $L$, the current observation is

$$\mathbf{I}_{t-L+1:t} = [\mathbf{I}_{t-L+1}, \ldots, \mathbf{I}_t]^\mathsf{T} \in \mathbb{R}^{L\times 6}. \tag{3.16}$$

If an episode has fewer than $L$ available samples near its start, padding must follow the same rule in training and deployment. The history length determines how much recorded context

is available, while the sampling rate determines the finest temporal change contained in that history. An encoder cannot reconstruct variations that were never captured by the logger.

Causal one-dimensional convolutions are a natural representation for such streams because each output depends only on the current and earlier samples [50]. Dilated convolutions increase the temporal receptive field without requiring a proportionate increase in depth. Pooling can then summarise local responses into a fixed-dimensional embedding. The particular history length, dilation pattern, and pooling operations used by the evaluated models are specified in Section 6.3.3 because they are implementation choices rather than general properties of temporal encoding.

## 3.9 Normalisation and Input Processing

Numerical scale affects optimisation even when two channels are both informative. If one input spans hundreds of native units and another spans a small interval, a shared linear projection can initially respond much more strongly to the larger-valued channel. Normalisation places heterogeneous features on comparable numerical scales and makes the meaning of a perturbation more consistent across channels.

For channel $j$, standardisation uses statistics fitted on the training split:

$$\tilde{x}_{t,j} = \frac{x_{t,j} - \mu_j}{\max(\sigma_j, \varepsilon)}, \tag{3.17}$$

where $\mu_j$ and $\sigma_j$ are the training mean and standard deviation and $\varepsilon$ prevents division by zero. Range-based scaling instead maps values relative to training minima and maxima or robust quantiles. Neither transformation removes noise or creates information; it changes the coordinates in which the optimiser sees the data.

Heavy-tailed register values can make observed extrema unstable. Clipping or winsorisation bounds values outside training-derived limits before or after scaling, depending on the model implementation. This operation is different from normalisation: clipping limits extreme values, while normalisation rescales the retained range. All fitted statistics and limits must come from the training data alone to prevent information from evaluation episodes leaking into the model. The same ordered transformation must then be applied during training, offline evaluation, and deployment. Section 6.2 specifies the training-derived caps and normalisation used by the evaluated policies.

## 3.10 Modality Utilisation and Collapse

Adding a modality to the input does not guarantee that the trained policy uses it. Multimodal networks can favour a representation that learns faster or already explains much of the target, leaving another modality weakly represented even when it contains complementary information [15, 16]. Representation collapse describes the related case in which the fused

model learns little useful signal from one modality because its learning signal is suppressed by the joint optimisation process [17].

Evaluating an added policy modality involves three distinct levels:

1. Does the current signal contain task-relevant information in the recorded data?
2. Does a trained policy change its predicted actions when that input changes?
3. Does including the signal improve closed-loop task performance?

These levels require different evidence: signal characterisation for recorded information, counterfactual input tests for model dependence, and matched real-robot comparisons for behavioural benefit.

A useful counterfactual keeps the image and joint positions fixed while replacing current with a reference value, such as its training mean. Comparing the two predicted action chunks then tests whether the model responds to the current input while the other recorded modalities remain unchanged. A larger value shows model sensitivity, but it does not by itself establish that the change is helpful or causally responsible for a successful rollout. Closed-loop evaluation is required to determine behavioural benefit. Section 6.5 defines the exact sensitivity metric used in the evaluation.

# 4 Effort-Signal Characterisation for Policy Learning

Before motor current can be added to a policy, it must first be shown to carry a repeatable and task-relevant signal. This chapter examines that question using the Waveshare ST3215 motor-current reading on the SO-101, exposed by the servo's "Present Current" register. The analysis considers whether the register can be sampled at the policy rate, how its values are distributed, how strongly they depend on pose, how noisy they are, and whether their temporal behaviour distinguishes contact-relevant conditions. Together, these results determine whether current is worth retaining as an observation and whether it should be encoded as an instantaneous value or as a temporal history. Conventional motor-current force estimation usually depends on motor constants, robot dynamics, friction models, or force/torque-labelled data [9, 11, 12]. Rather than treating the ST3215 register as a force sensor, this thesis asks a more modest question: does its uncalibrated output contain enough repeatable, contact-related structure to serve as a learnable policy input at the sampling rate and with the dataset used here? The ST3215 specification gives a gear ratio of 1:345, so the measured motor current is mediated by a high-ratio transmission rather than measured directly at the joint contact [51].

## 4.1 Candidate Observation and Decision Criteria

The decision to retain current as an observation rests on five empirical questions:

1. Can the relevant registers be acquired at the 30 Hz policy rate?
2. How does current vary with pose when there is no external contact?
3. Are known end-effector loads visible above drift and hysteresis?
4. Can gripper current distinguish between closing on an object and closing on empty space, both in controlled trials and during ordinary task motion?
5. Do arm-current events align with insertion, and do they last long enough to appear in the recorded temporal history?

As a practical screening rule, we treat this signal as clearly distinguishable when at least one comparison between conditions labelled without using that signal produces a standardised difference, Cohen's $d$, greater than 0.8. This cutoff was chosen for the exploratory analysis

rather than set before data collection. The decision to use current as a policy input therefore depends on three properties: it must be sampled at the policy rate, its contact-related changes must be large relative to ordinary variation, and the same six current channels must be available during both training and execution.

To characterise the signal, we carried out five controlled studies covering acquisition timing, static pose, known loads, scripted gripper contact, and natural task motion. We then reviewed insertion recordings to determine whether current excursions also appeared when the peg met the rim of the hole. Table 4.1 shows how each part of the characterisation contributes to the decision to use current as a policy input.

**Table 4.1:** Overview of the motor-current characterisation studies.

| Study | Evidence provided | Section |
|---|---|---|
| Logger timing and static noise | policy-rate acquisition and quiet-signal variation | Section 4.2 |
| Static pose | current variation with arm configuration | Section 4.3 |
| Known load | static-load sensitivity relative to drift | Section 4.3 |
| Scripted contact | separation of object contact from an empty close | Section 4.4 |
| Natural motion | contact separation during teleoperated motion | Section 4.5 |
| Insertion review | event timing, active joints, and temporal coverage | Section 4.6 |

## 4.2 Availability, Units, and Effective Sampling

The ST3215 provides load, position, speed, input-voltage, current, and temperature readings over its serial bus [8]. The Waveshare ST3215 control table specifies 6.5 mA per current count [8]. Multiplying every reading by the same constant would change neither the standardised effect sizes nor the per-channel normalisation used later, and the analysis therefore retains the original counts. In figures and tables, J1 to J6 denote shoulder pan, shoulder lift, elbow flexion, wrist flexion, wrist roll, and gripper, respectively.

The logger timing and static-noise study measured the practical acquisition rate by reading all six registers from all six servos in sequence for 60 s at each of three poses. The resulting timing and static-noise measurements appear in Table 4.2.

At 55.3 Hz, one complete sequential cycle takes approximately 18.1 ms. The dataset driver follows a different synchronised path for reading current, so the measured cycle time is not used as an estimate of that path's latency. The policy dataset stores a 30 Hz sample grid in its metadata and timestamps. Because those timestamps are generated from frame index, they do not measure bus-read latency and are not used to reinterpret the logger benchmark.

**Table 4.2:** Acquisition and static-noise measurements from the logger timing and static-noise study. The polling rate belongs to the sequential six-register logger and is not a protocol-level limit of the bus.

| Quantity | Value | Interpretation |
|---|---|---|
| Mean sequential poll rate | 55.3 Hz | adequate for one recorded state per 30 Hz policy tick |
| Variation across poses | less than 0.1 Hz | stable during the three static holds |
| Current standard deviation | 0.02–0.49 counts | sub-count static variation across joints |
| Mean J3 standard deviation | 0.389 counts | average across the three static poses; static-noise reference for task-aligned excursions |
| Largest within-hold voltage range | at most 16 counts | includes isolated read jumps |
| Largest within-hold temperature range | at most 37 °C | includes isolated read jumps |

## 4.3 Pose Dependence, Load Response, and Noise

The static-pose study examines the current baseline during 3 s holds at 20 arm poses, with three repetitions per pose. The heatmap in Figure 4.1 shows that the static baseline is not constant across poses. The visible changes in current are concentrated in J2, J3, and J4, while the current in J1, J5, and J6 remains close to zero throughout the study.

Without any external contact, J2 ranges from 0.1 count in a folded configuration to 25.8 counts in extended poses. No single current threshold can therefore identify contact throughout the workspace. Joint position remains part of the observation so that a policy can, in principle, account for this configuration dependence. The load register is less repeatable: its cross-repetition standard deviation reaches 73.5 counts for J2 at the home pose, whereas the corresponding variation in current usually remains below 5 counts.

The static-pose study also contains 28,080 temperature readings, with a median of 35 °C and a central 98% range from 33 °C to 37 °C. A few isolated read jumps extend the raw maximum to 107 °C. They are evidence of read quality rather than instantaneous heating, but the short duration of the experiment still leaves temperature and elapsed session time as possible validity factors later in the study.

The known-load study asks whether a known static load produces a change large enough to be useful. Four end-effector masses, 0, 51, 96, and 148 g, are tested at three poses before the arm returns to the unloaded condition. At each pose, the masses were applied in this fixed ascending order. The final unloaded hold was included to reveal any change in the baseline during the sequence. Figure 4.2 shows the suspended-load arrangement used for these trials.

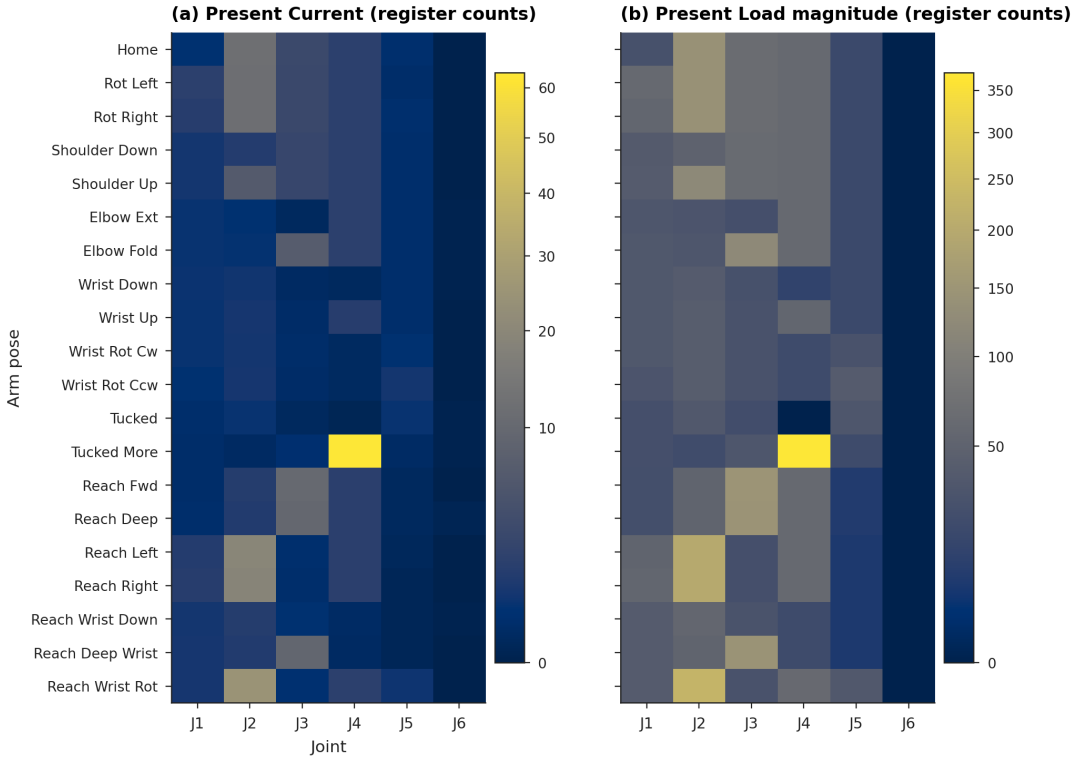



**Figure 4.1:** Mean static current and load across 20 arm poses and three repetitions. The colour scale uses square-root normalisation because a few large cells would otherwise obscure lower-magnitude differences; the colour-bar labels remain in register counts. Pose-related variation in current is concentrated in J2, J3, and J4.

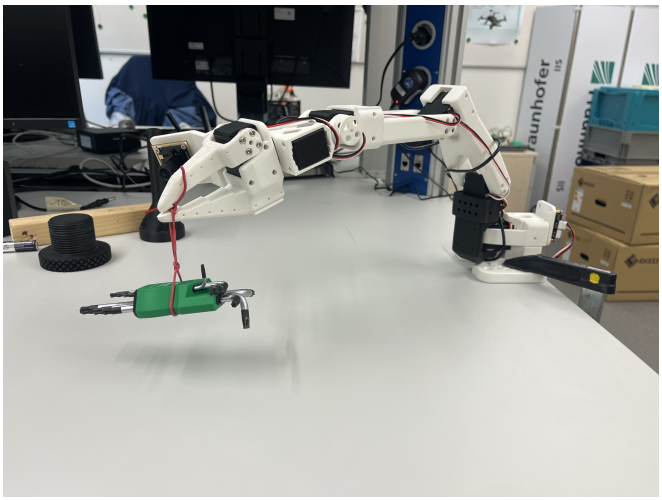


**Figure 4.2:** Known-load characterisation setup. The follower holds the suspended tool set used to vary the end-effector load while the arm posture is fixed.

Shoulder-lift current rises monotonically with added mass at all three tested poses. Relative to the first unloaded hold, the 148 g load increases its mean by 2.00 to 3.13 counts. Using the largest shoulder-lift standard deviation in the independent static-noise study as the noise scale gives a signal-to-noise ratio of 4.1 to 6.4. The samples within each hold are autocorrelated, so this ratio describes signal magnitude rather than an inferential effect size. After the mass is removed, shoulder-lift current retains an offset of 1.22 to 2.47 counts, which consumes much of the baseline-relative change. Because mass and elapsed time changed together, the study cannot separate the monotonic load response completely from drift during the sequence. It therefore shows a pose-conditioned load response rather than a pose-independent load estimate. For policy learning, this result supports combining current with joint position rather than interpreting current alone as force. The contact-specific contrasts in the next section are considerably larger.

The remaining static variation is small: every joint in the logger study has a standard deviation below one current count, and the readings are integer-quantised. Applying a low-pass or Kalman filter merely to suppress that variation could also attenuate brief task events. The dataset therefore keeps the measured current values and leaves their temporal structure to causal history encodings.

## 4.4 Gripper Contact Discrimination

The scripted-contact study provides the clearest independently labelled test of contact discrimination. At a gripper target of 1950 servo ticks, the protocol contains four sequential blocks of 15 scripted closes: an empty-close block, two repeated blocks using the same marker pen, and a folded-towel block. Two of these conditions are shown in Figure 4.3. Figure 4.4 then compares the measured responses across the object-contact and empty-close conditions.



**Figure 4.3:** Examples from the scripted gripper-contact study. The gripper closes on the marker used in both pen-contact blocks on the left and on the folded towel on the right.

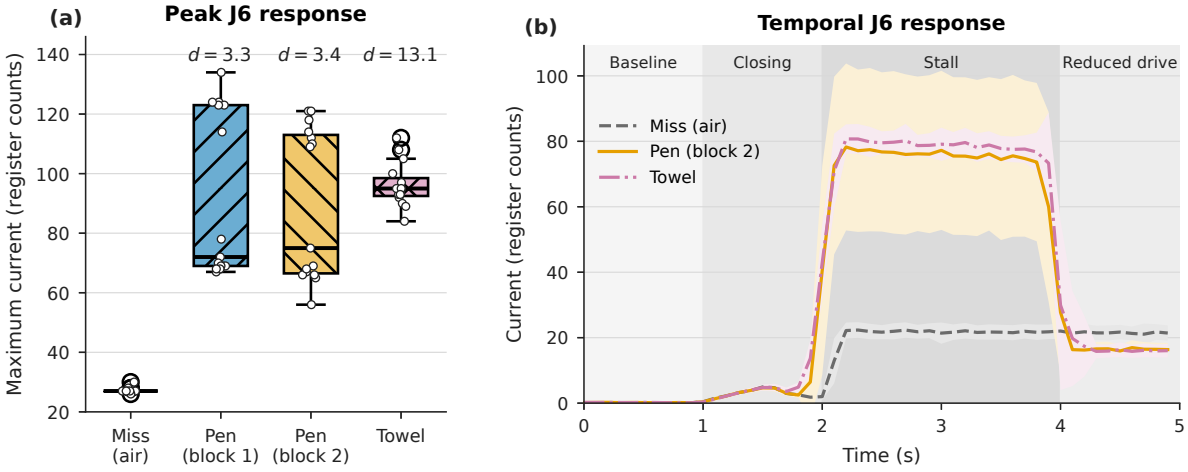



**Figure 4.4:** Scripted gripper-current response at target position 1950 servo ticks. The left panel shows the maximum J6 current in each trial, and the right panel shows temporal means with one-standard-deviation bands. During closing, the object-present traces separate from the misses.

Empty closes reach a maximum of 27.3 ± 0.9 counts. Closes in the first pen block, second pen block, and towel block reach 91.5 ± 27.6, 89.1 ± 25.5, and 96.1 ± 7.4 counts, respectively. Here and below, the reported dispersion is the sample standard deviation and Cohen's $d$ uses the pooled sample standard deviation of the two conditions. Treating each scripted close as one experimental unit, the corresponding differences from the empty-close condition

are $d = 3.29$, 3.42, and 13.06. Large-sample normal approximations give 95% intervals of $[2.2, 4.4]$, $[2.3, 4.5]$, and $[9.7, 16.4]$. The close- and episode-level effect-size intervals reported in this chapter are exploratory large-sample approximations from small samples; they do not quantify variation across independent hardware sessions. Each contrast clears the screening threshold by a wide margin. The especially large value for the towel condition comes mainly from its lower dispersion and should not be interpreted as evidence for a general material classifier.

The temporal traces add information that is lost in a single maximum. Current rises over roughly 100 ms in the second pen block and over roughly 300 ms in the towel block. The difference is descriptive because object geometry and placement were not controlled independently of material compliance. Both object-contact traces fall sharply about two seconds after the current reaches its stall plateau, while the empty-close trace remains nearly unchanged. The timing and residual level are consistent with overload protection. The retained follower configuration sets the gripper overload threshold to 25%, while the ST3215 control table specifies a 2 s delay and a protected output torque of 20% [8]. Overcurrent protection is less plausible because the configured threshold is 250 current counts and the largest recorded J6 current is 134 counts. The live protection registers and status bit were not logged, so the protection state is inferred from the recorded timing and configuration rather than observed directly.

Current and gripper position also describe different aspects of the close. With the same target command, the mean final position is 2002.5 for an empty close, 2057.9 for the towel block, and 2058.2 for the second pen block. A position that remains away from its target while current is elevated is consistent with a stalled close. Under this protocol, that combination gives the policy a useful contact cue without turning either register into a calibrated force measurement.

The load register can distinguish contact as well, with effect sizes ranging from $d = 1.23$ to 18.86 against the miss condition. However, it clips at 500 counts in some rigid-contact trials, was less repeatable in the static-pose study, and largely duplicates current in the static measurements. Across 51,474 repeated static samples, the two registers have a Spearman correlation of $\rho = 0.88$. Because those samples are not independent experimental units, the value describes co-variation within this arm rather than a population relationship. Load is excluded because it is the coarser and less repeatable of two strongly redundant inputs.

## 4.5 Behaviour During Natural Motion

The scripted closes establish separability under a controlled motion, but a policy must observe the same distinction during ordinary task behaviour. The natural-motion study therefore records five teleoperated pick-and-place episodes with a pen and five matched episodes in which the gripper closes on air, all at 30 Hz. As Figure 4.5 shows, variation in approach, speed, and grasp timing does not erase the difference in gripper current.

The per-episode maximum is $59.6 \pm 36.2$ counts with the object and $15.8 \pm 5.8$ without it. Taking one episode as the experimental unit, this contrast gives $d = 1.69$, with a large-sample

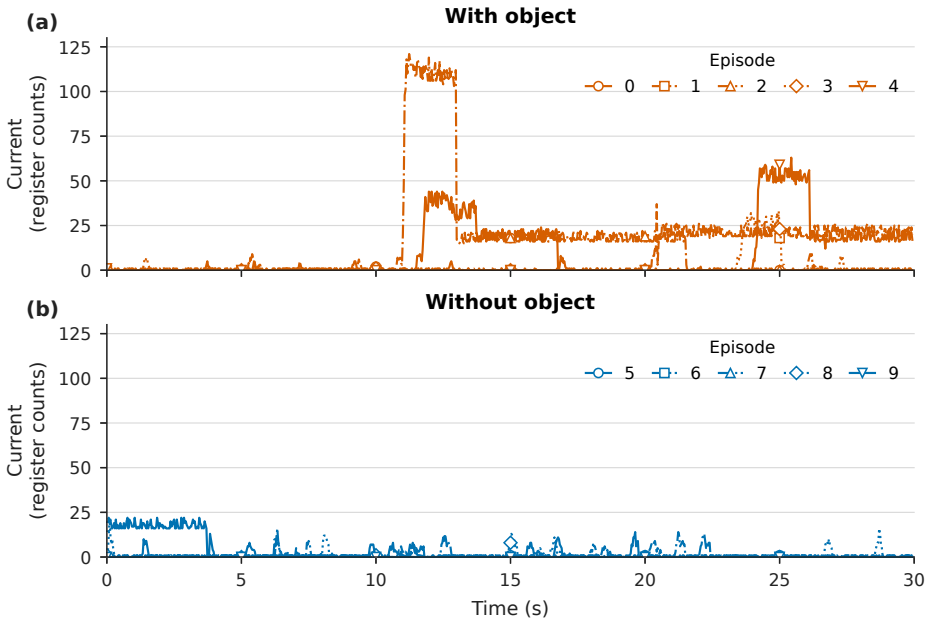



**Figure 4.5:** J6 current during five natural-motion episodes per condition. Object-present episodes show larger and longer current responses than empty-gripper episodes, although the timing varies across the teleoperated motions.

normal-approximation 95% interval of $[0.2, 3.1]$. The corresponding episode means are $6.3 \pm 4.0$ and $1.0 \pm 0.9$ counts, giving $d = 1.83$ with a 95% interval of $[0.4, 3.3]$. Current remains above 10 counts for an average of 6.5 s with the object, compared with 0.9 s without it. Both intervals are wide because each condition contains only five episodes.

The gripper channel remains discriminative even though one episode contains an arm-joint maximum of 137 counts. Its local response is useful during grasp and transport, but it is still a mixture of commanded motion, friction, stall, and possible firmware effects. The data provide no independently labelled contact event during approach, and release was not characterised separately.

## 4.6 Insertion-Phase Arm Current

During insertion, the relevant current response shifts from the gripper to the arm as the grasped peg approaches and enters the hole. The insertion-phase boundaries of the pool of episodes 1 to 30 were checked against video before this analysis. To compare episodes with different pose-dependent baselines, each current channel is referenced to its median over the 15 samples immediately before the annotated insertion phase, approximately 0.5 s at the recorded rate. For each joint, the phase excursion is the largest three-sample mean during insertion minus that channel-specific reference. The three-sample mean attenuates the contribution of an isolated one-sample excursion while retaining changes that persist over approximately 100 ms; it is a

smoothing statistic, not a persistence test.

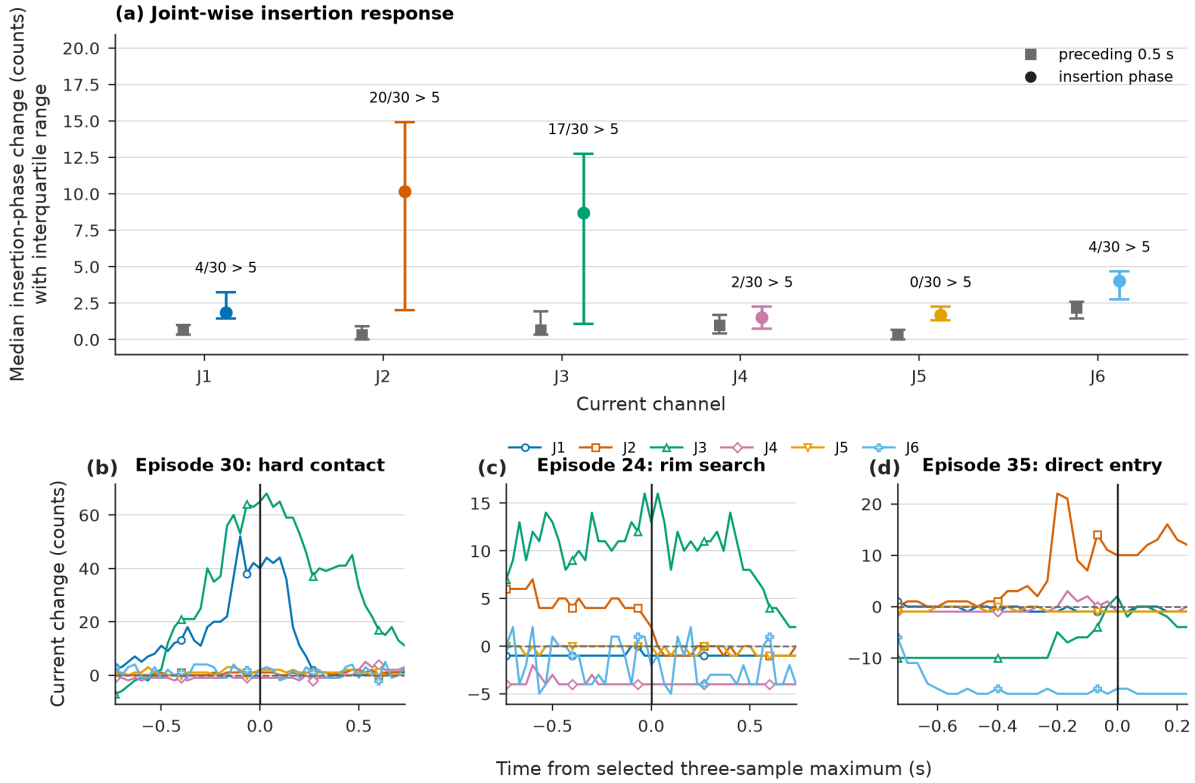



**Figure 4.6:** Baseline-corrected current during current-seeded, video-checked insertion phases. Panel (a) compares the median and interquartile range of the largest three-sample change during insertion with the same statistic in the preceding 0.5 s; the annotations give the number of insertion phases exceeding 5 counts. Panels (b) and (c) show all six current channels around the hard-contact and rim-search episodes from that pool, while panel (d) shows the separately reviewed episode 35 as a direct-entry illustration. Each trace is centred on the largest three-sample elbow-current mean and expressed relative to its own pre-insertion baseline.

The clearest recurring changes occur at the shoulder lift and elbow. The median baseline-corrected excursion is 10.17 counts at J2 and 8.67 counts at J3. J2 exceeds 5 counts in 20 of the 30 episodes, while J3 exceeds 5 counts in 17. In the preceding 0.5-s window sampled at the same rate, the corresponding medians are only 0.33 and 0.67 counts. The insertion statistic exceeds its paired preceding-window value in 25 of 30 episodes at J2 and 22 of 30 at J3. The remaining joints have median excursions between 1.50 and 4.00 counts. The hard-contact episode produces the largest coordinated response, reaching 22.7 counts at J2, 65.3 at J3, and 44.0 at J1 above their respective baselines. The rim-search example shows a moderate elbow response of 15.0 counts, while the separately reviewed direct-entry example has no positive elbow excursion but has a 17.3-count shoulder-lift change.

This distribution is more informative than comparing raw maxima because it accounts for the different current levels already present when each insertion begins. The result supports treating the six-channel current vector as a variable temporal input during insertion rather than as a fixed contact threshold. Vision and joint position provide the scene and configuration context needed to interpret that input.

The peak-centred traces contain both brief changes and longer plateaus. At 30 Hz, a ten-sample history covers nine intervals, or 0.30 s, and captures the local onset of short changes but not a multi-second insertion progression. Because samples are approximately 33 ms apart, changes that occur entirely between two samples are not recorded.

Taken together, the insertion recordings show a signal that is joint-specific and variable in duration and shape. All six current coordinates are therefore retained, allowing a learned representation to draw on gripper information during grasp and arm information during insertion.

## 4.7 Can Motor Current Serve as a Policy Input?

Table 4.3 summarises how each result determines the policy input.

**Table 4.3:** Summary of the signal evidence used to select the policy input.

| Evidence | Main result | Consequence for the policy input |
|---|---|---|
| Scripted gripper closes (Section 4.4) | Minimum $d = 3.29$ across 15 trials per condition | Include gripper current |
| Natural task motion (Section 4.5) | $d = 1.69$ to $1.83$ across five object and five empty-gripper episodes | Retain gripper current in task data |
| Known masses (Section 4.3) | At all three poses, current at the shoulder lift rises monotonically by 2.00–3.13 counts; offset after removal is 1.22–2.47 counts | Retain arm current with joint-position context |
| Current-seeded, video-checked insertion phases (Section 4.6) | Median excursions are 10.17 counts at the shoulder lift and 8.67 at the elbow, compared with 0.33 and 0.67 in the preceding window | Retain all arm-current channels with temporal context |
| Other registers (Sections 4.2 and 4.4) | Load is detectable but coarse, clipped, and redundant; temperature, voltage, and speed fall outside the current-only comparison or remain diagnostic metadata | Exclude them from the policy input |

On this arm and task, motor current is suitable for evaluation as an additional policy input because the ST3215 register distinguishes contact-relevant conditions. Gripper current separates closes on an object from closes on empty space in the controlled trials and remains discriminative in the small natural-motion sample. Arm current exhibits task-aligned insertion-phase excursions

of varying magnitude across the 30 current-seeded, video-checked demonstrations. At the same time, the signal is pose-dependent, zero-inflated in the training data, integer-quantised, and neither a calibrated force measurement nor a directional contact measurement.

The evidence therefore supports retaining all six current channels alongside the wrist image and six joint positions. Temperature remains available for diagnostics, while load, speed, and voltage are excluded from the policy input. The study deliberately isolates motor current as the added modality, so it neither derives a power proxy from voltage or speed nor supplies hand-designed current derivatives to the policies. Learned temporal variants instead receive causal histories of the six measured current channels, allowing temporal patterns to be learned without introducing another register or a manually specified derivative. Chapter 5 specifies the training-data transforms, and Chapter 6 defines the direct and history-based representations used in the policy comparisons.

This decision is tied to one SO-101, its particular ST3215 units and firmware state, and the task distribution recorded for this study.

# 5 Multimodal Dataset

A fair comparison between a vision-based policy and a current-aware policy requires more than recording both modalities. The policies must learn from the same demonstrations, images, joint positions, and action targets, while the additional current input is introduced through a reproducible transformation. The data used to fit that transformation must also remain separate from the closed-loop trials used to judge policy success.

These requirements define the peg-in-hole dataset used throughout the thesis. Its synchronised schema provides one common data source from which position-only and current-aware policies can be compared without changing the demonstrated behaviour.

## 5.1 Task and Demonstration Distribution

The final task is cylindrical peg-in-hole insertion with a peg diameter of 16.45 mm and a bore diameter of 16.8 mm, both measured directly with calipers during setup. This gives a diametral clearance of 0.35 mm, or $c/d = 0.021$, at an entry without a chamfer. Because 3D printing introduces dimensional variation, these measurements describe the actual fixture used in the experiments rather than the nominal CAD dimensions. The fixture is based on the printable NVIDIA IndustRealKit design [52].

We settled on the final task after collecting demonstrations for two earlier versions. The first collection used the flat configuration, a chamfered entry, and a wider clearance, which made the insertion comparatively forgiving. A second collection removed the chamfer and reduced the clearance while retaining the flat configuration. The final task kept this tighter geometry and used the vertical configuration, in which the peg began upright and the side grasp moved it directly in front of the wrist camera as the arm approached the fixture. The representative frames in Figure 5.1 show the resulting observation geometry near insertion: in the vertical configuration, the peg hides the peg-hole relation in the wrist view, while the side view retains the wider scene. The resulting task therefore preserved a meaningful vision-based baseline while making the close-contact stage more demanding. Chapter 7 compares the policies on this final task and reports the earlier task-development results.

The peg began upright at a fixed marked start position. The demonstrated task proceeded from approach and grasp through transport and insertion to release. Figure 5.2 shows these five stages in one reviewed demonstration and makes the two camera roles explicit. During collection, an episode was kept only when the peg appeared fully seated and repeated an attempt otherwise. The dataset stores the retained trajectories. Some retained demonstrations include corrections or repeated attempts, but recovery cases were not collected as a deliberately

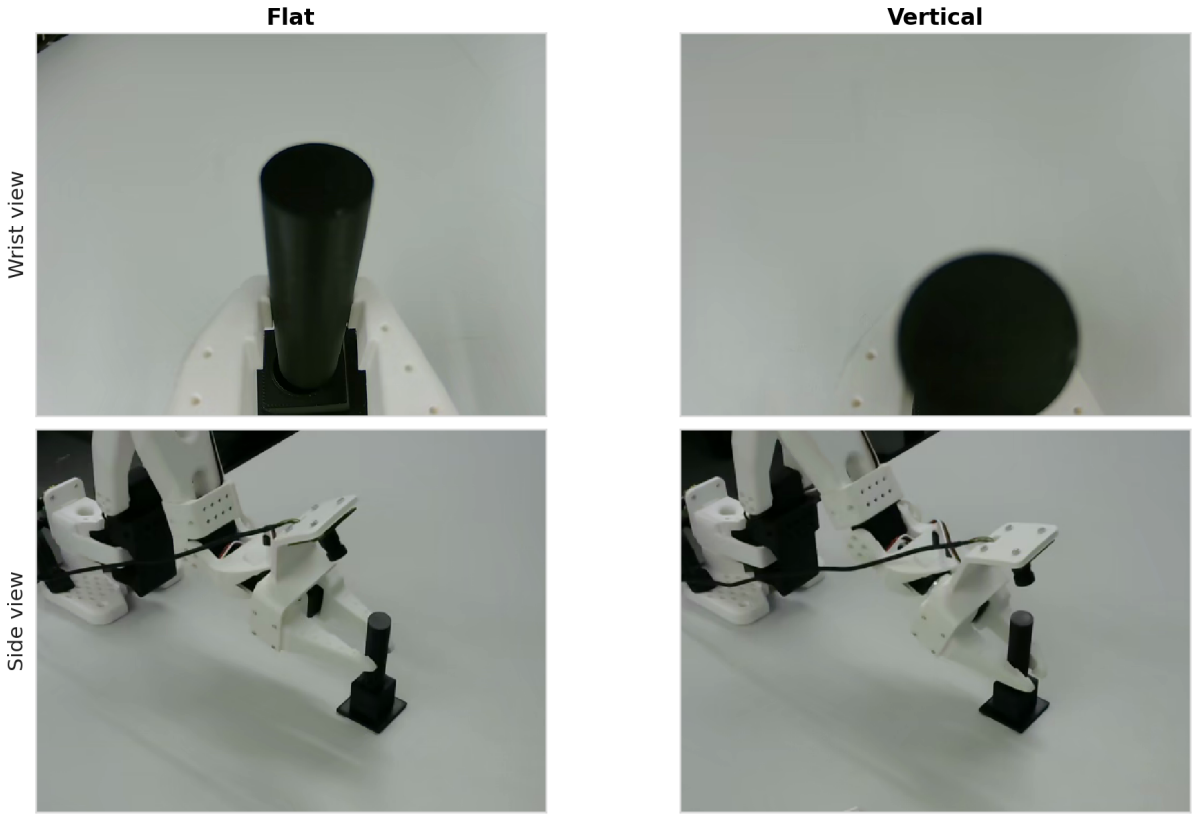



**Figure 5.1:** Flat and vertical task observations during insertion. The top row shows the complete wrist-camera policy observation; the bottom row shows synchronised side-camera frames with the same fixed crop for legibility.

balanced part of the dataset.

The wrist image and joint positions provide the main geometric description of the scene. Current is recorded for all six joints at every step, adding a low-dimensional measurement of motor effort alongside the image and joint-position observations. Using the same episodes for both observation spaces allows the later policy comparisons to ask whether this additional signal changes learning, rather than whether the operator demonstrated different motions.

## 5.2 Leader-Follower Collection and Alignment

Demonstrations were recorded with the LeRobot leader-follower interface [53] and two mechanically similar SO-101 arms. Passive leader was moved by hand while the follower tracked the six leader joint targets at a configured rate of 30 Hz. The two arms used separate LeRobot calibrations to express their five arm joints in degrees and the gripper on a scale from 0 to 100. The recovered calibration files are reproduced in Table 5.1. The same leader and follower calibration files were used from demonstration collection through all policy evaluations

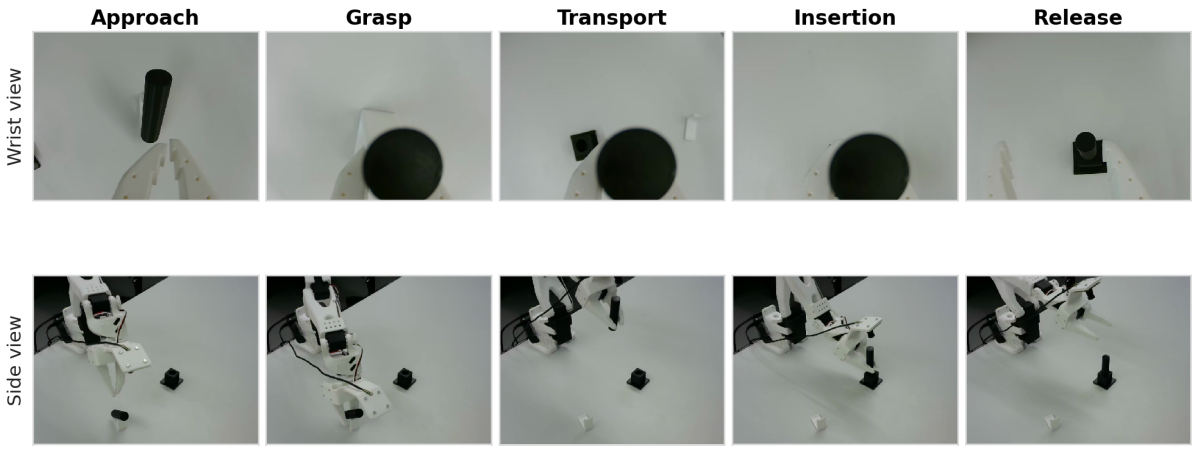



**Figure 5.2:** Approach, grasp, transport, insertion, and release in one reviewed vertical peg-in-hole demonstration. The top row shows the wrist-camera policy observation, and the bottom row shows the synchronised side-camera view used for inspection and annotation.

in this thesis. Their endpoint ranges and homing offsets are stated in native servo ticks, and every drive-mode flag is zero. The endpoint ranges define the coordinate conversion for each arm rather than a mechanical safety limit.

**Table 5.1:** Leader and follower calibration values used for demonstration collection. Ranges and offsets are in native servo ticks.

| | Leader | | | Follower | | |
|---|---|---|---|---|---|---|
| Joint | Minimum | Maximum | Offset | Minimum | Maximum | Offset |
| J1 shoulder pan | 620 | 3441 | -1749 | 670 | 3396 | 1944 |
| J2 shoulder lift | 866 | 3227 | -1277 | 983 | 3264 | 1906 |
| J3 elbow flexion | 877 | 3080 | 1431 | 793 | 2994 | 827 |
| J4 wrist flexion | 743 | 3166 | -1717 | 775 | 3200 | 1190 |
| J5 wrist roll | 0 | 4095 | -176 | 0 | 4095 | -1831 |
| J6 gripper | 2015 | 3161 | 1093 | 1909 | 3415 | -158 |

The stored action is the calibrated leader target requested for that control step, before any follower-side safety clipping. A maximum target change of 80 in these calibrated coordinates was configured as a safety limit, but the largest recorded difference between a target and the corresponding follower position was 21.63. No recorded target-to-position difference approaches the configured limit, which is consistent with the limiter remaining inactive. The post-limiter command was not stored separately, so non-activation cannot be verified directly from the dataset.

Figure 5.3 shows the hand-operated leader and the separate follower workspace used to record the demonstrations. Current was recorded from the follower but was not returned to



**Figure 5.3:** Leader and follower workspaces used for demonstration collection. The orange and white leader arm was moved by the operator, while the follower executed the corresponding targets in the peg-in-hole workspace. The follower carries the wrist camera, and the fixed side camera records the wider scene for inspection and annotation.

the leader, so the demonstrations were guided by sight and physical manipulation of the leader arm rather than current feedback. The final dataset contains two video streams at $640 \times 480$ pixels and a nominal 30 Hz. The wrist camera supplies the image used by the principal policies, while the fixed side camera supports annotation and the separate camera comparison reported in Chapter 7.

Each stored row pairs one observation vector and one leader action with a frame index on the configured 30 Hz grid. The camera and bus readings are acquired within the same control cycle. The alignment is therefore accurate to the recorded control step rather than simultaneous at the sensor level. Timestamps are generated as frame index divided by frame rate. These timestamps describe the intended sample grid.

## 5.3 Stored Schema and Controlled Observation Views

The retained schema extends the standard joint-state observation with six Waveshare ST3215 current values and six servo temperatures. Table 5.2 lists the stored features and explains how each is used.

The six joint coordinates follow the physical order shoulder pan, shoulder lift, elbow flexion, wrist flexion, wrist roll, and gripper. Current and temperature use the same order, which permits a direct association between each joint position and its local effort signal. Temperature is preserved because an unusual value can reveal a read or session-state problem, but it is not supplied to a learned policy.

Three derived views provide the inputs needed later. The position-only view retains the first six state coordinates. The raw current-aware view retains the first twelve coordinates. The bounded current-aware view contains the same positions and six current channels, but caps each current reading at a joint-specific upper value while it is still expressed in raw register counts.

**Table 5.2:** Stored multimodal schema and its role in policy learning.

| Stored feature | Contents and shape | Use in this thesis |
|---|---|---|
| Joint state [0:6] | six follower positions | every principal policy |
| Joint state [6:12] | six current registers | current-aware policies |
| Joint state [12:18] | six temperatures | session diagnostics only |
| Wrist image | RGB, $480 \times 640 \times 3$ | every principal policy |
| Side image | RGB, $480 \times 640 \times 3$ | annotation and camera study |
| Action | six leader joint targets | behavioural-cloning target |

This training-derived capping, also known as winsorisation, is distinct from normalisation, which is applied afterwards by the selected policy implementation. All three views preserve the same 100 episode identities, 99,831 sample indices, wrist images, timestamps, and action targets exactly. Their only intended difference is the selected state representation.

## 5.4 Dataset Composition and Lineage

The primary dataset, `peg-tight-vertical-100`, combines two 50-episode vertical-insertion batches recorded with the same task and schema. Each source batch contains two collection sessions of 25 episodes, giving four sessions in total. The merge reuses the source rows and videos and does not create additional demonstrations. Table 5.3 summarises the resulting training population.

**Table 5.3:** Composition of the primary multimodal training dataset.

| Property | Value |
|---|---:|
| Operator-retained episodes | 100 |
| Frames | 99,831 |
| Nominal frame rate | 30 Hz |
| Mean duration on the nominal grid | 33.3 s |
| Collection sessions | 4 |
| Recorded cameras | wrist and side |
| Full state dimension | 18 |
| Action dimension | 6 |

The earlier collections remain available as task-development records, but they are not mixed into the final policy comparison.

**Table 5.4:** Lineage of the retained insertion datasets.

| Collection | Episodes | Frames | Purpose |
|---|---|---|---|
| `peg-insertion-flat` | 50 | 50,263 | chamfered pilot |
| `peg-tight-flat` | 50 | 47,159 | tight-task development |
| `peg-tight-vertical` | 50 | 51,057 | first primary batch |
| `peg-tight-vertical-ext` | 50 | 48,774 | second primary batch |
| `peg-tight-vertical-100` | 100 | 99,831 | merged training view |
| Unique recorded data | 200 | 197,253 | excludes the merged row |

## 5.5 Current Distribution

Each current channel stores the uncalibrated current-register reading from one Waveshare ST3215 servo. The register table specifies 6.5 mA per count [8], so the observed maximum of 77 counts corresponds to approximately 500 mA. The policy input retains register counts because this global linear conversion does not add information and is absorbed by subsequent scaling. Every stored value in the primary dataset is a non-negative integer.

Figure 5.4 shows the distribution over all 99,831 training frames. Across the six channels, 44.9 % of the values are zero, and the per-joint zero fractions range from 34.6 % to 54.0 %. The shoulder-pan channel has a median of zero and a 99.5th-percentile value of 4 counts, but reaches 52 counts. Its skewness of 11.3 and excess kurtosis of 398.5 illustrate how strongly a small number of large readings extend the otherwise compact distribution. Chapter 4 showed that such changes can coincide with contact-relevant task phases, while this dataset-level view establishes the scale that the learning method must handle.

Ordinary mean and standard-deviation scaling maps a zero current reading to values between $-0.51$ and $-0.88$ across the joints. At the other end of the distribution, the observed maxima lie between +6.20 and +54.54 standard deviations above their channel means. This mismatch allows rare observations to dominate the numerical range after scaling. It motivates a bounded current representation before the family-specific policy normaliser is applied.

## 5.6 Training-Derived Capping and Leakage Prevention

The selected representation caps each joint independently at the 99.5th percentile of its training distribution. Figure 5.5 compares those caps with the complete observed ranges, while the transformation is defined as

$$\tilde{I}_j = \min(I_j, q_{99.5,j}). \tag{5.1}$$

The fitted caps are 4, 33, 32, 8, 3, and 33 counts for shoulder pan, shoulder lift, elbow flexion, wrist flexion, wrist roll, and gripper, respectively. Capping changes only values above these limits, corresponding to between 0.19 % and 0.50 % of the training values in each joint. The remaining values, including measurements equal to the cap, retain their original integer

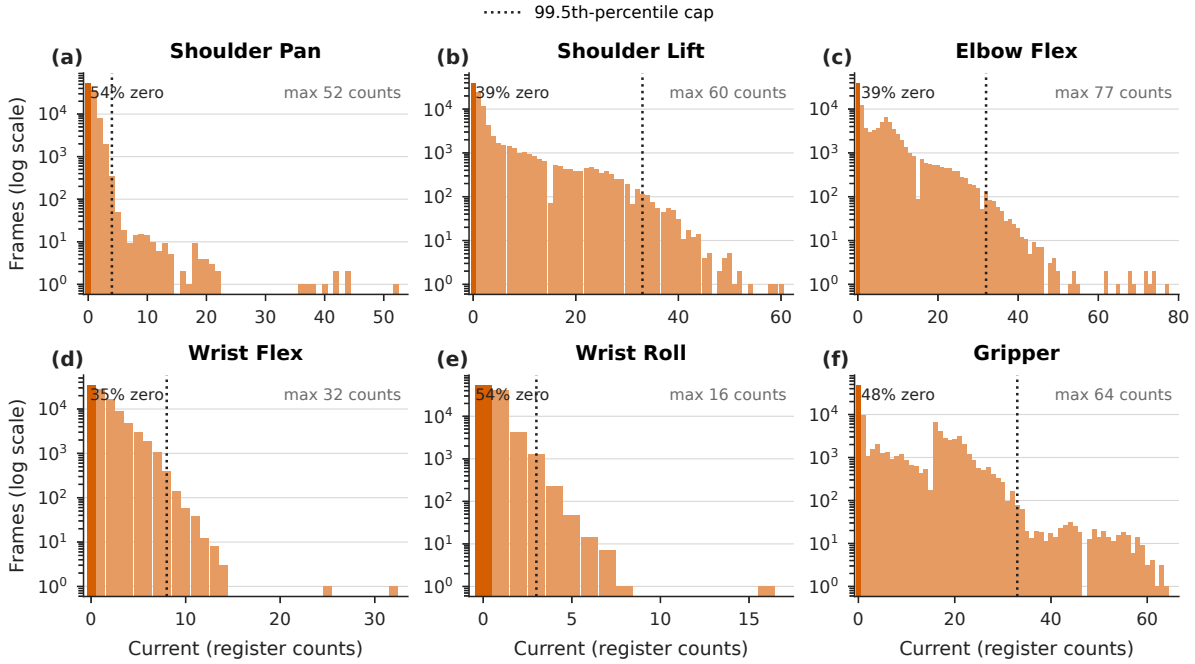



**Figure 5.4:** Per-joint current distributions over 99,831 training frames on a logarithmic frequency axis. The darker bar marks zero-valued readings, and the dotted line marks the training-set 99.5th-percentile cap used by the bounded current view.

resolution. The 99.5th percentile was fixed from the training distribution before the normalised and learned-current policy comparisons. Raw-current comparisons retain the unbounded register values and are identified explicitly where they are reported.

After capping, Action Chunking Transformer models apply their usual per-channel mean and standard-deviation normalisation. Diffusion Policy models instead apply the implementation's native per-channel minimum and maximum mapping. Because capping sets each current-channel maximum to its fitted limit, the Diffusion Policy mapping uses these values rather than the raw observed maxima. The cap therefore provides the common bounded representation, while each policy family retains its own established normalisation procedure. Chapter 6 specifies the corresponding deployment processor order and the causal current histories used by learned temporal variants. Chapter 7 reports the policy-level evidence used to compare alternative representations.

All 100 demonstrations form the training population, so the caps and normalisation statistics are fitted on those demonstrations only. There is no held-out demonstration split from which offline generalisation is claimed. Closed-loop evaluation episodes are recorded separately and never contribute to the fitted caps, normalisers, or training views. This separation prevents evaluation outcomes from influencing the observation representation and ensures that every policy condition receives the same demonstrated actions.

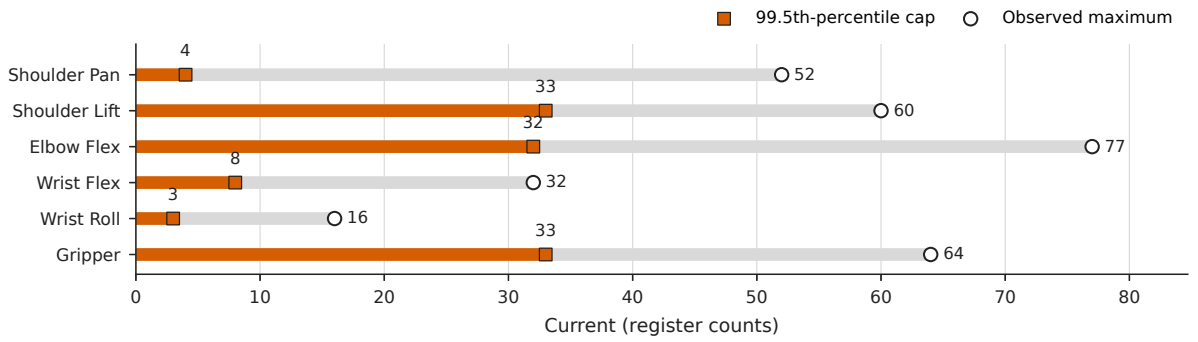



**Figure 5.5:** Observed current range and per-joint cap. Orange segments end at the 99.5th percentile, and open circles mark channel maxima. Larger values are capped before policy-family normalisation.

## 5.7 Quality Control and Annotation Provenance

Collection used a 45 s episode limit, a 15 s reset window, video recording at a nominal 30 Hz, and a scheduled ten-minute break between the two sessions in each batch. Temperature was stored continuously for later inspection of read quality and session state. The reproducibility check verifies the episode and frame counts, finite state and action values, dimensions, timestamp grid, dataset lineage, and exact equality of the controlled views. It also confirms that the generated bounded view equals the raw-current view after applying the six recorded caps, with positions, actions, and indices unchanged.

Phase annotations are stored separately and are never used as policy inputs. Chapter 4 uses a contiguous set of 30 reviewed insertion episodes for its insertion-phase analysis. These annotations support phase-specific inspection and the timing rule described in Chapter 6. The dataset represents one task, one follower arm and servo configuration, one operator, and four collection sessions. Its operator-retained demonstrations emphasise successful task execution and offer limited examples of autonomous failure recovery. Camera frames and bus values are aligned by control step rather than by hardware timestamps.

# 6 Multimodal Policy Learning Method

Using motor current as a policy observation requires a representation that preserves useful variation and combines it effectively with vision and joint position. The first comparison therefore adds the current measured at the present time step to an otherwise unchanged Action Chunking Transformer. A second ACT variant replaces this direct concatenation with learned current histories for the individual joints, allowing the policy to capture short temporal patterns. The final comparison adds a learned current route to Diffusion Policy and examines the same modality under a different approach to action generation.

The study separates three stages of evidence. The input pipeline specifies exactly which current measurements reach the model and how they are processed. Counterfactual tests on selected checkpoints show whether changing those measurements changes the predicted actions. Repeated trials on the robot test whether the resulting behaviour improves peg-in-hole performance.

For concise labels, ACT-V and DP-V denote the vision-and-position baselines, while ACT-M and DP-M denote policies that also receive motor current. ACT-M-norm identifies the direct ACT-M variant that applies training-derived current caps before normalisation.

## 6.1 Research Design and Controlled Contrasts

Every principal policy uses feature subsets of the same 100-demonstration vertical peg-in-hole dataset. The wrist image, six absolute joint-position targets, episode boundaries, and training seed are shared within each policy family. The side camera remains available for inspection and annotation but is not supplied to the principal policies. Feature subsetting therefore changes what a model can observe without changing the demonstrations from which it learns.

Table 6.1 states the interpretation of each comparison before any result is considered. The direct ACT comparison changes the current input and its preprocessing while keeping the remaining architecture and training settings fixed. The temporal ACT comparison deliberately changes both the representation of current and its fusion location. It is evaluated against its contemporaneous ACT-V control and therefore tests that complete learned representation within its own session, rather than ranking it directly against concatenation results obtained in another session. The final Diffusion Policy comparison holds the demonstrations, starting checkpoint, additional training updates and configured deployment procedure in common. It evaluates a complete current-aware design against a control matched for training budget. Its choice to freeze the current-aware base while updating the comparator base is part of that design, so the individual contribution of current is not isolated.

**Table 6.1:** Controlled policy comparisons used in the thesis.

| Comparison | Shared elements | Changed element | Interpretation |
|---|---|---|---|
| Direct ACT (Section 6.3.2) | Dataset, image, positions, ACT trunk, objective, seed, and paired rollout controller | Instantaneous current is absent, raw, or bounded | Direct test of current as an additional state input |
| Temporal ACT (Section 6.3.3) | Dataset, image, positions, ACT trunk, objective, seed, and paired rollout controller | Ten-sample per-joint history and six token routes replace direct current fusion | Test of a learned temporal representation and joint-selective fusion |
| Final Diffusion Policy (Section 6.4.2) | Dataset, warm-start source, image and position route, seed, update budget, weight selection, sampler, and execution queue | Current encoder and residual conditioning are added; the current-aware base is frozen while the comparator base is updated | Cross-architecture system comparison with a matched additional training budget |

## 6.2 Observations, Actions, and Preprocessing

At time $t$, the common visual observation is the wrist image $x_t$. The position-only state is

$$\mathbf{s}_t^V = \mathbf{q}_t = [q_{1,t}, \ldots, q_{6,t}] \in \mathbb{R}^6, \tag{6.1}$$

and the current-aware state is

$$\mathbf{s}_t^M = [\mathbf{q}_t, \mathbf{I}_t] = [q_{1,t}, \ldots, q_{6,t}, I_{1,t}, \ldots, I_{6,t}] \in \mathbb{R}^{12}. \tag{6.2}$$

Here $I_{j,t}$ is the motor-current register value for joint $j$. It is treated as an uncalibrated observation coordinate, not as force or torque. The action $\mathbf{a}_t \in \mathbb{R}^6$ contains the six calibrated leader targets requested for the follower at the same indexed frame.

The image reference, position vector, current vector, and action share the camera-paced frame index described in Chapter 5. Load, speed, and voltage are excluded so that the multimodal comparison concerns the current signal selected in Chapter 4.

The preprocessing order is fixed by the saved policy processor. For the bounded current view, each current channel is first clipped at the 99.5th percentile fitted on the training data. The policy family then applies its native feature normaliser. ACT uses the training mean and standard deviation for state and action coordinates. Diffusion Policy uses mean and standard deviation normalisation for images and min-max normalisation for state and action. The same current transform runs before the saved normaliser during deployment.

Temporal variants use only past and present measurements. The ACT encoder receives ten current samples, $\mathbf{I}_{t-9:t} \in \mathbb{R}^{10\times6}$, which span nine intervals or 0.30 s at 30 Hz. The selected Diffusion Policy receives two observation times at 10 Hz and a nine-sample current history ending at each one. Each Diffusion history spans eight intervals or 0.8 s. At the start of an episode, unavailable preceding samples are left-padded after normalisation.

## 6.3 Action Chunking Transformer Policies

### 6.3.1 Vision-and-Position Baseline

The ACT baseline follows the conditional variational action-chunking formulation of Zhao et al. [2], implemented in LeRobot. The $640 \times 480$ wrist image passes through a ResNet-18 backbone [42]. The backbone is initialised with ImageNet-1K weights. Its final $512 \times 15 \times 20$ feature map is projected to 300 visual tokens of width 256. The six joint positions are linearly projected to one state token of the same width. A four-layer, eight-head Transformer encoder combines these tokens with a 32-dimensional latent token. A one-layer Transformer decoder uses 50 learned queries to predict 50 absolute six-joint targets.

During training, a separate variational encoder infers the latent distribution from the robot state and target action chunk. The loss is the padding-masked $L_1$ action error plus the Kullback–Leibler term with weight 10. At inference, the implementation supplies a zero latent vector,

making the ACT prediction deterministic for a fixed observation.

### 6.3.2 Direct Current Concatenation

Direct ACT-M changes the state from six positions to the twelve coordinates $[\mathbf{q}_t, \mathbf{I}_t]$. One linear layer maps the complete vector to a single state token. The model therefore receives current, but it has no separate current encoder, token, or loss. This is early fusion in its simplest form.

Two preprocessing versions are retained because they answer different practical questions. The first supplies raw register counts. The second applies the training-only percentile caps before normalisation and tests the same architecture without allowing rare current values to determine the input scale. Within each paired session, the ACT-V and current-aware checkpoints remain fixed, and both policies use the same rollout controller.

### 6.3.3 Joint-Selective Temporal Representation

The learned ACT representation removes current from the aggregate state token and exposes it through six separate temporal tokens. For each joint, the same one-dimensional CNN processes that joint's ten normalised current samples. The three convolutional layers have output widths $(16, 16, 8)$, kernel widths $(3, 3, 3)$, dilations $(1, 2, 4)$, unit stride, and a rectified linear unit (ReLU) after each layer. Mean, maximum, and latest-value pooling summarise the learned eight-channel sequence into 24 values. The matching joint position is appended, giving a 25-dimensional per-joint representation. A shared linear projection maps it to width 256, and a learned joint identity preserves the J1 to J6 ordering.

The resulting encoder sequence contains one latent token, one position token, six joint-specific temporal tokens, and 300 visual tokens. The ordinary position token remains six-dimensional before projection, so the current history has no direct-concatenation bypass. The variational posterior likewise receives position and target actions but not current. Current can influence the predicted action only through the six learned tokens. Figure 6.1 shows the shared ACT baseline and the two current routes side by side.

### 6.3.4 Training and Execution

Table 6.2 gives the ACT training settings. ACT-V, raw direct ACT-M, ACT-M-norm, and the joint-selective temporal model are trained for 100,000 optimiser steps with seed 1000.

The historical raw-current comparison executes the complete 50-action chunk before obtaining a new policy prediction. The normalised direct and temporal ACT comparisons use one shared phase-adaptive controller. Before a grasp is detected, ACT predicts a fresh chunk at 30 Hz and temporally ensembles overlapping predictions with $m = 0.01$ in the weighting convention defined in Equation 3.6. A causal detector observes only normalised gripper position. Its parameters were fixed by replaying the detector over video-reviewed demonstration annotations described in Section 5.7. After a verified open-close cycle remains stable for 15 frames, the transition latches and temporal ensembling is disabled for the rest of the episode. The controller

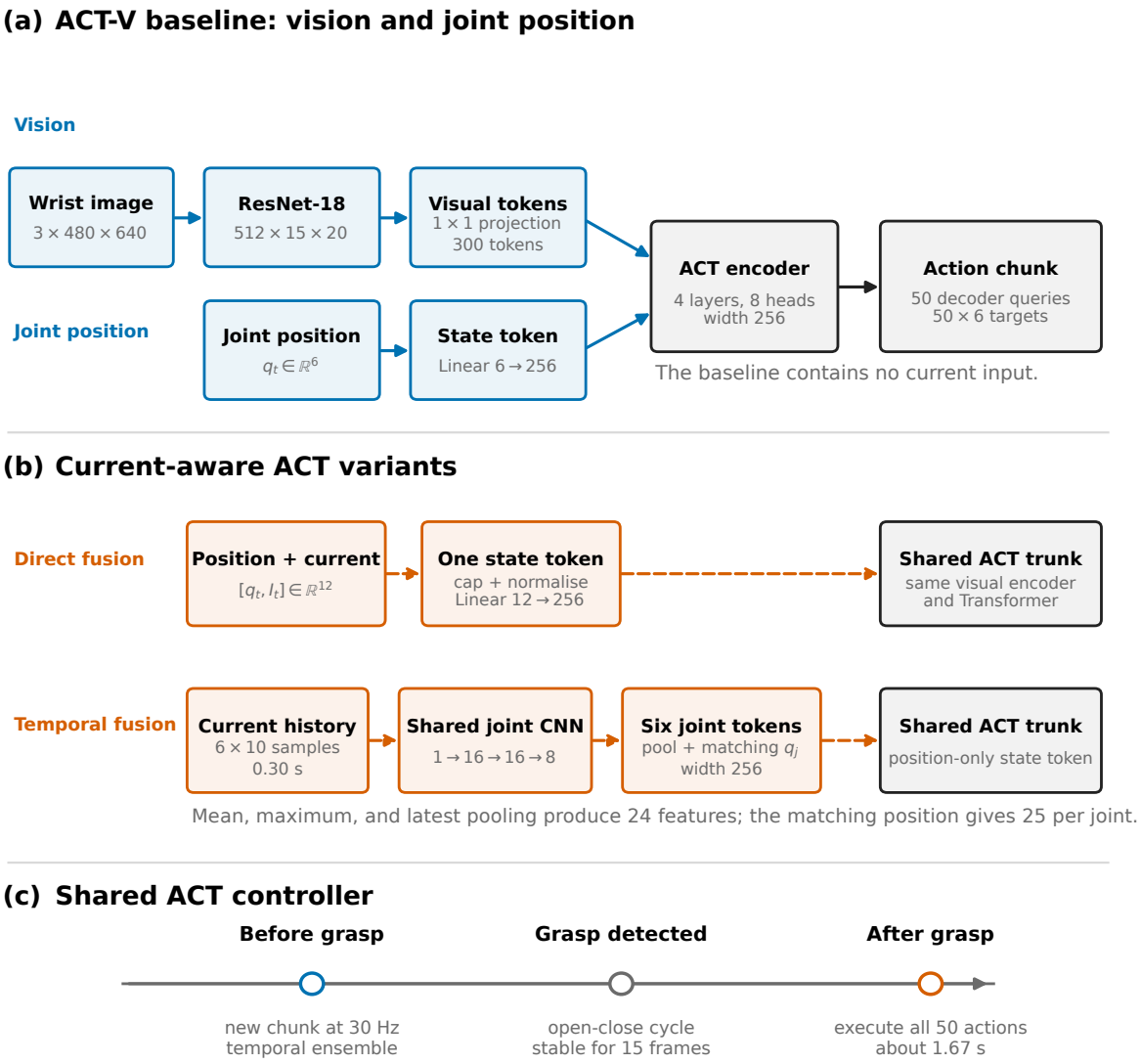



**Figure 6.1:** ACT-V baseline, current-aware ACT routes, and the shared rollout controller. Panel (a) shows the complete vision-and-position baseline before any current input is added. Direct ACT-M mixes instantaneous position and current in one state token; the raw comparison uses unbounded register counts, while the later direct comparison uses the bounded current view. The learned variant instead removes current from that token and exposes six identity-coded temporal joint representations. Both retain the same wrist-vision and ACT action-chunk trunk. The lower panel shows the controller shared within the paired normalised comparisons.

then executes the complete 50-action chunk, about 1.67 s, before asking the policy for another prediction. Episode reset clears the latch and action queue but retains no information from the preceding trial.

This controller is applied identically to both policies in a paired comparison. Its purpose is

**Table 6.2:** ACT configuration used by the completed policy checkpoints.

| Component | Configuration |
|---|---|
| Input image | wrist RGB, $640 \times 480$, 30 Hz |
| Visual backbone | ResNet-18, ImageNet initialisation |
| Action chunk | 50 absolute targets, six coordinates each |
| Transformer | width 256, eight heads, four encoder and one decoder layer |
| Feed-forward width / dropout | 3200 / 0.1 |
| Variational encoder | latent width 32, four layers, KL weight 10 |
| Optimiser | AdamW, learning rate $10^{-5}$, weight decay $10^{-4}$ |
| Batch / updates / seed | 16 / 100,000 / 1000 |
| Checkpoint interval | 20,000 updates |

to preserve trajectory coherence after grasping rather than to act as a high-frequency contact controller. Images or current measurements acquired while a full chunk is queued cannot change the actions already being executed.

## 6.4 Diffusion Policy Models

Diffusion Policy generates an action trajectory by iteratively denoising a noisy sequence conditioned on the robot observation [3]. During training, one diffusion step is sampled and the model minimises the mean-squared error between injected noise and the network prediction. Its one-dimensional conditional U-Net uses widths $(512, 1024, 2048)$, kernel width 5, group normalisation (GroupNorm) with eight groups, skip connections, and a 128-dimensional diffusion-step embedding.

### 6.4.1 Vision-and-Position Route

The evaluated DP-V receives two observation times separated by 0.1 s. Each wrist image is resized to $144 \times 192$ without cropping and encoded by a ResNet-18 with GroupNorm. A spatial-softmax layer returns 32 keypoints, or 64 visual coordinates. Appending six joint positions gives 70 values per observation and a 140-dimensional global condition after the two observations are flattened. Together with the diffusion-step embedding, each conditional residual block receives 268 nominal conditioning values.

The model predicts a horizon of 16 six-dimensional absolute actions. Training uses a 100-step DDPM process. Deployment uses the exponential moving average (EMA) weights and ten DDIM reverse steps. This deployment choice was validated against 100-step DDPM inference on 64 noise-matched demonstration observations for each of three deployment-relevant checkpoints. DDIM-10 produced mean action cosines from 0.994 to 0.999 and relative $L_2$ differences from 0.035 to 0.075 against that reference; increasing the sampler to 16 steps did not improve any of these measures. The fidelity probe compared 10 and 16 steps, while the separate

timing benchmark included 20 steps to test the real-time boundary. On the deployment host, the final DP-V and DP-M checkpoints required 61.4 ms and 66.7 ms at the 95th percentile, whereas the 20-step benchmark required 125.8 ms and exceeded the 100 ms control period. DDIM-10 was therefore the high-fidelity, deadline-compliant choice among the schedules tested, and the same sampler was held fixed for DP-V and DP-M. The policy executes horizon indices 1 through 8 (excluding index 0) and replans when the queue is empty. Images, positions, and current histories are nevertheless updated at every control tick; observations acquired while actions remain queued affect the next plan rather than the commands already in the queue.

### 6.4.2 Direct and Learned Current Routes

The first DP-M implementation appends instantaneous current to the ordinary global condition. This preserves the native conditioning mechanism and supplies a direct-concatenation counterpart to ACT-M. Its bounded version applies the same training-derived current caps as the ACT model, followed by Diffusion Policy's own min-max state normalisation.

The current-aware policy selected for the final comparison uses a separate learned route. For each of the two observation times, a shared per-joint CNN receives nine current samples. It has two convolutional layers of width 16, kernel width 3, unit dilation, unit stride, and ReLU activations. Mean, maximum, and latest-value pooling produce 48 values per joint, which a linear layer maps to eight. The two observation times and six joints therefore form a 96-dimensional current condition. Raw current is removed from the nominal global condition.

Every one of the U-Net's twelve conditional residual blocks receives the same current condition through its own adapter:

$$\mathbf{c}_I \xrightarrow{\text{LayerNorm}} \mathbb{R}^{96} \xrightarrow{\text{Linear, Mish}} \mathbb{R}^{64} \xrightarrow{\text{Linear}} \mathbb{R}^{2C}, \tag{6.3}$$

where $C$ is the output width of that residual block. The final adapter layer is initialised to zero. Its scale residual is bounded to $[-0.5, 0.5]$ by a hyperbolic tangent. Its bias residual is bounded by one detached, per-sample feature-channel root mean square (RMS) computed across the local temporal axis. These bounds limit feature modulation inside a block; they are not limits on robot motion or action magnitude. The construction is an identity-initialised residual form of feature-wise linear modulation [46]. Figure 6.2 shows the complete visual baseline and the additional residual current route used by the evaluated DP-M system.

### 6.4.3 Warm Start, Training, and Deployment

Both targets start from the same 200,000-step source checkpoint and receive 40,000 further updates with batch size 16 and seed 1000. Each uses a fresh optimiser, learning-rate schedule and EMA state. The additional training budget is therefore matched; the control is not merely the unchanged source policy. The DP-V comparator updates the complete base network. The current-aware target copies that base, freezes it, and trains only the current encoder and residual

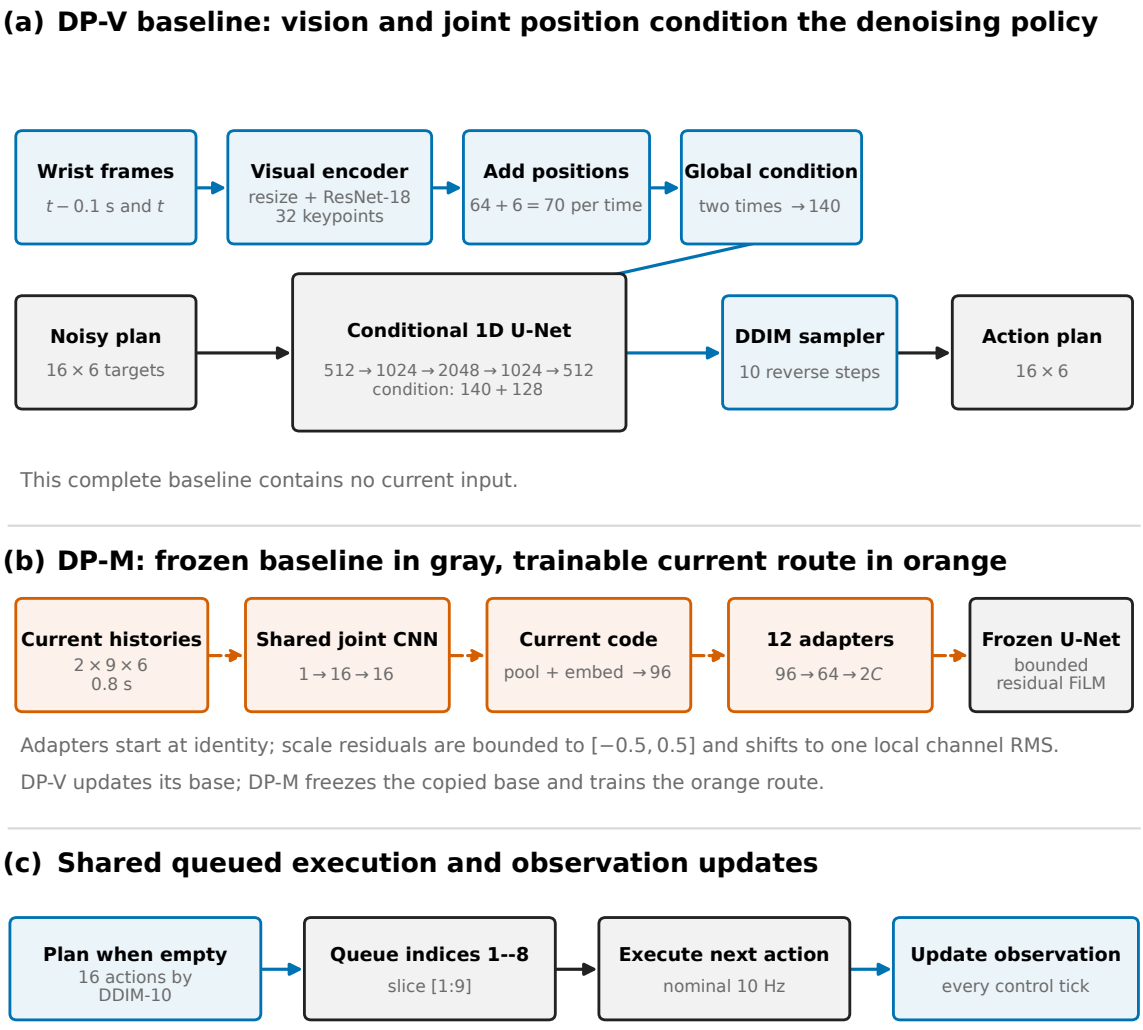



**Figure 6.2:** DP-V baseline and the evaluated current-aware Diffusion Policy. Panel (a) shows the complete vision-and-position baseline, from two observations to the denoised action plan. Both policies form the nominal condition from two wrist and position observations and denoise a 16-step absolute-action trajectory. Panel (b) shows the additional current route: the current-aware policy encodes two nine-sample per-joint histories into a 96-dimensional condition and applies zero-initialised bounded residual modulation at each of twelve U-Net residual blocks. Both designs share their starting weights and additional update budget; the current-aware base is frozen while the DP-V comparator updates its base, so the comparison evaluates the resulting systems. Panel (c) shows that observations and current histories are updated at every control tick, while a new plan is generated only when the eight-action queue is empty.

adapters. Because every adapter is initially zero, the current-aware target implements the source policy exactly at initialisation for matched diffusion noise.

Freezing retains the learned base weights while the added current route can modify its action predictions. This tests whether current conditioning with that training strategy preserves or improves task behaviour relative to a visual policy given the same additional training budget. A favourable difference supports the complete design; its attribution to current, to freezing rather than fine-tuning the base, or to their interaction remains unresolved. Table 6.3 records the common training, denoising, and execution settings of the final pair.

**Table 6.3:** Configuration of the final Diffusion Policy pair.

| Component | Configuration |
|---|---|
| Observations / action horizon / executed prefix | 2 / 16 / 8 |
| Data and deployment rate | stride 3 from 30 Hz / native 10 Hz |
| Image transform | resize to $144 \times 192$, no crop |
| U-Net | widths 512, 1024, 2048; kernel 5; GroupNorm 8 |
| Diffusion embedding / training process | 128 / DDPM with 100 steps |
| Final sampler / weights | DDIM with 10 reverse steps / EMA |
| Optimiser | AdamW, learning rate $10^{-4}$, weight decay $10^{-6}$ |
| Optimiser betas / warm-up / schedule | $(0.95, 0.999)$ / 500 updates / cosine |
| Fresh updates / batch / seed | 40,000 / 16 / 1000 |
| Current history / encoder | nine samples; shared $1 \to 16 \to 16$ CNN |
| Current pooling / embedding | mean, maximum, latest / eight per joint and time |
| Residual adapter | $96 \to 64 \to 2C$, zero initialised |

The final rollout is synchronous and nominally scheduled at 10 Hz. When the queue is empty, chunk generation consumes part of that control tick's 100 ms budget; an overrun delays the next command rather than adding a separate fixed delay before a full 0.8 s execution interval. A deployment-host benchmark checks the isolated DDIM-10 chunk-generation time before hardware evaluation, with recorded 95th- and 99th-percentile values below one nominal control period. This benchmark verifies model-side feasibility on the deployment host but does not measure end-to-end rollout jitter, including camera, bus, and operating-system delays.

## 6.5 Modality-Use Diagnostic

A multimodal input can be present without affecting the policy. To test whether the trained model responds to current, the same preprocessed observation is evaluated twice. The first pass uses the recorded current. The second replaces every instantaneous and temporal current coordinate with the corresponding training-set mean in the checkpoint's normalised space. Positions and images remain unchanged.

Figure 6.3 shows the paired intervention and the aggregation used for the reported diagnostic.

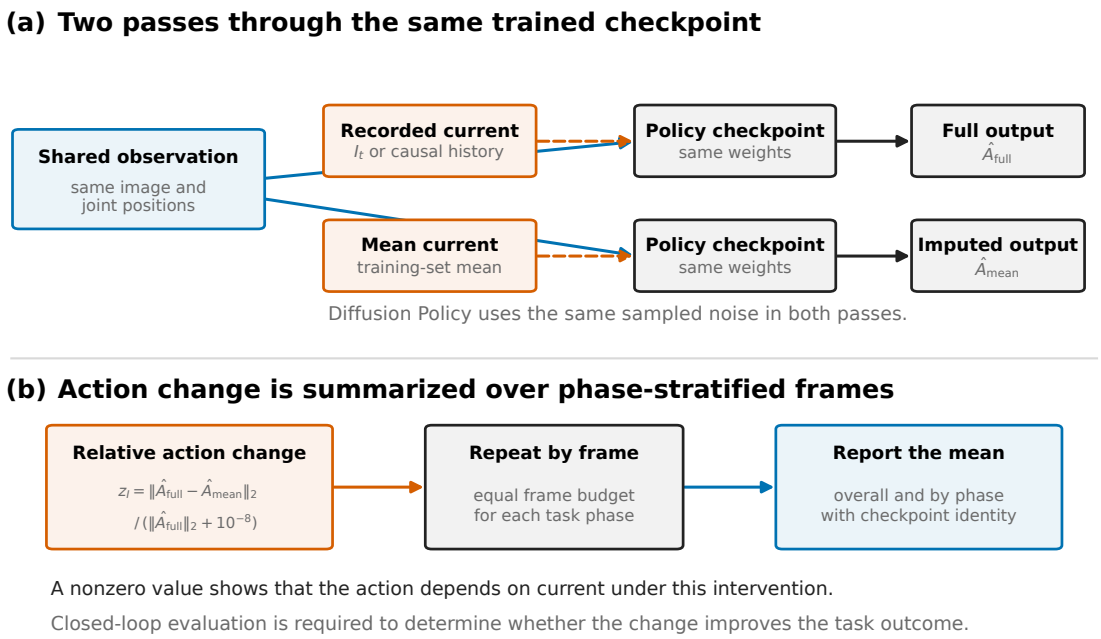



**Figure 6.3:** Paired mean-imputation diagnostic for current use. The same checkpoint receives identical images and joint positions in both passes, while the recorded current is replaced by its training-set mean in the second pass. Diffusion Policy also receives identical sampled noise. The relative action change is first computed per frame and then averaged over equally sampled task phases.

For observation $i$, current sensitivity is

$$z_I^{(i)} = \frac{\left\| \hat{\mathbf{A}}_{\text{full}}^{(i)} - \hat{\mathbf{A}}_{\text{mean current}}^{(i)} \right\|_2}{\left\| \hat{\mathbf{A}}_{\text{full}}^{(i)} \right\|_2 + 10^{-8}}, \tag{6.4}$$

where the norm covers every returned chunk position and all six action coordinates in normalised action space, before postprocessing or rollout ensembling and queueing. The reported value is the mean across training-demonstration frames selected at equal budgets from the annotated task phases. The phase assignments combine video-reviewed boundaries with current-based heuristic estimates for unreviewed episodes, so phase-stratified sensitivity is descriptive rather than an independently labelled contact test. The principal direct and joint-selective temporal ACT analyses use 102 frames per phase, or 510 in total; the Diffusion Policy analysis uses 30 per phase, or 150 in total. ACT is deterministic for a fixed observation. For Diffusion Policy, both passes use a newly initialised generator with the same seed, so the initial and scheduler noise are identical and do not contribute to the difference.

This counterfactual is applicable to direct state fusion, temporal tokens, and residual feature conditioning because it removes every current route represented in the batch. A direct comparison of attention weights would not provide a common measure across these architectures because current enters through one state token, six temporal tokens, or convolutional residual adapters. A nonzero value means that current changes the predicted action under this intervention. Closed-loop trials are still required to determine whether that change is useful. The diagnostic results are reported in Chapter 7.

The physical evaluation protocol, outcome definition and statistical analysis are specified in Sections 7.1 and 7.2. Additional conditioning methods and development outcomes are reported in Appendix B.

# 7 Evaluation

## 7.1 Evaluation Criteria and Evidence

The evaluation follows three steps. It first checks whether a trained policy changes its action when current is removed from the observation. It then compares the closed-loop success of each current-aware policy with a contemporaneous vision-and-position control. Finally, it locates failures within the task to determine whether any difference is associated with grasping, transport, or insertion. These steps distinguish action sensitivity from behavioural benefit.

Binary task success is the primary endpoint. A trial succeeds when the peg is fully seated and flush with the fixture surface before the timeout; every other outcome is a failure. The operator records the failure category and, where available, the phase in which the rollout ends.

Success rates are reported with Wilson 95% confidence intervals [54]; narrower intervals indicate more precise estimates. Differences between the current-aware and vision-and-position success rates use Newcombe–Wilson intervals [55], with positive values favouring the current-aware policy. An interval spanning zero leaves the direction of the difference unresolved; it does not establish that the policies are equivalent.

Relative change is $100(p_M/p_V - 1)$, where $p_M$ and $p_V$ are the observed current-aware and vision-and-position success proportions. This expresses the observed gain relative to the control rate, whereas the absolute difference is reported in percentage points. A ten-percentage-point difference was fixed before the normalised ACT comparison as the smallest practically relevant change and serves as a descriptive reference for the other comparisons.

Two-sided Fisher exact tests assess evidence against equal success probabilities [56]. Smaller $p$-values indicate stronger evidence of a difference, without measuring its size or practical importance. A post-hoc Holm correction accounts for testing four principal comparisons [57]; the adjusted $p$-value for raw ACT is 0.0355, while the other comparisons remain unresolved. The Diffusion Policy block analysis uses the Cochran–Armitage test for a trend across ordered blocks [58], separately from the comparison between policies. These trial-level calculations treat outcomes as independent successes or failures within each comparison.

Table 7.1 summarises the completed policy and representation comparisons.

## 7.2 Experimental Protocol

All principal comparisons use the same SO-101 follower arm, peg, insertion fixture, and 100-demonstration dataset described in Chapter 5. The peg begins at one marked position for both policies. The wrist camera supplies $640 \times 480$ images at 30 Hz; the synchronised side

**Table 7.1:** Evaluation coverage across policy families and current representations.

| Policy family | Direct current fusion | Learned current representation |
|---|---|---|
| ACT | Raw and normalised comparisons (Sections 7.5.1 and 7.5.2) | Joint-selective temporal comparison (Section 7.6) |
| Diffusion Policy | Initial screens (Appendix B.2) | Residual-conditioning system comparison (Section 7.7) |

camera is retained for inspection but is not a policy input. One operator resets and scores every trial.

The principal comparisons use interleaved blocks to expose both policies to similar conditions within a session. A 30-trial comparison contains five blocks, each with six trials per policy. The first policy alternates between blocks in the normalised ACT, joint-selective temporal ACT, and final Diffusion Policy comparisons. Only the raw ACT comparison and the explicitly exploratory screens use a fixed order.

The completed ACT sessions use a 60-second pause between policy phases and 180 seconds between blocks. The final Diffusion Policy session uses 60 seconds between phases and 120 seconds between blocks. Blocks start only when all reported servo temperatures are at most 50 °C; five consecutive readings at or above 65 °C require cooling before evaluation continues. The timeout is 45 s, and the nominal reset time between trials is 15 seconds.

The ACT pairs share the dataset, seed, model family, and rollout controller within each comparison. The Diffusion Policy systems share the dataset, temporal sampling, action horizon, and hardware execution settings. The final Diffusion pair also shares its starting checkpoint and additional training budget; its different base-freezing strategies are specified in Section 6.4. Vision-and-position policies disable current-register acquisition, while current-aware policies add a synchronised read of all six current registers. Both policies use the same configured controller and target rate within a comparison.

## 7.3 Evaluation Task Selection

Preliminary trials established the task and sensing configuration used for the principal policy comparisons. Clearance, entry chamfer, peg orientation, camera count, and demonstration count changed across this development sequence. The rows in Table 7.2 therefore identify the selected operating point rather than estimate isolated effects of those factors.

The final setting therefore uses a vertical peg, no chamfer, 0.35 mm diametral clearance, 100 demonstrations, and one wrist camera. It places the vision baseline in an intermediate range and directs many failures towards insertion, where effort information is most relevant to the thesis question. Every principal comparison uses this setting and includes a control evaluated in the same session.

**Table 7.2:** Task-development results used to select the final evaluation setting.

| Condition | Demonstrations | Cameras | Success | Main recorded failure |
|---|---|---|---|---|
| Flat, 1.4 mm clearance, chamfer | 50 | wrist | 25/30 | mixed |
| Flat, 0.55 mm clearance, chamfer | 50 | wrist | 23/30 | unsuccessful grasp |
| Flat, 0.55 mm clearance, no chamfer | 50 | wrist | 14/30 | missed insertion |
| Flat, 0.35 mm clearance, no chamfer, retrained | 50 | wrist | 19/30 | dropped peg |
| Vertical, 0.35 mm clearance, no chamfer | 50 | wrist | 5/30 | dropped peg |
| Vertical, 0.35 mm clearance, no chamfer | 50 | wrist and side | 0/30 | unsuccessful grasp / dropped peg |
| Vertical, 0.35 mm clearance, no chamfer | 100 | wrist | 15/30 | missed insertion / dropped peg |
| Vertical, 0.35 mm clearance, no chamfer | 100 | wrist and side | 14/30 | missed insertion / unsuccessful grasp |

## 7.4 Counterfactual Modality-Use Analysis

The mean-imputation diagnostic defined in Section 6.5 evaluates each current-aware checkpoint twice on the same observation. One pass uses the recorded current, while the other replaces current with its training-set mean. Images and joint positions remain unchanged, and Diffusion Policy receives identical sampled noise in both passes. A nonzero relative action change shows that the checkpoint responds to current under this intervention.

The normalised direct ACT-M checkpoint gives a mean relative action change of 0.0795 over 510 phase-stratified demonstration frames. Its insertion-phase mean is 0.0856. A post-hoc analysis applies the same intervention to the joint-selective temporal ACT-M checkpoint evaluated in the principal trials. It gives a mean relative action change of 0.0729 over 510 frames and 0.0843 during insertion. The residual-conditioned DP-M checkpoint gives 0.00649 over 150 frames and 0.00754 during insertion. Figure 7.1 shows that the response is not confined to a single annotated phase, although its magnitude is checkpoint dependent. The current encoders and input-history lengths differ, so these magnitudes do not rank representations by task-relevant reliance on current.

For context, replacing joint position by its training mean and setting the image to zero produce relative action changes of 0.403 and 0.445 for ACT-M-norm. The corresponding values for DP-M are 0.681 and 0.113. These interventions are not physically equivalent and do not rank the intrinsic value of the modalities. Temporal ACT-M likewise responds less to current than to the tested position and image interventions, whose mean changes are 0.573 and 0.435. Excluding seven frames with inconsistent phase boundaries gives a mean of 0.0732 over 503 frames. The protocol, frame list and diagnostic code are included in the accompanying code package (Appendix A).

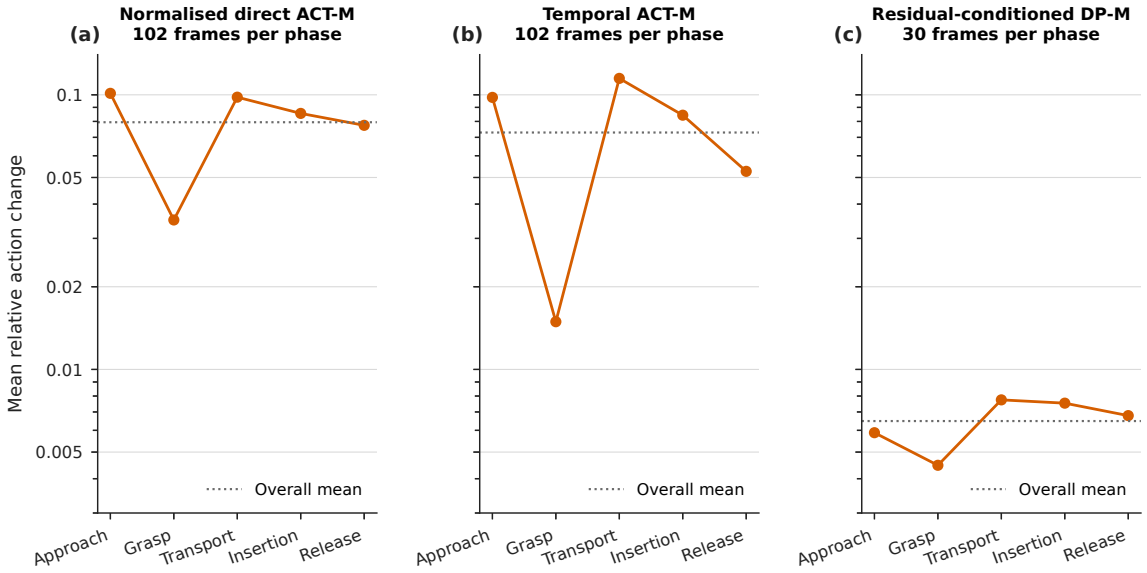



**Figure 7.1:** Current sensitivity over five task phases for the normalised direct ACT-M, joint-selective temporal ACT-M and residual-conditioned DP-M checkpoints. Each point is the mean relative change of the predicted action chunk when current is replaced by its training-set mean. The logarithmic vertical scale accommodates the different magnitudes, and the dotted line gives the checkpoint's overall mean. The frames come from training demonstrations, so the figure measures model response rather than phase-specific rollout success.

The diagnostic establishes that current reaches the action output in the three principal current-aware checkpoints tested here. For temporal ACT, it does not separate past-current information from present-current information, nor establish that current caused the five additional rollout successes. It does not show whether the induced action change is useful. That question requires the closed-loop comparisons below.

## 7.5 Direct Current Fusion in ACT

### 7.5.1 Raw Direct Concatenation

The first comparison appends the six raw current values to the six joint positions. ACT-V succeeds in $19/30$ trials, or 63.3% with a Wilson 95% interval of $[45.5, 78.1]\%$. Direct ACT-M succeeds in $8/30$, or 26.7% with an interval of $[14.2, 44.4]\%$. The current-aware difference is $-36.7$ percentage points with a Newcombe–Wilson 95% interval of $[-56.0, -11.5]$ points, and Fisher's exact test gives $p = 0.0089$. Relative to the contemporaneous ACT-V success rate, the raw-current system records 57.9% lower success.

The observed result favours the vision-and-position policy in this comparison.

The failure distribution shows that the degradation was not limited to hole alignment. ACT-M

records eight unsuccessful grasps compared with two for ACT-V. The raw model therefore introduced errors before insertion as well as during the contact-rich endpoint.

### 7.5.2 Normalised Direct Concatenation

The revised input clips each current channel at its training-set 99.5th percentile and then applies the saved normalisation transform. The raw and normalised contrasts use different controllers and hardware sessions, so the difference between those contrasts does not isolate the effect of current capping. The execution controller is identical for ACT-V and ACT-M-norm, and policy order alternates between blocks.

ACT-V succeeds in $17/30$ trials and ACT-M-norm in $15/30$. This is an 11.8% relative decrease, or $-6.7$ percentage points with a Newcombe–Wilson 95% interval of $[-29.9, 17.6]$ points and Fisher's exact $p = 0.796$. The observed result favours the vision-and-position policy, although the interval leaves the difference unresolved. Table 7.3 gives the success rates for the two policies.

**Table 7.3:** Normalised direct-current ACT comparison with 30 trials per policy.

| Policy | Successes | Rate | Wilson 95% CI |
|---|---|---|---|
| ACT-V | 17/30 | 56.7% | [39.2%, 72.6%] |
| ACT-M-norm | 15/30 | 50.0% | [33.2%, 66.8%] |

All 28 failed trials end during insertion. ACT-M-norm drops the peg less often than ACT-V, but this does not translate into a higher overall success rate.

## 7.6 Joint-Selective Temporal Current Fusion in ACT

The joint-selective temporal policy represents current with six joint histories, a shared temporal convolution, and one token per joint. The vision-and-position control and the current-aware model are evaluated in five blocks with alternating order.

ACT-V succeeds in $7/30$ trials, or 23.3% with a Wilson 95% interval of $[11.8, 40.9]$%. The joint-selective ACT-M policy succeeds in $12/30$, or 40.0% with an interval of $[24.6, 57.7]$%. The current-aware setting therefore records a 71.4% relative increase in success over its contemporaneous baseline, corresponding to five additional successes and an absolute gain of 16.7 percentage points. This is the largest favourable relative and absolute point estimate among the four principal comparisons, and its absolute magnitude exceeds the ten-point relevance criterion introduced for the normalised comparison. The Newcombe–Wilson 95% interval spans $-6.7$ to $+37.8$ percentage points and Fisher's exact test gives $p = 0.267$. The observed result favours the current-aware policy, while the interval leaves superiority unresolved.

The current-aware policy scores higher in three blocks and ties its control in two. These blocks describe the same checkpoint pair within one logical evaluation session, completed over two dates; they are not independent replications.

The two policies reach the insertion phase at different rates. ACT-V reaches insertion in $29/30$ trials and succeeds in seven, whereas joint-selective ACT-M reaches insertion in $23/30$ and succeeds in twelve. The current-aware policy therefore loses six more trials before insertion but converts more of the trials that reach insertion into success. The conditional counts, $7/29$ and $12/23$, are descriptive because reaching insertion is itself affected by the policy. The unconditioned $7/30$ versus $12/30$ comparison remains the primary result. The post-hoc diagnostic shows that this policy responds to current, without identifying the cause of the additional successes (Section 7.4). Among the recorded failure categories, joint-selective ACT-M drops the peg six times, compared with eleven for ACT-V. This count motivates a retention hypothesis without establishing a current-mediated mechanism.

## 7.7 Residual-Conditioned Current Fusion in Diffusion Policy

Direct current fusion performed poorly in initial screening and was not pursued further. The selected current-aware system instead uses joint-selective temporal encoding and bounded residual conditioning, initialised from an operational vision-and-position policy. Development results are reported in Appendix B.

The final DP-V and DP-M systems share the dataset, 200,000-step source checkpoint, temporal sampling, action horizon, eight-action execution queue, EMA weight selection and DDIM-10 sampler. Both receive 40,000 further training updates with batch size 16 and seed 1000. DP-V updates its base network, while DP-M freezes the copied base and trains the added current route. Five blocks with alternating order provide 30 trials per policy.

DP-V succeeds in $9/30$ trials, or 30.0% with a Wilson 95% interval of $[16.7, 47.9]\%$. DP-M succeeds in $12/30$, or 40.0% with an interval of $[24.6, 57.7]\%$. The observed difference is $+10.0$ percentage points, with a Newcombe–Wilson interval of $[-13.6, +32.1]$ points and Fisher's exact test $p = 0.589$. The observed result favours the current-aware design by 33.3% relative to its contemporaneous control matched for training budget. The uncertainty leaves superiority unresolved, and the comparison does not separate current conditioning from the strategy used to train it.

Both policies reach insertion in $28/30$ trials. Missed insertion is the largest failure category for both policies, with five dropped pegs for DP-M and five wrong approaches for DP-V among the remaining recorded categories. More importantly, combined success falls monotonically from $8/12$ in the first block to $6/12$, $4/12$, $2/12$, and $1/12$. The descriptive Cochran–Armitage trend test gives $p = 0.0006$. Start temperatures plateau after the second block, and the combined decline spans blocks in which policy order alternates. This is an observational session trend rather than evidence for a particular cause. The shared within-session deterioration is therefore stronger than the observed between-policy difference and limits interpretation of either absolute rate.

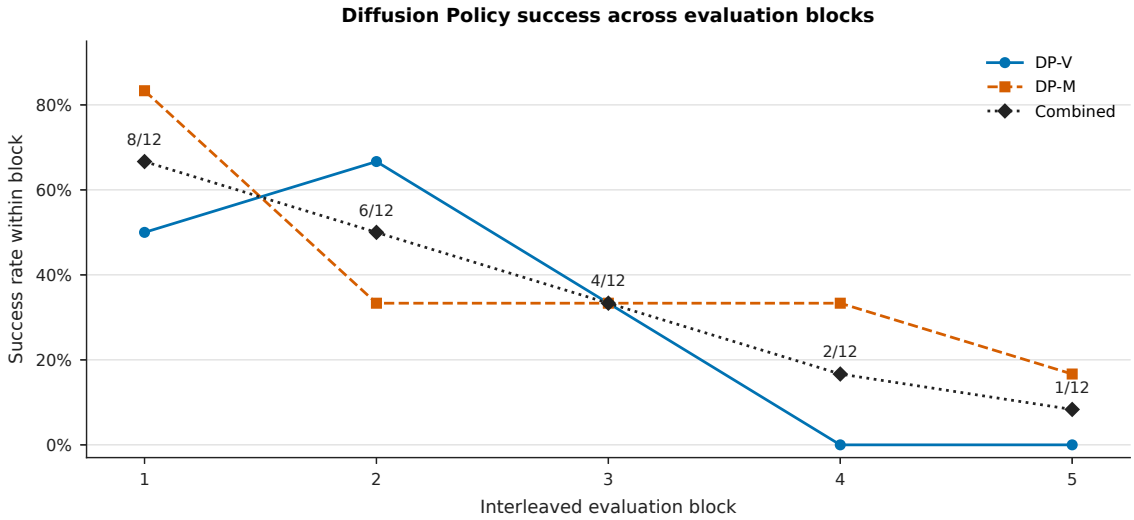



**Figure 7.2:** Success rates for the two Diffusion Policy systems and their combined rate across five interleaved blocks. Each policy contributes six trials per block; the combined rate therefore uses twelve trials.

## 7.8 Failure-Phase Analysis

Figure 7.4 follows each policy from the start of a trial to reaching insertion and then to final success.

**Table 7.4:** Successes and failure phases in the policy comparisons.

| Comparison | Policy | Success | Approach | Grasp | Transport | Insertion |
|---|---|---|---|---|---|---|
| Normalised direct ACT | ACT-V | 17 | 0 | 0 | 0 | 13 |
| Normalised direct ACT | ACT-M-norm | 15 | 0 | 0 | 0 | 15 |
| Temporal-current comparison | ACT-V control | 7 | 1 | 0 | 0 | 22 |
| Temporal-current comparison | Joint-selective ACT-M | 12 | 2 | 5 | 0 | 11 |
| Residual-conditioning comparison | DP-V control | 9 | 0 | 2 | 0 | 19 |
| Residual-conditioning comparison | DP-M | 12 | 0 | 2 | 0 | 16 |

Both policies in the normalised direct ACT comparison reach insertion in every trial, where all failures occur. The joint-selective ACT-M policy has more failures before insertion but more final successes than its control. The two Diffusion Policy systems reach insertion equally often and differ by three final successes. No pattern repeats across policy families strongly enough to support a phase-specific benefit from current. The phase evidence instead identifies where each system loses trials and keeps that question separate from whether its action responds to current.

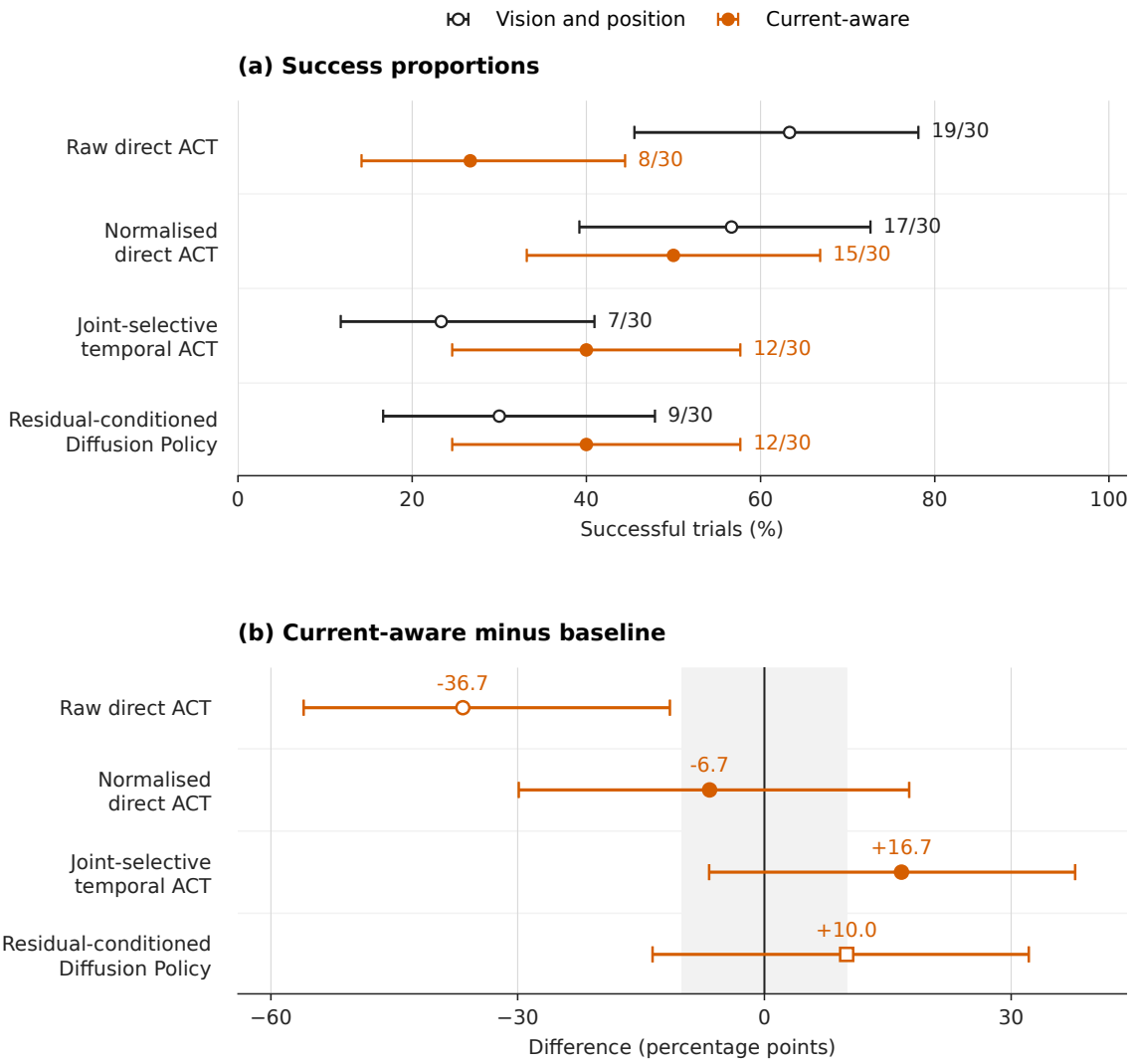



**Figure 7.3:** Principal closed-loop comparisons, with 30 trials per policy in each row. Panel (a) shows vision-and-position success proportions (open black circles) and current-aware proportions (filled orange circles), with Wilson 95% intervals and successful-trial counts. Panel (b) shows current-aware minus baseline differences with Newcombe–Wilson 95% intervals; the shaded band marks absolute differences within ten percentage points of zero. In panel (b), the open circle identifies fixed-order raw ACT and the square identifies the Diffusion Policy system comparison, matched for additional training updates but using different base-freezing strategies.

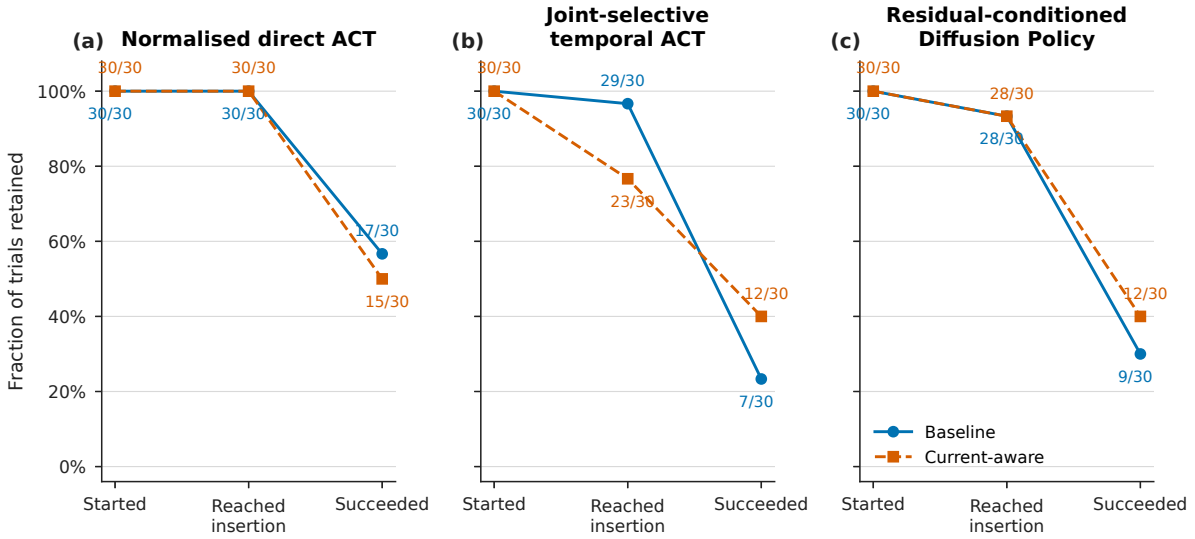



**Figure 7.4:** Trial funnels for the normalised ACT, temporal ACT and Diffusion Policy comparisons. Each line begins with all trials, retains those that reach insertion, and ends with successful trials. Each panel shows one completed interleaved comparison with 30 trials per policy. Reaching insertion is a policy-dependent intermediate outcome, so the final segment is not interpreted as a separate causal comparison.

## 7.9 Evaluation Summary

The learned temporal ACT and residual-conditioned Diffusion Policy designs have favourable observed differences, while both direct ACT comparisons have adverse differences. Table 7.5 collects the counts, relative changes and absolute-difference intervals; the raw ACT comparison uses fixed within-block order. Figure 7.3 shows the two success proportions alongside the absolute difference and its uncertainty for each comparison.

The favourable temporal ACT result is of practically relevant observed size, although its difference interval includes zero and does not establish superiority. The Diffusion Policy estimate has the same positive direction for a complete current-conditioning and training design under a matched update budget; the contribution of current alone remains unresolved.

The discussion considers which aspects of representation, demonstrations and execution could account for these different outcomes.

**Table 7.5:** Observed relative success changes and absolute differences for the principal comparisons. Relative change uses the contemporaneous vision-and-position success rate as its denominator; confidence intervals refer to the absolute difference.

| Comparison | Vision and position | Current aware | Relative change | Absolute difference (pp; 95% CI) |
|---|---|---|---|---|
| Raw direct ACT | 19/30 | 8/30 | −57.9% | −36.7 [−56.0, −11.5] |
| Normalised direct ACT | 17/30 | 15/30 | −11.8% | −6.7 [−29.9, 17.6] |
| Joint-selective temporal ACT | 7/30 | 12/30 | +71.4% | +16.7 [−6.7, 37.8] |
| Residual-conditioned Diffusion Policy | 9/30 | 12/30 | +33.3% | +10.0 [−13.6, 32.1] |

*Note.* Each current-aware policy should be compared with its own contemporaneous vision-and-position baseline. The comparisons took place in different sessions, with varying baseline performance, so this table does not establish a ranking of fusion methods. For Diffusion Policy, DP-V updates the base network, whereas DP-M freezes it and trains the added current branch. The observed difference therefore reflects both current conditioning and the training strategy and cannot be attributed to current alone.

# 8 Discussion

Existing motor-current readings on the evaluated SO-101 contain contact-related information and can influence learned action predictions. The strongest favourable task result comes from joint-selective temporal ACT: success increases from 23.3% to 40.0%, a 71.4% relative improvement over its contemporaneous vision-and-position control. This observation supports further investigation of the current-aware setting, while its uncertainty leaves superiority unresolved. The scientific contribution lies in following the connection between available effort information, its influence on a learned policy, and the task outcome achieved when that policy controls the robot.

## 8.1 Interpretation of the Main Findings

The characterisation establishes a reason to include current in the policy observation. Controlled gripper closes distinguish object-present contact from misses, and the insertion recordings show task-aligned changes across several arm joints (Chapter 4). The signal nevertheless combines loading with pose, motion, friction and the internal servo controller. It is quantised, zero-inflated and dependent on the robot configuration. These properties support treating current as a contact-related learning input whose meaning must be inferred together with the other observations. They do not supply a calibrated force measurement or a complete description of the contact state.

The policy diagnostics establish a second part of the connection. Replacing current with its training mean changes the predicted actions of the retained normalised direct ACT, joint-selective temporal ACT and residual-conditioned Diffusion Policy checkpoints (Section 7.4). Current therefore reaches the action output under this intervention, although the response is smaller than under the tested position and vision interventions. Those interventions have different physical meanings and cannot provide an equivalent removal of each modality. The diagnostic shows that these models respond to current.

The temporal ACT result has a practically relevant observed magnitude relative to the ten-point reference introduced for normalised ACT (Section 7.6). Its interval spans benefit and harm, so the favourable direction motivates investigation rather than establishing superiority.

The other comparisons qualify the scope of that observation. Residual-conditioned Diffusion Policy also has a favourable point estimate, with a 33.3% relative increase over a contemporaneous control matched for additional training updates, whereas direct concatenation shows no observed benefit in either principal ACT comparison (Table 7.5). The raw ACT contrast is the only principal comparison whose difference interval excludes zero, in the adverse direction,

although its fixed within-block order limits attribution to current. The pattern motivates examining whether, and under which learning and execution conditions, current can help. Since visual-baseline success varies across sessions, it does not establish a controlled ranking of the current representations. The study thus separates signal informativeness, policy sensitivity and observed task benefit without establishing that learned fusion is superior to direct concatenation.

## 8.2 Relationship to Contact-Aware Imitation Learning

The present study examines a different input assumption from methods that first estimate a mechanically interpreted contact quantity. For example, Wahrburg et al. recover Cartesian contact forces and torques from motor-side measurements using a robot dynamics model [9]. Here, the policy receives magnitude-only bus-servo current without force calibration or force-labelled supervision. The characterisation supports the presence of contact-related structure in that input on the tested platform. The remaining learning problem is to identify which changes are relevant to the demonstrated action when the same register also responds to ordinary robot motion. Avoiding an explicit force estimator makes the observation accessible on this hardware, but leaves that separation to the learned representation and the information available in the dataset.

FTACT concatenates wrist force/torque measurements with joint state before the ACT encoder [6]. This provides a direct architectural precedent for the state-fusion route used here, while the physical meaning of its input differs from the bus-servo register. The adverse direct ACT comparisons show that the same simple fusion pattern did not produce an observed benefit for uncalibrated current under the evaluated settings. They leave open whether a matched change in representation could convert that input into useful control. FACTR 2 reports improved NIST Insertion task progress but unchanged Cap Screwing performance when raw current is added to ACT [14, Appendix D.6]. This reinforces the need to interpret the present direct-fusion results within their task and session conditions.

FACTR addresses a related learning difficulty by reducing visual fidelity early in training so that external joint torque acquires a predictive role [5]. Its analysis of visual dominance is relevant to the weaker current responses found in the retained checkpoints here. However, these responses do not establish modality collapse: they are nonzero, and a small action change may still matter at a critical contact event. The adverse direct-fusion outcomes also cannot identify whether current was underused, used at an unhelpful moment, or allowed to disturb the visual policy. The connection to FACTR is therefore a question about learning opportunity and useful dependence, rather than confirmation of a shared failure mechanism.

FoAR predicts impending contact to guide force fusion and reactive refinement, while PhaForce coordinates slow chunk-level visual–force planning with faster phase-guided force correction [7], [28]. These designs make explicit both when interaction information is relevant and how promptly it can alter execution. The policies studied here receive current through an instantaneous state vector, temporal tokens or residual conditioning, while action changes remain subject to their queued execution interfaces. This difference provides a concrete context

for interpreting an informative input that produces limited or inconsistent task benefit. It does not demonstrate that phase scheduling or a faster correction path would improve the present task; neither alternative was evaluated as a matched trained condition.

Together, these comparisons place the thesis at the boundary between an available interaction signal and an effective contact-aware controller. Its contribution is an empirical examination of that boundary using uncalibrated effort readings, shared demonstrations and explicit modality-use diagnostics. The direct-fusion results and favourable temporal observation show why input availability alone is insufficient to predict the outcome.

## 8.3 Representation, Demonstrations and Execution

Several explanations are consistent with the observed outcomes. They concern different links between the recorded signal and the executed action, and the retained evidence does not select a single mechanism.

One possibility is competition between representations during behavioural cloning. Images and joint positions are available throughout the demonstrations, whereas current is unevenly informative across the trajectory. The model may reduce prediction loss mainly through the visual and positional routes, leaving effort poorly aligned with the moments at which it could improve control. Normalisation addresses channel scale but does not change how frequently current supplies information beyond those observations. This account is consistent with the retained sensitivity measurements and the learning issue studied by FACTR [5]. Establishing it for the evaluated policies would require evidence about learning dynamics or a matched intervention on the training procedure.

A second possibility is that current reveals loading without specifying the required correction. An unsigned increase does not uniquely identify contact location or the direction in which the peg should move to find the hole. The gripper characterisation establishes contact discrimination under controlled closes, while insertion also requires geometric alignment. Temporal ACT's failure distribution offers a possible behavioural clue: it drops the peg less often than its control, but loses more trials before insertion (Sections 7.6 and 7.8). This motivates a retention or action-selection hypothesis without establishing that current caused either change. The policies reach insertion on different subsets of trials, so their insertion outcomes cannot independently isolate a contact-phase mechanism.

A third possibility concerns when newly observed contact can influence control. After the grasp latch, ACT executes a complete chunk before predicting again, while Diffusion Policy replans when its eight-action queue is empty (Sections 6.3.4 and 6.4). For temporal ACT, the 0.30 s current history is shorter than the 1.67 s execution interval, so a brief event may leave the history before the next prediction. Maintaining responsiveness during chunked execution remains an active research challenge. Related work released during this thesis, including ARLI, investigates asynchronous execution and latency-aware adaptation of generalist robot policies [59].

The demonstration process supplies a fourth possible constraint. The operator controlled the

leader arm visually and did not receive the follower's current as feedback (Chapter 5). The recordings can contain visually guided contact corrections, but the study did not establish how consistently current predicts those corrections beyond images and joint positions. A learned effort representation can only support behavioural cloning when the demonstrations contain a corresponding relationship to the action target. Temporal encoding may expose a useful history, yet its presence alone cannot create a corrective dependence that is absent or inconsistent in the demonstrations.

These explanations connect representation, supervision and execution rather than assigning the outcomes to one architecture choice. The contrasting outcomes identify the need to distinguish useful current dependence from sensitivity that has little task value.

## 8.4 Methodological Contributions and Research Implications

The framework enables weak servo-current observations to be studied with established imitation-learning models. The synchronised multimodal dataset records camera observations, joint positions, actions and current within the same demonstrations (Chapter 5). Feature subsetting permits policies with and without effort to learn from that shared source, so a different demonstration collection is not required merely to change the observation modality.

The common counterfactual diagnostic provides a further research capability. Current enters the evaluated architectures through different state, token and feature-conditioning routes, making attention weights difficult to compare directly. Replacing every current route in normalised input space tests whether the supplied current changes the predicted action (Section 6.5). Structurally different models can therefore be examined through their outputs using the same intervention.

## 8.5 Limitations

**Scope and precision.** The findings describe one SO-101, one peg-in-hole task and one demonstration dataset, with one training seed per condition and one operator. Each principal comparison contains 30 trials per policy. The favourable temporal ACT and Diffusion Policy estimates therefore provide evidence for these evaluated settings; their intervals show how precisely the success differences are estimated. Transfer to other tasks, robots and training runs remains outside the scope of these results. The decline in Diffusion Policy success across blocks also shows that performance can change within an evaluation session (Figure 7.2).

**Attribution.** The mean-imputation diagnostic establishes that the selected policies respond to current; it does not identify which current-dependent actions account for the additional successes. These boundaries distinguish the observed performance differences from a causal explanation of them.

**Measurement.** Motor current combines external loading with pose, motion, friction and internal servo control, so its interpretation depends on the surrounding observations. The strongest controlled contact evidence comes from gripper closes, and the insertion analysis using video-reviewed boundaries. Task success measures whether the peg is seated, without measuring contact force or motion efficiency. The phase and failure analyses consequently describe where the policies respond or lose trials, while leaving the physical causes open.

Local contact and slip remain further observability limits. An object may move within the gripper without producing a distinctive recorded current event. Tactile sensing provides contact-surface information, while PolyTouch combines tactile, acoustic and peripheral visual observations [22], [60]. The present results establish the value of examining existing actuator telemetry as part of the policy observation, with its control benefit depending on the task, representation and execution conditions.

# 9 Conclusion and Future Work

## 9.1 Conclusion

This thesis investigates whether motor-current readings already available on a low-cost robot can provide useful contact information for visuomotor imitation learning. The aim is to learn from weak, uncalibrated signals without adding sensing hardware or first converting them into force estimates. The investigation combines signal characterisation on an SO-101, a synchronised 100-demonstration peg-in-hole dataset, current-aware ACT and Diffusion Policy implementations, modality-use diagnostics and repeated physical trials. Together, these address whether the signal is informative, whether policies respond to it and whether that response improves task performance.

The characterisation establishes that current contains contact-related information on the evaluated platform (Chapter 4). Controlled gripper closes distinguish contact with an object from an empty close, while insertion recordings reveal task-aligned changes across several arm joints. These findings justify including current among the policy observations. Its dependence on pose and motion also makes the surrounding visual and positional context important: the register provides an effort observation whose meaning must be learned in context.

The modality analysis shows that this additional observation influences the predicted actions of the retained normalised direct ACT, joint-selective temporal ACT and residual-conditioned Diffusion Policy checkpoints (Section 7.4). Replacing current with its training mean changes their action predictions while the other observations are held fixed. The policies therefore exhibit measurable sensitivity to current.

The strongest favourable task result is obtained with joint-selective temporal ACT. It succeeds in $12/30$ trials against $7/30$ for its contemporaneous vision-and-position control, a 71.4% relative improvement and a gain of 16.7 percentage points. This is a practically relevant observed difference, although its 95% interval spans $-6.7$ to $+37.8$ percentage points and leaves superiority unresolved. Direct current concatenation produces lower observed success in both the raw and normalised ACT comparisons (Table 7.5). The contrast motivates further study of temporal representation, while variation between evaluation sessions prevents a controlled ranking of the fusion methods.

The Diffusion Policy comparison also favours the current-aware design, with $12/30$ successes against $9/30$, a 33.3% relative improvement and a gain of 10 percentage points. The comparison evaluates the two systems as a whole, and the confidence interval includes zero (Section 7.7).

The thesis thus establishes contact-related information in current and measurable policy sensitivity to it on this platform, with favourable observed outcomes for the evaluated temporal

ACT and residual-conditioned Diffusion Policy settings. This answers the original question at three distinct levels: contact-related information is present, selected policies respond to it, and the task evidence identifies promising settings whose benefits require confirmation. These conclusions apply to the studied task, robot and training runs; the broader interpretation and validity limits are discussed in Chapter 8.

The thesis also provides an extended LeRobot-compatible dataset schema and 100 multimodal demonstrations, together with the evaluated ACT and Diffusion Policy implementations. The accompanying software covers signal characterisation, data collection, training, physical evaluation and the diagnostic for measuring current dependence. Appendix A explains how these resources are organised and how to run the main stages of the study.

## 9.2 Future Work

Motor current and vision provide information at different time scales. The 30 Hz recordings used here reveal contact-related variation, but asynchronous multimodal learning could make use of a more detailed current stream, for example at around 100 Hz, alongside slower visual observations. The research question is how to represent that history so that brief interaction patterns contribute useful information to the policy. The favourable temporal ACT result provides a basis for pursuing this direction, while leaving the benefit of higher-rate input to be established.

Contact-aware control also requires a connection between the learned representation and the opportunity to act on it. The policies in this thesis predict action chunks, which limits when newly observed contact can affect motion. A broader direction is to develop policies that coordinate visual planning with responsive effort-based feedback throughout execution. This would address how weak current signals can support corrective behaviour during contact-rich manipulation.

Generalisation beyond the evaluated setup is another open direction. Current depends on both the task interaction and the robot's mechanics, so a representation useful on one platform may contain features specific to that setting. Research across different manipulation tasks, geometries and actuator designs could examine which interaction patterns transfer and how much new demonstration data is needed. This would connect the platform-specific evidence here to the wider goal of reusable contact representations for low-cost robots.

# A Reproducibility

The accompanying USB stick contains the full code and data package, including recorded videos and the retained trained models. The CD provides the code and a smaller dataset and model subset for the principal temporal ACT comparison. On either medium, start with `README.txt` for installation and dataset setup and `CODE-AND-DATA.md` for the software overview. The project source is in `src/`; the commands below are run from that directory after activating the Python environment described in `so101-workspace/README.md`.

The main folders are:

- `experiments/configs/`: robot connections, cameras, task settings and dataset views.
- `experiments/phase-1-characterisation/`: signal-measurement scripts, recorded CSV files and analysis scripts for Chapter 4.
- `experiments/datasets/`: demonstrations, metadata and phase annotations. The principal training dataset is `peg-tight-vertical-100`.
- `experiments/runs/`: training configurations, saved commands and model diagnostics; `experiments/evals/`: evaluation sessions, trial outcomes and recording information.
- `experiments/scripts/`: collection, dataset preparation, training, evaluation and plotting programs.
- `so101-workspace/`: the LeRobot implementation and robot software.

For signal characterisation, the scripts in `experiments/phase-1-characterisation/scripts/` cover static noise, arm poses, known loads, gripper contact and teleoperated motion. For example, the following records the static-noise study and analyses its output:

```
cd experiments/phase-1-characterisation
python scripts/exp_e_noise.py
python scripts/analyze_exp_e.py
cd ../..
```

The folder's README gives the remaining measurement procedures and analysis commands. The measurement scripts require the connected arm; the analysis scripts can use the supplied recordings.

Demonstrations are collected through the leader–follower interface. Set the device ports, cameras and calibration in `experiments/configs/robot.yaml`, and select a task from `experiments/configs/tasks.yaml`. For the vertical peg task, collection is started with:

```
python experiments/scripts/collect.py --task peg-tight-vertical
```

The collector saves the sessions and prepares the dataset views using `dataset.py`. The position-only and current-aware views contain the same demonstrations. The supplied 100-demonstration dataset combines the two vertical batches described in Chapter 5.

Training uses `train.py`; variant names select the policy and its input representation. For example, the visual ACT baseline and the normalised direct-current model are trained with:

```
python experiments/scripts/train.py \
  --dataset peg-tight-vertical-100 --variants actv actm-norm \
  --chunk-size 50 --resolution 640 --steps 100000 \
  --batch-size 16 --seed 1000
```

The other ACT and Diffusion Policy variants are listed in `experiments/README.md`. Use the saved training configurations and launch arguments for the reported runs, including the warm-start settings of the final Diffusion Policy models. Trained models are supplied with the package; their locations are listed in `CODE-AND-DATA.md`.

Physical evaluation uses `evaluate.py`, which runs the selected policies in blocks and asks the operator to score each trial. A basic invocation is:

```
python experiments/scripts/evaluate.py \
  --dataset peg-tight-vertical-100 --variants actv actm-norm \
  --session-id example-act --execution-mode pilot \
  --blocks 1 --trials-per-block 5
```

The saved session manifests in `experiments/evals/` contain the complete commands used for the reported comparisons, including controller settings, policy order, timing and checkpoint selection. Each session's `results.csv` contains its scored outcomes. Collection and physical evaluation require the calibrated robot and the task setup described in Chapters 5 and 7.

# B Auxiliary Experiments

This appendix records experiments that informed model selection or tested an additional policy family but were not part of the principal ACT and Diffusion Policy comparisons. Their outcomes are reported descriptively.

## B.1 Bounded ACT Feature Conditioning

A further ACT variant tests whether the same learned current representation is more effective when it modifies visual features instead of becoming six Transformer tokens. It reuses the six ordered 25-dimensional joint representations from the temporal-token model, preserving the current history, CNN, pooling, and matching joint position. The resulting $6 \times 25$ tensor is flattened to 150 values and passed through LayerNorm and a $150 \to 64 \to 1024$ head with a ReLU between the linear layers. The 1024 outputs form one scale and one shift value for each of the 512 channels in the final ResNet feature map.

For feature channel $c$, the modulation is

$$\mathbf{v}'_c = \mathbf{v}_c \, 2\sigma(\gamma_c) + \tanh(\beta_c) \, \mathrm{RMS}(\mathbf{v}_c), \tag{B.1}$$

where the RMS is calculated across the spatial extent of that channel. The multiplicative scale therefore lies between zero and two, while the absolute shift is limited to one channel RMS. The last layer of the conditioning head is initialised to zero, giving a scale of one and a shift of zero before training. This makes the added route an identity-initialised form of feature-wise linear modulation [46].

Current is absent from both the ordinary state token and the Transformer token sequence in this variant. It can affect the policy only by modulating the final visual feature map before its $1 \times 1$ projection. The behaviour-preserving version copies and freezes a trained ACT-V base, then trains the current encoder and conditioning head for 40,000 updates. An optional causal latch holds the modulation at its identity until the gripper-position detector in Section 6.3.4 has observed a verified open-close cycle. These variants were assessed in the development screens summarised in Section B.2. Figure B.1 shows how the behaviour-preserving variant conditions the copied visual feature map without altering the state-token route.

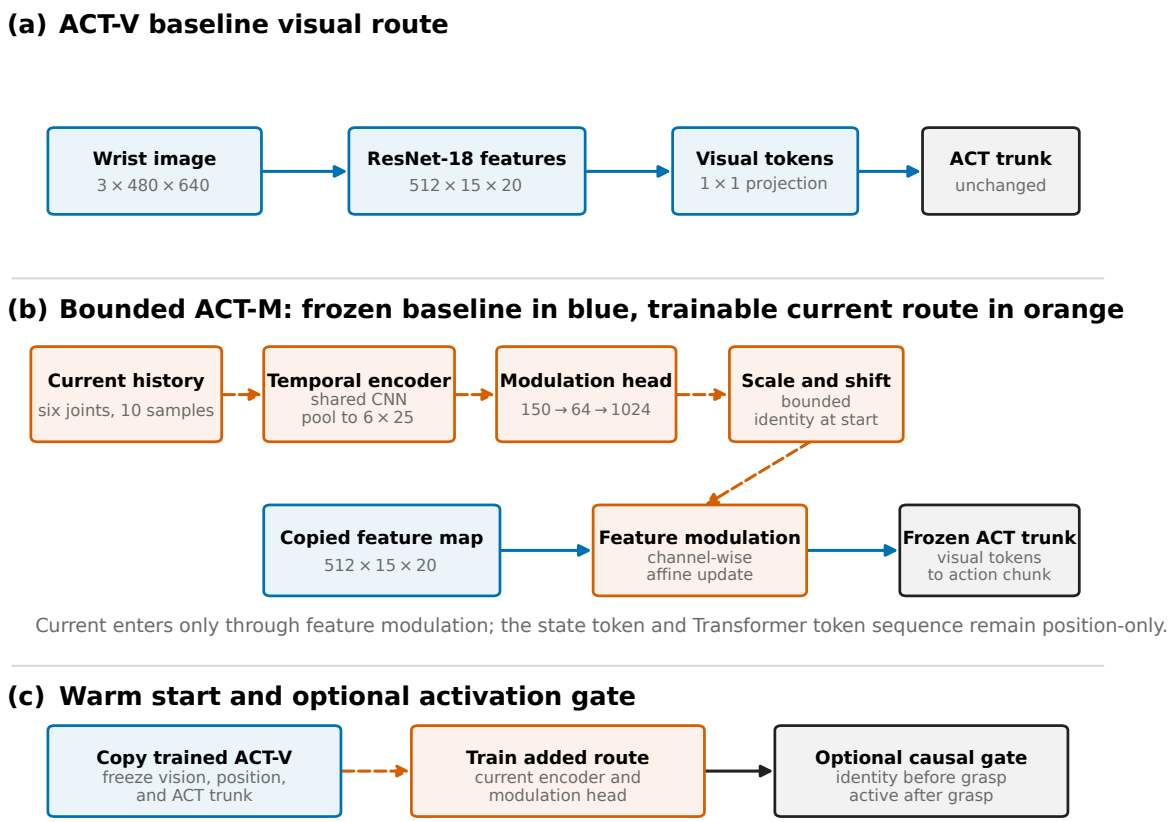



**Figure B.1:** Bounded ACT feature conditioning. The upper panel first shows the ACT-V visual route that is copied and frozen. The route reuses the six joint-wise temporal representations but sends them to an identity-initialised modulation head rather than to Transformer tokens. The trained vision-and-position base can remain frozen while the added current route learns bounded scale and shift values for the final ResNet feature map.

## B.2 Policy Development Screens

Tables B.1 and B.2 summarise the descriptive development outcomes used during model selection.

**Table B.1:** ACT development experiments retained outside the principal comparisons. In the bounded screen, concat denotes direct concatenation; scratch and warm distinguish training from random initialisation and a trained visual base, and gated adds the causal latch.

| Experiment | Compared outcomes | Success |
|---|---|---|
| Historical temporal and FiLM fusion | temporal CNN; FiLM | 6/30; 0/30 |
| Current transform pilot | winsorised linear; binned | 3/12; 2/12 |
| Static joint-token session | ACT-V; joint tokens | 5/12; 1/12 |
| Bounded feature-conditioning screen | control; concat; scratch FiLM; warm FiLM; gated FiLM | 3/5; 2/5; 1/5; 1/5; 3/5 |
| No-ensemble probe | joint-selective temporal ACT-M | 0/5 |
| Future-current pilot | ACT-V; current-aware | 3/5; 3/5 |
| Stopped cross-session extension | ACT-V; joint-selective temporal ACT-M | 3/5; not run |

**Table B.2:** Diffusion Policy development experiments retained outside the principal comparisons.

| Experiment | Compared outcomes | Success |
|---|---|---|
| Initial baseline and current screen | DP-V; direct DP-M; learned DP-M | 0/5; 0/5; 0/5 |
| Recovered vision baseline | DP-V | 17/30 |
| Recovered direct-current pilots | direct DP-M, EMA; online weights | 0/5; 0/5 |
| Direct-current session | direct DP-M | 1/6 |
| Behaviour-preserving screen | DP-V; concat; instantaneous residual; temporal residual; gated residual | 4/5; 0/5; 3/5; 4/5; 1/5 |
| Final candidate screen | control; joint-selective residual; cross-attention | 1/5; 3/5; 2/5 |
| Future-current pilot | DP-V; current-aware | 2/5; 0/5 |
| Stopped cross-session extension | DP-V; DP-M | 0/5; 1/10 |

## B.3 SmolVLA Evaluation

I tested whether the same demonstrations supported a pretrained vision–language–action policy family. It compared a merged 16-layer SmolVLA teacher with depth-distilled 6-layer and 4-layer students using wrist images, joint positions, and a fixed instruction. This evaluation aims to understand the vision (not current aware) capability of the model, impact of distillation on performance and generalisation capability after training.

**Table B.3:** Outcomes for the auxiliary 100-trial SmolVLA evaluation. Partial insertions are not counted as successes.

| Policy | Success | Partial | Failure | Total |
|---|---|---|---|---|
| Merged 16-layer teacher | 34 | 32 | 34 | 100 |
| Distilled 6-layer student | 11 | 28 | 61 | 100 |
| Distilled 4-layer student | 7 | 28 | 65 | 100 |

A later 20-trial probe crossed the 9,000-step and 30,000-step teacher checkpoints with the trained peg position and a second, 5 millimetre displaced peg as in Figure B.2. The 30,000-step checkpoint succeeded once in ten trials and the 9,000-step checkpoint did not succeed.

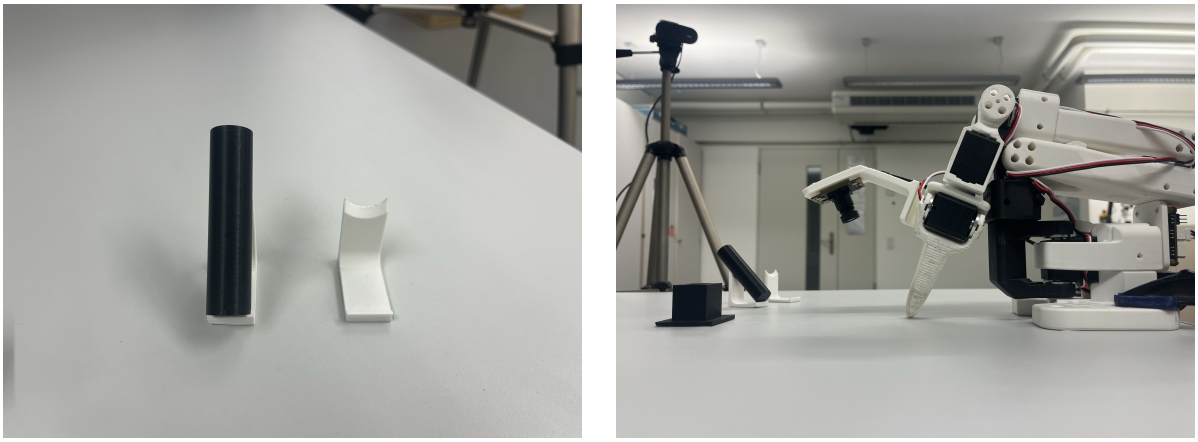


**Figure B.2:** Components and workspace arrangement for the second peg position used in the auxiliary SmolVLA probe.

# Bibliography


[1] T. Tsuji, Y. Kato, G. Solak, H. Zhang, T. Petrič, F. Nori, and A. Ajoudani, "A Survey on Imitation Learning for Contact-Rich Tasks in Robotics", *The International Journal of Robotics Research*, 2026. DOI: 10.1177/02783649261417694

[2] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware", in *Proc. Robotics: Science and Systems (RSS)*, Daegu, Republic of Korea, 2023. DOI: 10.15607/RSS.2023.XIX.016

[3] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, "Diffusion Policy: Visuomotor Policy Learning via Action Diffusion", in *Proc. Robotics: Science and Systems (RSS)*, Daegu, Republic of Korea, 2023. DOI: 10.15607/RSS.2023.XIX.026

[4] N. Funk, C. Chen, T. Schneider, G. Chalvatzaki, R. Calandra, and J. Peters, *On the Importance of Tactile Sensing for Imitation Learning: A Case Study on Robotic Match Lighting*, 2025. arXiv: 2504.13618 [cs.RO].

[5] J. J. Liu, Y. Li, K. Shaw, T. Tao, R. Salakhutdinov, and D. Pathak, "FACTR: Force-Attending Curriculum Training for Contact-Rich Policy Learning", in *Proc. Robotics: Science and Systems (RSS)*, Los Angeles, CA, USA, 2025. DOI: 10.15607/RSS.2025.XXI.079

[6] R. Watanabe, M. Alvarez, P. Ferreiro, P. Savkin, and G. Sano, *FTACT: Force Torque Aware Action Chunking Transformer for Pick-and-Reorient Bottle Task*, 2025. arXiv: 2509.23112 [cs.RO].

[7] Z. He, H. Fang, J. Chen, H.-S. Fang, and C. Lu, "FoAR: Force-Aware Reactive Policy for Contact-Rich Robotic Manipulation", *IEEE Robotics and Automation Letters*, vol. 10, no. 6, pp. 5625–5632, 2025. DOI: 10.1109/LRA.2025.3560871

[8] Waveshare, *ST3215 Serial Bus Servo Memory Register Map (Control Table)*, Version 3.7, Waveshare Electronics, 2023. Accessed: Sep. 5, 2026. [Online]. Available: https://files.waveshare.com/upload/2/27/ST3215%20memory%20register%20map-EN.xls

[9] A. Wahrburg, J. Bös, K. D. Listmann, F. Dai, B. Matthias, and H. Ding, "Motor-Current-Based Estimation of Cartesian Contact Forces and Torques for Robotic Manipulators and Its Application to Force Control", *IEEE Transactions on Automation Science and Engineering*, vol. 15, no. 2, pp. 879–886, 2018. DOI: 10.1109/TASE.2017.2691136

[10] A. del Prete, N. Mansard, O. E. Ramos, O. Stasse, and F. Nori, "Implementing Torque Control with High-Ratio Gear Boxes and Without Joint-Torque Sensors", *International Journal of Humanoid Robotics*, vol. 13, no. 1, 2016, Art. no. 1550044. DOI: 10.1142/S0219843615500449

[11] S. Shan and Q.-C. Pham, "Fine Robotic Manipulation Without Force/Torque Sensor", *IEEE Robotics and Automation Letters*, vol. 9, no. 2, pp. 1206–1213, 2024. DOI: 10.1109/LRA.2023.3341770

[12] J. Lu, X. Jiang, and T. Zou, "Learning-Driven Sensorless Interaction Force Estimation for Low-Cost Robot Arm with Limited Dynamic Features", *Mechatronics*, vol. 111, 2025, Art. no. 103396. DOI: 10.1016/j.mechatronics.2025.103396

[13] Z. Zhang, H. Xu, Z. Yang, C. Yue, Z. Lin, H.-a. Gao, Z. Wang, and H. Zhao, *TA-VLA: Elucidating the Design Space of Torque-aware Vision-Language-Action Models*, 2025. arXiv: 2509.07962 [cs.RO].

[14] S. Oh, J. J. Liu, T. Tao, P. Han, K. Shaw, S. Funabashi, R. Salakhutdinov, and D. Pathak, *FACTR 2: Learning External Force Sensing for Commodity Robot Arms Improves Policy Learning*, 2026. arXiv: 2606.12406 [cs.RO].

[15] W. Wang, D. Tran, and M. Feiszli, "What Makes Training Multi-Modal Classification Networks Hard?", in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 12 695–12 705. DOI: 10.1109/CVPR42600.2020.01271

[16] X. Peng, Y. Wei, A. Deng, D. Wang, and D. Hu, "Balanced Multimodal Learning via On-the-fly Gradient Modulation", in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 8238–8247. DOI: 10.1109/CVPR52688.2022.00806

[17] A. Chaudhuri, A. Dutta, T. Bui, and S. Georgescu, "A Closer Look at Multimodal Representation Collapse", in *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267, PMLR, 2025, pp. 7555–7577. arXiv: 2505.22483.

[18] A. De Luca and R. Mattone, "Sensorless Robot Collision Detection and Hybrid Force/Motion Control", in *IEEE Proc. Intl. Conf. on Robotics and Automation (ICRA)*, Barcelona, Spain, 2005, pp. 999–1004. DOI: 10.1109/ROBOT.2005.1570247

[19] S. Haddadin, A. De Luca, and A. Albu-Schäffer, "Robot Collisions: A Survey on Detection, Isolation, and Identification", *IEEE Transactions on Robotics*, vol. 33, no. 6, pp. 1292–1312, 2017. DOI: 10.1109/TRO.2017.2723903

[20] K. Yamane, Y. Li, M. Konosu, K. Inami, J. Oaki, T. Tsuji, and S. Sakaino, *Design and Experimental Validation of Sensorless 4-Channel Bilateral Teleoperation for Low-Cost Manipulators*, 2025. arXiv: 2507.06174 [cs.RO].

[21] T. Buamanee, M. Kobayashi, Y. Uranishi, and H. Takemura, "Bi-ACT: Bilateral Control-Based Imitation Learning via Action Chunking with Transformer", in *IEEE Proc. Intl. Conf. on Advanced Intelligent Mechatronics (AIM)*, Boston, MA, USA, 2024, pp. 410–415. DOI: 10.1109/AIM55361.2024.10637173

[22] J. Zhao, N. Kuppuswamy, S. Feng, B. Burchfiel, and E. H. Adelson, "PolyTouch: A Robust Multi-Modal Tactile Sensor for Contact-Rich Manipulation Using Tactile-Diffusion Policies", in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 104–110. DOI: 10.1109/ICRA55743.2025.11128816

[23] B. Zhou, R. Jiao, Y. Li, X. Yuan, F. Fang, and S. Li, "Admittance Visuomotor Policy Learning for General-Purpose Contact-Rich Manipulations", *IEEE Transactions on Industrial Electronics*, vol. 72, no. 11, pp. 11 520–11 529, 2025. DOI: 10.1109/TIE.2025.3561816

[24] B. Huang, Y. Wang, X. Yang, Y. Luo, and Y. Li, "3D-ViTac: Learning Fine-Grained Manipulation with Visuo-Tactile Sensing", in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270, PMLR, 2025, pp. 2557–2578. arXiv: 2410.24091.

[25] K. Yu, Y. Han, Q. Wang, V. Saxena, D. Xu, and Y. Zhao, "MimicTouch: Leveraging Multi-modal Human Tactile Demonstrations for Contact-rich Manipulation", in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270, PMLR, 2025, pp. 4844–4865. arXiv: 2310.16917.

[26] P. M. Uriguen Eljuri, H. Shibata, K. Maeyama, Y. Jia, and T. Taniguchi, *Haptic-Informed ACT with a Soft Gripper and Recovery-Informed Training for Pseudo Oocyte Manipulation*, 2025. arXiv: 2506.18212 [cs.RO].

[27] E. Helmut, N. Funk, T. Schneider, C. de Farias, and J. Peters, *Tactile-Conditioned Diffusion Policy for Force-Aware Robotic Manipulation*, 2025. arXiv: 2510.13324 [cs.RO].

[28] M. Wang, Z. Yue, R. Lu, et al., *PhaForce: Phase-Scheduled Visual–Force Policy Learning with Slow Planning and Fast Correction for Contact-Rich Manipulation*, 2026. arXiv: 2603.08342 [cs.RO].

[29] H. Xue, J. Ren, W. Chen, G. Zhang, F. Yuan, G. Gu, H. Xu, and C. Lu, "Reactive Diffusion Policy: Slow-Fast Visual-Tactile Policy Learning for Contact-Rich Manipulation", in *Proc. Robotics: Science and Systems (RSS)*, Los Angeles, CA, USA, 2025. DOI: 10.15607/RSS.2025.XXI.052

[30] W. Chen, H. Xue, Y. Wang, F. Zhou, J. Lv, Y. Jin, S. Tang, C. Wen, and C. Lu, *ImplicitRDP: An End-to-End Visual-Force Diffusion Policy with Structural Slow-Fast Learning*, 2025. arXiv: 2512.10946.

[31] J. Yu, H. Liu, Q. Yu, et al., *ForceVLA: Enhancing VLA Models with a Force-Aware MoE for Contact-Rich Manipulation*, 2025. arXiv: 2505.22159 [cs.RO].

[32] Y. Li, Zhaxizhuoma, H. Jiang, et al., "ForceVLA2: Unleashing Hybrid Force-Position Control with Force Awareness for Contact-Rich Manipulation", in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2026, pp. 8911–8920. arXiv: 2603.15169 [cs.RO].

[33] W. Xie and N. Correll, *Towards Forceful Robotic Foundation Models: A Literature Survey*, 2025. arXiv: 2504.11827 [cs.RO].

[34] S. Shan, C. Zhou, R. Wang, et al., *Learning Physical Interaction: A Survey of Tactile- and Force-aware Robot Learning*, 2026. arXiv: 2608.07558 [cs.RO].

[35] D. E. Whitney, "Quasi-Static Assembly of Compliantly Supported Rigid Parts", *ASME Journal of Dynamic Systems, Measurement, and Control*, vol. 104, no. 1, pp. 65–77, 1982. DOI: 10.1115/1.3149634

[36] M. T. Mason, "Compliance and Force Control for Computer Controlled Manipulators", *IEEE Transactions on Systems, Man and Cybernetics*, vol. 11, no. 6, pp. 418–432, 1981. DOI: 10.1109/TSMC.1981.4308708

[37] N. Hogan, "Impedance Control: An Approach to Manipulation: Part I: Theory", *ASME Journal of Dynamic Systems, Measurement, and Control*, vol. 107, no. 1, pp. 1–7, 1985. DOI: 10.1115/1.3140702

[38] T. Lozano-Pérez, M. T. Mason, and R. H. Taylor, "Automatic Synthesis of Fine-Motion Strategies for Robots", *The International Journal of Robotics Research*, vol. 3, no. 1, pp. 3–24, 1984. DOI: 10.1177/027836498400300101

[39] L. P. Kaelbling, M. L. Littman, and A. R. Cassandra, "Planning and Acting in Partially Observable Stochastic Domains", *Artificial Intelligence*, vol. 101, no. 1–2, pp. 99–134, 1998. DOI: 10.1016/S0004-3702(98)00023-X

[40] D. A. Pomerleau, "ALVINN: An Autonomous Land Vehicle in a Neural Network", in *Advances in Neural Information Processing Systems*, vol. 1, 1988. [Online]. Available: https://proceedings.neurips.cc/paper/1988/hash/812b4ba287f5ee0bc9d43bbf5bbe87fb-Abstract.html

[41] S. Ross, G. J. Gordon, and J. A. Bagnell, "A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning", in *Proceedings of the 14th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2011, pp. 627–635. arXiv: 1011.0686.

[42] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition", in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778. DOI: 10.1109/CVPR.2016.90

[43] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention Is All You Need", in *Advances in Neural Information Processing Systems*, vol. 30, 2017. arXiv: 1706.03762.

[44] K. Sohn, H. Lee, and X. Yan, "Learning Structured Output Representation Using Deep Conditional Generative Models", in *Advances in Neural Information Processing Systems*, vol. 28, 2015. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2015/hash/8d55a249e6baa5c06772297520da2051-Abstract.html

[45] J. Ho, A. Jain, and P. Abbeel, "Denoising Diffusion Probabilistic Models", in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 6840–6851. arXiv: 2006.11239.

[46] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville, "FiLM: Visual Reasoning with a General Conditioning Layer", in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, 2018. DOI: 10.1609/aaai.v32i1.11671

[47] J. Song, C. Meng, and S. Ermon, "Denoising Diffusion Implicit Models", in *Proc. International Conference on Learning Representations (ICLR)*, 2021. arXiv: 2010.02502.

[48] T. Baltrušaitis, C. Ahuja, and L.-P. Morency, "Multimodal Machine Learning: A Survey and Taxonomy", *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 2, pp. 423–443, 2019. DOI: 10.1109/TPAMI.2018.2798607

[49] S. Li and H. Tang, *Multimodal Alignment and Fusion: A Survey*, 2024. arXiv: 2411.17040.

[50] S. Bai, J. Z. Kolter, and V. Koltun, *An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling*, 2018. arXiv: 1803.01271 [cs.LG].

[51] Feetech, *ST-3215-C018 Product Specification*, A/0, Shenzhen Feetech RC Model Co., Ltd., 2023. [Online]. Available: https://www.feetechrc.com/Data/feetechrc/upload/file/20260622/639177251992311307585485 1.pdf

[52] B. Tang, M. A. Lin, I. A. Akinola, A. Handa, G. S. Sukhatme, F. Ramos, D. Fox, and Y. S. Narang, "IndustReal: Transferring Contact-Rich Assembly Tasks from Simulation to Reality", in *Proc. Robotics: Science and Systems (RSS)*, 2023. DOI: 10.15607/RSS.2023.XIX.039

[53] R. Cadene, S. Aliberts, F. Capuano, et al., "LeRobot: An Open-Source Library for End-to-End Robot Learning", in *Proc. Intl. Conf. on Learning Representations (ICLR)*, 2026. DOI: 10.48550/arXiv.2602.22818

[54] E. B. Wilson, "Probable Inference, the Law of Succession, and Statistical Inference", *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927. DOI: 10.1080/01621459.1927.10502953

[55] R. G. Newcombe, "Interval Estimation for the Difference Between Independent Proportions: Comparison of Eleven Methods", *Statistics in Medicine*, vol. 17, no. 8, pp. 873–890, 1998. DOI: 10.1002/(SICI)1097-0258(19980430)17:8<873::AID-SIM779>3.0.CO;2-I

[56] R. A. Fisher, "On the Interpretation of $\chi^2$ from Contingency Tables, and the Calculation of $P$", *Journal of the Royal Statistical Society*, vol. 85, no. 1, pp. 87–94, 1922. DOI: 10.1111/j.2397-2335.1922.tb00768.x

[57] S. Holm, "A Simple Sequentially Rejective Multiple Test Procedure", *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979. [Online]. Available: https://www.jstor.org/stable/4615733

[58] P. Armitage, "Tests for Linear Trends in Proportions and Frequencies", *Biometrics*, vol. 11, no. 3, pp. 375–386, 1955. DOI: 10.2307/3001775

[59] B. Zhu, M. Khalil, E. Harrison, et al., *Learning to Act While Waiting: RL Finetuning of Generalist Robot Policies Under Inference Latency*, 2026. arXiv: 2608.23831 [cs.RO].

[60] R. Calandra, A. Owens, M. Upadhyaya, W. Yuan, J. Lin, E. H. Adelson, and S. Levine, "The Feeling of Success: Does Touch Sensing Help Predict Grasp Outcomes?", in *Proceedings of the 1st Annual Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 78, PMLR, 2017, pp. 314–323. arXiv: 1710.05512.